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EDITED BY
J. MCKEEN CATTELL

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MAY TO OCTOBER, 1900

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OCTOBER, 1900.

ADDRESS OF THE PRESIDENT BEFORE THE BRITISH ASSOCIATION.^A

BY SIR WILLIAM TURNER, F. R. S.,
UNIVERSITY OF EDINBURGH.

^A Given at Bradford on September 5, 1900.

TWENTY-SEVEN years ago the British Association met in Bradford, not at that time raised to the dignity of a city. The meeting was very successful, and was attended by about two thousand persons—a forecast, let us hope, of what we may expect at the present assembly. A distinguished chemist, Prof. A. W. Williamson, presided. On this occasion the association has elected for the presidential chair one whose attention has been given to the study of an important department of biological science. His claim to occupy, however unworthily, the distinguished position in which he has been placed, rests, doubtless, on the fact that, in the midst of the engrossing duties devolving on a teacher in a great university and school of medicine, he has endeavored to contribute to the sum of knowledge of the science which he professes. It is a matter of satisfaction to feel that the success of a meeting of this kind does not rest upon the shoulders of the occupant of the presidential chair, but is due to the eminence and active coöperation of the men of science who either preside over or engage in the work of the nine or ten sections into which the association is divided, and to the energy and ability for organization displayed by the local secretaries and committees. The programme prepared by the general and local officers of the association shows that no efforts have been spared to provide an ample bill of fare, both in its scientific and social aspects. Members and associates will, I feel sure, take away from the Bradford meeting as pleasant memories as did our

colleagues of the corresponding Association Française, when, in friendly collaboration at Dover last year, they testified to the common citizenship of the Universal Republic of Science. As befits a leading center of industry in the great county of York, the applications of science to the industrial arts and to agriculture will form subjects of discussion in the papers to be read at the meeting.

Since the association was at Dover a year ago, two of its former presidents have joined the majority. The Duke of Argyll presided at the meeting in Glasgow so far back as 1855. Throughout his long and energetic life, he proved himself to be an eloquent and earnest speaker, one who gave to the consideration of public affairs a mind of singular independence, and a thinker and writer in a wide range of human knowledge. Sir J. Wm. Dawson was president at the meeting in Birmingham in 1886. Born in Nova Scotia in 1820, he devoted himself to the study of the Geology of Canada, and became the leading authority on the subject. He took also an active and influential part in promoting the spread of scientific education in the Dominion, and for a number of years he was Principal and Vice-Chancellor of the McGill University, Montreal.

SCIENTIFIC METHOD.

Edward Gibbon has told us that diligence and accuracy are the only merits which an historical writer can ascribe to himself. Without doubt they are fundamental qualities necessary for historical research, but in order to bear fruit they require to be exercised by one whose mental qualities are such as to enable him to analyze the data brought together by his diligence, to discriminate between the false and the true, to possess an insight into the complex motives that determine human action, to be able to recognize those facts and incidents which had exercised either a primary or only a secondary influence on the affairs of nations, or on the thoughts and doings of the person whose character he is depicting.

In scientific research, also, diligence and accuracy are fundamental qualities. By their application new facts are discovered and tabulated, their order of succession is ascertained and a wider and more intimate knowledge of the processes of nature is acquired. But to decide on their true

significance a well-balanced mind and the exercise of prolonged thought and reflection are needed. William Harvey, the father of exact research in physiology, in his memorable work, 'De Motu Cordis et Sanguinis,' published more than two centuries ago, tells us of the great and daily diligence which he exercised in the course of his investigations, and the numerous observations and experiments which he collated. At the same time he refers repeatedly to his cogitations and reflections on the meaning of what he had observed, without which the complicated movements of the heart could not have been analyzed, their significance determined and the circulation of the blood in a continuous stream definitely established. Early in the present century, Carl Ernst von Baer, the father of embryological research, showed the importance which he attached to the combination of observation with meditation by placing side by side on the title page of his famous treatise 'Ueber Entwicklungsgeschichte der Thiere' (1828) the words *Beobachtung und Reflexion*.

Though I have drawn from biological science my illustrations of the need of this combination, it must not be inferred that it applies exclusively to one branch of scientific inquiry; the conjunction influences and determines progress in all the sciences, and when associated with a sufficient touch of imagination, when the power of seeing is conjoined with the faculty of foreseeing, of projecting the mind into the future, we may expect something more than the discovery of isolated facts; their coördination and the enunciation of new principles and laws will necessarily follow.

Scientific method consists, therefore, in close observation, frequently repeated so as to eliminate the possibility of erroneous seeing; in experiments checked and controlled in every direction in which fallacies might arise; in continuous reflection on the appearances and phenomena observed, and in logically reasoning out their meaning and the conclusions to be drawn from them. Were the method followed out in its integrity by all who are engaged in scientific investigations, the time and labor expended in correcting errors committed by ourselves or by other observers and experimentalists would be saved, and the volumes devoted annually to scientific literature would be materially diminished in size. Were it applied, as far as the conditions of life admit, to the conduct and management of human affairs, we should not require to be told, when critical periods in our

welfare as a nation arise, that we shall muddle through somehow. Recent experience has taught us that wise discretion and careful provision are as necessary in the direction of public affairs as in the pursuit of science, and in both instances, when properly exercised, they enable us to reach with comparative certainty the goal which we strive to attain.

IMPROVEMENTS IN MEANS OF OBSERVATION.

While certain principles of research are common to all the sciences, each great division requires for its investigation specialized arrangements to insure its progress. Nothing contributes so much to the advancement of knowledge as improvements in the means of observation, either by the discovery of new adjuncts to research, or by a fresh adaptation of old methods. In the industrial arts, the introduction of a new kind of raw material, the recognition that a mixture or blending is often more serviceable than when the substances employed are uncombined, the discovery of new processes of treating the articles used in manufactures, the invention of improved machinery, all lead to the expansion of trade to the occupation of the people, and to the development of great industrial centers. In science, also, the invention and employment of new and more precise instruments and appliances enable us to appreciate more clearly the signification of facts and phenomena which were previously obscure, and to penetrate more deeply into the mysteries of nature. They mark fresh departures in the history of science, and provide a firm base of support from which a continuous advance may be made and fresh conceptions of nature can be evolved.

It is not my intention, even had I possessed the requisite knowledge, to undertake so arduous a task as to review the progress which has recently been made in the great body of sciences which lie within the domain of the British Association. As my occupation in life has required me to give attention to the science which deals with the structure and organization of the bodies of man and animals—a science which either includes within its scope or has intimate and widespread relations to comparative anatomy, embryology, morphology, zoölogy, physiology and anthropology—I shall limit myself to the attempt to bring before you some of the more important

observations and conclusions which have a bearing on the present position of the subject. As this is the closing year of the century it will not, I think, be out of place to refer to the changes which a hundred years have brought about in our fundamental conceptions of the structure of animals. In science, as in business, it is well from time to time to take stock of what we have been doing, so that we may realize where we stand and ascertain the balance to our credit in the scientific ledger.

So far back as the time of the ancient Greeks it was known that the human body and those of the more highly organized animals were not homogeneous, but were built up of parts, the *partes dissimilares* (τὰ ἀνόμοια μέρη {ta anomoia merê}) of Aristotle, which differed from each other in form, color, texture, consistency and properties. These parts were familiarly known as the bones, muscles, sinews, blood-vessels, glands, brain, nerves and so on. As the centuries rolled on, and as observers and observations multiplied, a more and more precise knowledge of these parts throughout the animal kingdom was obtained, and various attempts were made to classify animals in accordance with their forms and structure. During the concluding years of the last century and the earlier part of the present, the Hunters, William and John, in our country, the Meckels in Germany, Cuvier and St. Hilaire in France, gave an enormous impetus to anatomical studies, and contributed largely to our knowledge of the construction of the bodies of animals. But whilst by these and other observers the most salient and, if I may use the expression, the grosser characters of animal organization had been recognized, little was known of the more intimate structure or texture of the parts. So far as could be determined by the unassisted vision, and so much as could be recognized by the use of a simple lens, had indeed been ascertained, and it was known that muscles, nerves and tendons were composed of threads or fibers, and the blood and lymph-vessels were tubes, that the parts which we call fasciæ and aponeuroses were thin membranes and so on.

Early in the present century, Xavier Bichat, one of the most brilliant men of science during the Napoleonic era in France, published his 'Anatomie Générale,' in which he formulated important general principles. Every animal is an assemblage of different organs, each of which discharges a function, and acting together, each in its own way, assists in the preservation of the whole. The organs are, as it were, special machines

situated in the general building which constitutes the factory or body of the individual. But, further, each organ or special machine is itself formed of tissues which possess different properties. Some, as the blood-vessels, nerves, fibrous tissues, etc., are generally distributed throughout the animal body, whilst others, as bones, muscles, and cartilage, etc., are found only in certain definite localities. While Bichat had acquired a definite philosophical conception of the general principles of construction and of the distribution of the tissues, neither he nor his pupil Béclard was in a position to determine the essential nature of the structural elements. The means and appliances at their disposal and at that of other observers in their generation were not sufficiently potent to complete the analysis.

Attempts were made in the third decennium of this century to improve the methods of examining minute objects by the manufacture of compound lenses, and, by doing away with chromatic and spherical aberration, to obtain, in addition to magnification of the object, a relatively large flat field of vision with clearness and sharpness of definition. When in January, 1830, Joseph Jackson Lister read to the Royal Society his memoir "On Some Properties in Achromatic Object-Glasses Applicable to the Improvement of Microscopes," he announced the principles on which combinations of lenses could be arranged, which would possess these qualities. By the skill of our opticians, microscopes have now for more than half a century been constructed which, in the hands of competent observers, have influenced and extended biological science with results comparable to those obtained by the astronomer through improvements in the telescope.

In the study of the minute structure of plants and animals, the observer has frequently to deal with tissues and organs, most of which possess such softness and delicacy of substance and outline that, even when microscopes of the best construction are employed, the determination of the intimate nature of the tissue, and the precise relation which one element of an organ bears to the other constituent elements, is, in many instances, a matter of difficulty. Hence additional methods have had to be devised in order to facilitate study and to give precision and accuracy to our observations. It is difficult for one of the younger generation of biologists, with all the appliances of a well-equipped laboratory at his command, with experienced teachers to direct him in his work, and with excellent text-books, in which the modern methods are described, to realize the conditions under which his

predecessors worked half a century ago. Laboratories for minute biological research had not been constructed, the practical teaching of histology and embryology had not been organized, experience in methods of work had not accumulated; each man was left to his individual efforts, and had to puzzle his way through the complications of structure to the best of his power. Staining and hardening reagents were unknown. The double-bladed knife invented by Valentin, held in the hand, was the only improvement on the scalpel or razor for cutting thin, more or less translucent slices suitable for microscopic examination; mechanical section-cutters and freezing arrangements had not been devised. The tools at the disposal of the microscopist were little more than knife, forceps, scissors, needles; with acetic acid, glycerine and Canada balsam as reagents. But in the employment of the newer methods of research care has to be taken, more especially when hardening and staining reagents are used, to discriminate between appearances which are to be interpreted as indicating natural characters, and those which are only artificial productions.

Notwithstanding the difficulties attendant on the study of the more delicate tissues, the compound achromatic microscope provided anatomists with an instrument of great penetrative power. Between the years 1830 and 1850 a number of acute observers applied themselves with much energy and enthusiasm to the examination of the minute structure of the tissues and organs in plants and animals.

CELL THEORY.

It had, indeed, long been recognized that the tissues of plants were to a large extent composed of minute vesicular bodies, technically called cells (Hooke, Malpighi, Grew). In 1831 the discovery was made by the great botanist, Robert Brown, that in many families of plants a circular spot, which he named areola or nucleus, was present in each cell; and in 1838 M. J. Schleiden published the fact that a similar spot or nucleus was a universal elementary organ in vegetables. In the tissues of animals also structures had begun to be recognized comparable with the cells and nuclei of the vegetable tissues, and in 1839 Theodore Schwann announced the important generalization that there is one universal principle of development

for the elementary part of organisms, however different they may be in appearance, and that this principle is the formation of cells. The enunciation of the fundamental principle that the elementary tissues consisted of cells constituted a step in the progress of biological science which will forever stamp the century now drawing to a close with a character and renown equalling those which it has derived from the most brilliant discoveries in the physical sciences. It provided biologists with the visible anatomical units through which the external forces operating on, and the energy generated in, living matter come into play. It dispelled forever the old mystical idea of the influence exercised by vapors or spirits in living organisms. It supplied the physiologist and pathologist with the specific structures through the agency of which the functions of organisms are discharged in health and disease. It exerted an enormous influence on the progress of practical medicine. A review of the progress of knowledge of the cell may appropriately enter into an address on this occasion.

STRUCTURE OF CELLS.

A cell is a living particle, so minute that it needs a microscope for its examination; it grows in size, maintains itself in a state of activity, responds to the action of stimuli, reproduces its kind and in the course of time it degenerates and dies.

Let us glance at the structure of a cell to determine its constituent parts and the rôle which each plays in the function to be discharged. The original conception of a cell, based upon the study of the vegetable tissues, was a minute vesicle inclosed by a definite wall, which exercised chemical or metabolic changes on the surrounding material and secreted into the vesicle its characteristic contents. A similar conception was at first also entertained regarding the cells of animal tissues; but as observations multiplied, it was seen that numerous elementary particles, which were obviously in their nature cells, did not possess an inclosing envelope. A wall ceased to have a primary value as a constituent part of a cell, the necessary vesicular character of which therefore could no longer be entertained.

The other constituent parts of a cell are the cell plasm, which forms the body of the cell, and the nucleus embedded in its substance.

Notwithstanding the very minute size of the nucleus, which even in the largest cells is not more than one-five-hundredth of an inch in diameter, and usually is considerably smaller, its almost constant form, its well-defined sharp outline and its power of resisting the action of strong reagents when applied to the cell, have from the period of its discovery by Robert Brown caused histologists to bestow on it much attention. Its structure and chemical composition; its mode of origin; the part which it plays in the formation of new cells, and its function in nutrition and secretion have been investigated.

When examined under favorable conditions in its passive or resting state, the nucleus is seen to be bounded by a membrane which separates it from the cell plasma and gives it the characteristic sharp contour. It contains an apparently structureless nuclear substance, nucleoplasm or enchylema, in which are embedded one or more extremely minute particles called nucleoli, along with a network of exceedingly fine threads or fibers, which in the active living cell play an essential part in the production of new nuclei within the cell. In its chemical composition the nuclear substance consists of albuminous plastin and globulin; and of a special material named nuclein, rich in phosphorus and with an acid reaction. The delicate network within the nucleus consists apparently of the nuclein, a substance which stains with carmine and other dyes, a property which enables the changes, which take place in the network in the production of young cells, to be more readily seen and followed out by the observer.

The mode of origin of the nucleus and the part which it plays in the production of new cells have been the subject of much discussion. Schleiden, whose observations, published in 1838, were made on the cells of plants, believed that within the cell a nucleolus first appeared, and that around it molecules aggregated to form the nucleus. Schwann again, whose observations were mostly made on the cells of animals, considered that an amorphous material existed in organized bodies, which he called cytoblastema. It formed the contents of cells, or it might be situated free or external to them. He figuratively compared it to a mother liquor in which crystals are formed. Either in the cytoblastema within the cells or in that situated external to them, the aggregation of molecules around a nucleolus to form a nucleus might occur, and, when once the nucleus had been formed, in its turn it would serve as a center of aggregation of additional

molecules from which a new cell would be produced. He regarded, therefore, the formation of nuclei and cells as possible in two ways—one within preëxisting cells (endogenous cell-formation), the other in a free blastema lying external to cells (free cell-formation). In animals, he says, the endogenous method is rare, and the customary origin is in an external blastema. Both Schleiden and Schwann considered that after the cell was formed the nucleus had no permanent influence on the life of the cell, and usually disappeared.

Under the teaching principally of Henle, the famous Professor of Anatomy in Göttingen, the conception of the free formation of nuclei and cells in a more or less fluid blastema, by an aggregation of elementary granules and molecules, obtained so much credence, especially amongst those who were engaged in the study of pathological processes, that the origin of cells within preëxisting cells was to a large extent lost sight of. That a parent cell was requisite for the production of new cells seemed to many investigators to be no longer needed. Without doubt this conception of free cell-formation contributed in no small degree to the belief, entertained by various observers, that the simplest plants and animals might arise, without preëxisting parents, in organic fluids destitute of life, by a process of spontaneous generation; a belief which prevailed in many minds almost to the present day. If, as has been stated, the doctrine of abiogenesis cannot be experimentally refuted, on the other hand it has not been experimentally proved. The burden of proof lies with those who hold the doctrine, and the evidence that we possess is all the other way.

MULTIPLICATION OF CELLS.

Although von Mohl, the botanist, seems to have been the first to recognize (1835) in plants a multiplication of cells by division, it was not until attention was given to the study of the egg in various animals and to the changes which take place in it, attendant on fertilization, that in the course of time a much more correct conception of the origin of the nucleus and of the part which it plays in the formation of new cells was obtained. Before Schwann had published his classical memoir in 1839, von Baer and other observers had recognized within the animal ovum the germinal

vesicle, which obviously bore to the ovum the relation of a nucleus to a cell. As the methods of observation improved, it was recognized that, within the developing egg, two vesicles appeared where one only had previously existed, to be followed by four vesicles, then eight, and so on in multiple progression until the ovum contained a multitude of vesicles, each of which possessed a nucleus. The vesicles were obviously cells which had arisen within the original germ-cell or ovum. These changes were systematically described by Martin Barry so long ago as 1839 and 1840 in two memoirs communicated to the Royal Society of London, and the appearance produced, on account of the irregularities of the surface occasioned by the production of new vesicles, was named by him the mulberry-like structure. He further pointed out that the vesicles arranged themselves as a layer within the envelope of the egg or zona pellucida, and that the whole embryo was composed of cells filled with the foundations of other cells. He recognized that the new cells were derived from the germinal vesicle or nucleus of the ovum, the contents of which entered into the formation of the first two cells, each of which had its nucleus, which in its turn resolved itself into other cells, and by a repetition of the process into a greater number. The endogenous origin of new cells within a preëxisting cell and the process which we now term the segmentation of the yolk were successfully demonstrated. In a third memoir, published in 1841, Barry definitely stated that young cells originated through division of the nucleus of the parent cell, instead of arising, as a product of crystallization, in the fluid cytoblastema of the parent cell or in a blastema situated external to the cell.

In a memoir published in 1842, John Goodsir advocated the view that the nucleus is the reproductive organ of the cell, and that from it, as from a germinal spot, new cells were formed. In a paper, published three years later, on nutritive centers, he described cells, the nuclei of which were the permanent source of successive broods of young cells, which from time to time occupied the cavity of the parent cell. He extended also his observations on the endogenous formation of cells to the cartilage cells in the process of inflammation and to other tissues undergoing pathological changes. Corroborative observations on endogenous formation were also given by his brother, Harry Goodsir, in 1845. These observations on the part which the nucleus plays by cleavage in the formation of young cells by endogenous development from a parent center—that an organic continuity

existed between a mother cell and its descendants through the nucleus—constituted a great step in advance of the views entertained by Schleiden and Schwann, and showed that Barry and the Goodsirs had a deeper insight into the nature and functions of cells than was possessed by most of their contemporaries, and are of the highest importance when viewed in the light of recent observations.

In 1841 Robert Remak published an account of the presence of two nuclei in the blood corpuscles of the chick and the pig, which he regarded as evidence of the production of new corpuscles by division of the nucleus within a parent cell; but it was not until some years afterwards (1850 to 1855) that he recorded additional observations and recognized that division of the nucleus was the starting-point for the multiplication of cells in the ovum and in the tissues generally. Remak's view was that the process of cell division began with the cleavage of the nucleolus, followed by that of the nucleus, and that again by cleavage of the body of the cell and its membrane. Kölliker had previously, in 1843, described the multiplication of nuclei in the ova of parasitic worms, and drew the inference that in the formation of young cells within the egg the nucleus underwent cleavage, and that each of its divisions entered into the formation of a new cell. By these observations, and by others subsequently made, it became obvious that the multiplication of animal cells, either by division of the nucleus within the cell, or by the budding off of a part of the protoplasm of the cell, was to be regarded as a widely spread and probably a universal process, and that each new cell arose from a parent cell.

Pathological observers were, however, for the most part inclined to consider free cell-formation in a blastema or exudation by an aggregation of molecules, in accordance with the views of Henle, as a common phenomenon. This proposition was attacked with great energy by Virchow in a series of memoirs published in his 'Archiv,' commencing in Vol. 1, 1847, and finally received its death-blow in his published lectures on Cellular Pathology, 1858. He maintained that in pathological structures there was no instance of cell development *de novo*; where a cell existed, there one must have been before. Cell-formation was a continuous development by descent, which he formulated in the expression *omnis cellula e cellula*.

KARYOKINESIS.

While the descent of cells from preëxisting cells by division of the nucleus during the development of the egg, in the embryos of plants and animals, and in adult vegetable and animal tissues, both in healthy and diseased conditions, had now become generally recognized, the mechanism of the process by which the cleavage of the nucleus took place was for a long time unknown. The discovery had to be deferred until the optician had been able to construct lenses of a higher penetrative power, and the microscopist had learned the use of coloring agents capable of dyeing the finest elements of the tissues. There was reason to believe that in some cases a direct cleavage of the nucleus, to be followed by a corresponding division of the cell into two parts, did occur. In the period between 1870 and 1880 observations were made by Schneider, Strasburger, Bütschli, Fol, van Beneden and Flemming, which showed that the division of the nucleus and the cell was due to a series of very remarkable changes, now known as indirect nuclear and cell division, or karyokinesis. The changes within the nucleus are of so complex a character that it is impossible to follow them in detail without the use of appropriate illustrations. I shall have to content myself, therefore, with an elementary sketch of the process.

I have previously stated that the nucleus in its passive or resting stage contains a very delicate network of threads or fibers. The first stage in the process of nuclear division consists in the threads arranging themselves in loops and forming a compact coil within the nucleus. The coil then becomes looser, the loops of threads shorten and thicken, and somewhat later each looped thread splits longitudinally into two portions. As the threads stain when coloring agents are applied to them, they are called chromatin fibers, and the loose coil is the chromosome (Waldeyer).

As the process continues, the investing membrane of the nucleus disappears, and the loops of threads arrange themselves within the nucleus so that the closed ends of the loops are directed to a common center, from which the loops radiate outwards and produce a starlike figure (aster). At the same time clusters of extremely delicate lines appear both in the nucleoplasm and in the body of the cell, named the achromatic figure, which has a spindle-like form with two opposite poles, and stains much more feebly than the chromatic fibers. The loops of the chromatic star then

arrange themselves in the equatorial plane of the spindle, and bending round turn their closed ends towards the periphery of the nucleus and the cell.

The next stage marks an important step in the process of division of the nucleus. The two longitudinal portions, into which each looped thread had previously split, now separate from each other, and whilst one part migrates to one pole of the spindle, the other moves to the opposite pole, and the free ends of each loop are directed toward its equator (metakinesis). By this division of the chromatin fibers, and their separation from each other to opposite poles of the spindle, two starlike chromatin figures are produced (dyaster).

Each group of fibers thickens, shortens, becomes surrounded by a membrane, and forms a new or daughter nucleus (dispirem). Two nuclei therefore have arisen within the cell by the division of that which had previously existed, and the expression formulated by Flemming—*omnis nucleus e nucleo*—is justified. Whilst this stage is in course of being completed, the body of the cell becomes constricted in the equatorial plane of the spindle, and, as the constriction deepens, it separates into two parts, each containing a daughter nucleus, so that two nucleated cells have arisen out of a preëxisting cell.

A repetition of the process in each of these cells leads to the formation of other cells, and, although modifications in details are found in different species of plants and animals, the multiplication of cells in the egg and in the tissues generally on similar lines is now a thoroughly established fact in biological science.

In the study of karyokinesis, importance has been attached to the number of chromosomes in the nucleus of the cell. Flemming had seen in the Salamander twenty-four chromosome fibers, which seems to be a constant number in the cells of epithelium and connective tissues. In other cells, again, especially in the ova of certain animals, the number is smaller, and fourteen, twelve, four and even two only have been described. The theory formulated by Boveri that the number of chromosomes is constant for each species, and that in the karyokinetic figures corresponding numbers are found in homologous cells, seems to be not improbable.

In the preceding description I have incidentally referred to the appearance in the proliferating cell of an achromatic spindle-like figure.

Although this was recognized by Fol in 1873, it is only during the last ten or twelve years that attention has been paid to its more minute arrangements and possible signification in cell-division.

The pole at each end of the spindle lies in the cell plasm which surrounds the nucleus. In the center of each pole is a somewhat opaque spot (central body) surrounded by a clear space, which, along with the spot, constitutes the centrosome of the sphere of attraction. From each centrosome extremely delicate lines may be seen to radiate in two directions. One set extends towards the pole at the opposite end of the spindle, and, meeting or coming into close proximity with radiations from it, constitutes the body of the spindle, which, like a perforated mantle, forms an imperfect envelope around the nucleus during the process of division. The other set of radiations is called the polar and extends in the region of the pole towards the periphery of the cell.

The question has been much discussed whether any constituent part of the achromatic figure, or the entire figure, exists in the cell as a permanent structure in its resting phase; or if it is only present during the process of karyokinesis. During the development of the egg the formation of young cells, by division of the segmentation nucleus, is so rapid and continuous that the achromatic figure, with the centrosome in the pole of the spindle, is a readily recognizable object in each cell. The polar and spindle-like radiations are in evidence during karyokinesis, and have apparently a temporary endurance and function. On the other hand, van Beneden and Boveri were of opinion that the central body of the centrosome did not disappear when the division of the nucleus came to an end, but that it remained as a constituent part of a cell lying in the cell plasm, near to the nucleus. Flemming has seen the central body with its sphere in leucocytes, as well as in epithelial cells and those of other tissues. Subsequently Heidenhain and other histologists have recorded similar observations. It would seem, therefore, as if there were reason to regard the centrosome, like the nucleus, as a permanent constituent of a cell. This view, however, is not universally entertained. If not always capable of demonstration in the resting stage of a cell, it is doubtless to be regarded as potentially present, and ready to assume, along with the radiations, a characteristic appearance when the process of nuclear division is about to begin.

One can scarcely regard the presence of so remarkable an appearance as the achromatic figure without associating with it an important function in the economy of the cell. As from the centrosome at the pole of the spindle both sets of radiations diverge, it is not unlikely that it acts as a center or sphere of energy and attraction. By some observers the radiations are regarded as substantive fibrillar structures, elastic or even contractile in their properties. Others, again, look upon them as morphological expressions of chemical and dynamical energy in the protoplasm of the cell body. On either theory we may assume that they indicate an influence, emanating, it may be, from the centrosome and capable of being exercised both on the cell plasm and on the nucleus contained in it. On the contractile theory, the radiations which form the body of the spindle, either by actual traction of the supposed fibrillæ or by their pressure on the nucleus which they surround, might impel during karyokinesis the dividing chromosome elements toward the poles of the spindle, to form there the daughter nuclei. On the dynamical theory, the chemical and physical energy in the centrosome might influence the cell plasm and the nucleus and attract the chromosome elements of the nucleus to the poles of the spindle. The radiated appearance would therefore be consequent and attendant on the physico-chemical activity of the centrosome. One or other of these theories may also be applied to the interpretation of the significance of the polar radiations.

CELL PLASM.

In the cells of plants, in addition to the cell wall, the cell body and the cell juice require to be examined. The material of the cell body, or the cell contents, was named by von Mohl (1846) protoplasm, and consisted of a colorless tenacious substance which partly lined the cell wall (primordial utricle) and partly traversed the interior of the cell as delicate threads inclosing spaces (vacuoles) in which the cell juice was contained. In the protoplasm the nucleus was embedded. Nägeli, about the same time, had also recognized the difference between the protoplasm and the other contents of vegetable cells, and had noticed its nitrogenous composition.

Though the analogy with a closed bladder or vesicle could no longer be sustained in the animal tissues, the name 'cell' continued to be retained for

descriptive purposes, and the body of the cell was spoken of as a more or less soft substance inclosing a nucleus (Leydig). In 1861 Max Schultze adopted for the substance forming the body of the animal cell the term 'protoplasm.' He defined a cell to be a particle of protoplasm in the substance of which a nucleus was situated. He regarded the protoplasm, as indeed had previously been pointed out by the botanist Unger, as essentially the same as the contractile sarcode which constitutes the body and pseudopodia of the *Amœba* and other Rhizopoda. As the term 'protoplasm,' as well as that of 'bioplasm' employed by Lionel Beale in a somewhat similar though not precisely identical sense, involves certain theoretical views of the origin and function of the body of the cell, it would be better to apply to it the more purely descriptive term 'cytoplasm' or 'cell plasm.'

Schultze defined protoplasm as a homogeneous, glassy, tenacious material, of a jelly-like or somewhat firmer consistency, in which numerous minute granules were embedded. He regarded it as the part of the cell especially endowed with vital energy, whilst the exact function of the nucleus could not be defined. Based upon this conception of the jelly-like character of protoplasm, the idea for a time prevailed that a structureless, dimly granular, jelly or slime destitute of organization, possessed great physiological activity, and was the medium through which the phenomena of life were displayed.

More accurate conceptions of the nature of the cell plasm soon began to be entertained. Brücke recognized that the body of the cell was not simple, but had a complex organization. Hemming observed that the cell plasm contained extremely delicate threads, which frequently formed a network, the interspaces of which were occupied by a more homogeneous substance. Where the threads crossed each other, granular particles (milkrosomen) were situated. Bütschli considered that he could recognize in the cell plasm a honeycomb-like appearance, as if it consisted of excessively minute chambers in which a homogeneous more or less fluid material was contained. The polar and spindle-like radiations visible during the process of karyokinesis, which have already been referred to, and the presence of the centrosome, possibly even during the resting stage of the cell, furnished additional illustrations of differentiation within the cell plasm. In many cells there appears also to be a difference in the character of the cell plasm which immediately surrounds the nucleus and that which lies at and near the

periphery of the cell. The peripheral part (ektoplasma) is more compact and gives a definite outline to the cell, although not necessarily differentiating into a cell membrane. The inner part (endoplasma) is softer and is distinguished by a more distinct granular appearance and by containing the products specially formed in each particular kind of cell during the nutritive process.

By the researches of numerous investigators on the internal organization of cells in plants and animals, a large body of evidence has now been accumulated, which shows that both the nucleus and the cell plasm consist of something more than a homogeneous, more or less viscid, slimy material. Recognizable objects in the form of granules, threads, or fibers can be distinguished in each. The cell plasm and the nucleus respectively are therefore not of the same constitution throughout, but possess polymorphic characters, the study of which in health and the changes produced by disease will for many years to come form important matters for investigation.

(To be concluded.)

THE BUBONIC PLAGUE.

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THE province of Yunnan in China adjoins French Tonkin and British Burmah. It is of interest to the student of epidemiology because from this mountainous and difficultly accessible region there has issued but recently a disease which has been considered as practically extinct. Frightful as have been the ravages of the pest in the middle ages, it is noteworthy that during the past hundred years, with the exception of two slight outbreaks (Noja in Italy in 1815, and Vetlianka in Russia in 1878), the disease has been unknown in Europe. During this time the pest has not been extinct, but has existed to a greater or less extent in certain parts of Asia and in Africa. Four and possibly five of these endemic foci are known to-day. The province of Yunnan is one of these regions. The mountainous district of Gurhwal, lying along the southern slope of the Himalayas, is another center where the pest has continued to prevail. The recent travels of Koch in eastern Africa have brought to light a third region about Lake Victoria, in the British province of Uganda, and the German Kisiba, where the plague has existed from time immemorial, cut off as it were from the outer world. Only last year Sakharoff called attention to a fourth focus in northeastern China, and it is quite likely that a fifth focus exists in Arabia. These regions are of great importance in so far as the existence of permanent endemic foci sheds not a little light upon the development and spread of those great epidemics which, like great tidal waves, have in the past swept over whole countries and even continents.

It is not known when or from whence the pest was first introduced into Yunnan. Unquestionably, it has existed in the extreme western parts of the province for many decades. Eventually the disease spread throughout the

province, and frightful ravages are known to have occurred in 1871–73. Repeated visitations of this dread disease have taught the natives of Yunnan, as well as those of Gurhwal and of Uganda, to desert their villages as soon as an unusual mortality is found to prevail among the rats. In spite of the frequent recurrence of the plague, it did not spread to neighboring provinces, largely because of the fact that little or no communication exists between Yunnan and the adjoining Chinese states. Recently, however, the plague did succeed in crossing the frontier, and, in so doing, it has given rise to an epidemic which, as will be presently seen, has already made an unenviable record and has a future that no one can foretell.

The way in which the disease spread from Yunnan has been quite clearly established. Along the Tonkin frontier, throughout the provinces of Quan-si and Yunnan, the Chinese maintain a large number of military posts. Mule supply-trains for these posts passed from province to province over the difficult mountain paths. The mule-drivers were natives of Yunnan. In 1892 the plague existed in Yunnan, and it was in the summer of 1893 that the disease appeared at Long-Cheou in Quansi among the Yunnan mule-drivers. These drivers arriving at the post of Lieng-Cheng, after one of their journeys from Yunnan, repaired to the city of Long-Cheou, about ten miles distant. During their sojourn in this city the muleteers developed the first known cases of the plague. From these men the disease spread throughout the city and to the neighboring posts and villages.

From Long-Cheou the plague descended the Canton River and reached Naning-Phu. From thence it followed overland to the seaport Pakhoi, some hundred and fifty miles distant. A few months later, in February, 1894, it reached Canton, either by descending the river from Naning-Phu or by boat from Pakhoi. That the plague at Canton, in 1894, had not lost any of its old-time destructiveness is seen in the fact that it is estimated to have caused not less than one hundred thousand deaths in Canton in the short space of two months.

From Canton the plague spread to Hong Kong in April, 1894. It was during the existence of this epidemic that the first bacteriological studies of the disease were made and resulted in the discovery of the plague bacillus. In the fall of 1894, the disease died out in Hong Kong, but it reappeared in 1895 and 1896. Considering the fact that Hong Kong is one of the most important maritime centers, it is not surprising to find that in the spring of

1896 the plague was carried by shipping to the Island of Formosa. It is quite certain that about the same time the plague was carried from Hong Kong to Bombay. At all events, the existence of this disease was recognized in Bombay in September, 1896, by Doctor Viegas. Previous to this date, the mortality in Bombay was abnormally high, undoubtedly due to the very unsanitary condition of the overcrowded city.

The existence of famine in India, together with the filthy, overcrowded condition of the population, enabled the plague to gain a firm foothold in a relatively short time. Indeed, there can be no doubt but that the disease was well established at the time it was first recognized. It is no wonder, then, that in spite of the most stringent precautions, it spread like wildfire, so that in a short time the weekly deaths from the plague rose to nearly 2,000. In the face of such a relentless enemy, it is but natural that a large proportion of the population should seek safety in flight. It is believed that fully 300,000 people left Bombay shortly after the plague developed. There can be no doubt but that these refugees, directly or indirectly, carried the disease to the neighboring villages, and thus contributed to the enormous dissemination of the pest throughout Western India. In the Presidency of Bombay there were reported, in less than three years, more than 220,000 cases, with more than 164,000 deaths. When it is furthermore recognized that the natives concealed the existence of the disease as much as possible, it will be evident that these figures reveal a partial but, nevertheless, a grim truth.

With Bombay and the surrounding country thus seriously infected, it became merely a question of time when the disease would be carried to other ports and countries, by vessels and by overland routes. In spite of the sanitary perfection which we may flatter ourselves on having attained in recent years, it is nevertheless a fact that the disease is slowly but steadily and, as it were, stealthily invading port after port. That the sanitary methods, however, are not at fault is seen in the fact that when an early and prompt recognition occurred, the disease has been held in check. The insidious spread of the disease is rather due to the enormous development of commerce and to the rapid means of communication with distant countries.

From Bombay the plague has spread to ports on the Persian Gulf, on the Red Sea, and has reached Alexandria. Aden, Djeddah, Port Said, Cairo, have all had outbreaks of the disease. Beirut and Smyrna have each developed straggling cases. Isolated cases have been met with in London, at St.

Petersburg and in Vienna. However, only three appreciable outbreaks have as yet occurred on European soil. The first was that at Oporto in Portugal, where one hundred and sixty cases, with fifty-five deaths, have developed up to the present time. The second outbreak occurred at Kolobovka, a village near Astrakhan. Of the twenty-four cases that developed there in July and August, 1899, twenty-three died. The last outbreak is that at Glasgow, where the disease made its appearance but a few weeks ago.

In addition to following the great international highway of Suez, the disease has insidiously spread to the countries of East Africa. Mauritius and Madagascar, with the adjoining mainland of Mozambique and Lorenzo Marquez, have become more or less infected, and, if reports are to be credited, it has also appeared in one of the Boer towns and also on the Ivory Coast in Western Africa. Last fall the disease reached South America. It apparently was first recognized at Santos, in Brazil, during October, although early in September, according to reports, a peculiar disease, causing swelling of the glands and death within forty-eight hours, was reported at Asunción, the capital of Paraguay. At the present time Rio Janeiro is infected. The sanitary condition of these South American cities is far from being the best, and, consequently, there is but little hope that the disease will be eradicated or even held in check. With South America more or less thoroughly infected, it is evident that the United States, as well as Europe, are now threatened from all sides. The gravity of the situation is seen in the fact that already last November two cases of the plague were found in New York harbor aboard a coffee ship from Santos. Several cases have also developed on ships bound from the latter city for Mediterranean ports.

The United States is threatened not merely from the East Atlantic and South Atlantic, but also from the Pacific. As a matter of fact, the danger to our Pacific ports is greater, owing to the direct communication with the Orient. It has been already indicated that Hong Kong has continued to be infected ever since 1894. On several occasions it disappeared during the winter months, only to reappear in spring. With the more or less constant prevalence of the plague at this great seaport, it necessarily will lead directly or indirectly to a dissemination of the disease along the entire Pacific. Already it has prevailed at Amoy, and has even extended to other Chinese ports as far as Niu-Chwang. For several years it has already persisted on the

island of Formosa. Japan was invaded last fall at Kobe and at Osaka, and although it disappeared during the winter, yet only a few weeks ago it has reappeared at the latter city. Sidney in Australia, and Noumea in New Caledonia, are also infected at the present time.

Manila, Honolulu and San Francisco have successively become infected. In all these places the disease, with but very few exceptions, has attacked the native or Oriental population. The extinction of the plague in the Hawaiian Islands since the end of March is a splendid demonstration of what energetic, vigorous measures can accomplish. The presence of the plague since March 8 in Chinatown, in San Francisco, is readily recognized as a most serious condition, especially after the courts have granted an injunction restraining the health officers from carrying out the necessary vigorous preventive measures.

A few words should be given here to the overland dissemination of the disease. Europe is not merely threatened by infected ships which may come from China, India, Eastern Africa or South America. The overland routes from China and India are fully as grave a source of danger. Indeed, as will be presently shown, these are the routes along which the great epidemics of cholera and plague have always traveled in the past.

One of these great caravan routes leads from Lahore in Punjab through Afghanistan into the Russian province of Turkestan, where it meets the Trans-Caspian railway. This railway begins at Samarcand in Turkestan, and passes through Bokhara, Merv, Askabad and ends at Uzun Ada on the Caspian Sea opposite Baku. Early in 1899 an outbreak of the plague occurred near Samarcand, undoubtedly brought up from India. The precautions taken to prevent the spread were entirely successful, and although no accounts have been officially published as to the means employed, nevertheless it will be seen that the radical procedure employed by Loris Melikoff some twenty years ago was again resorted to. Inasmuch as the entire village was said to be afflicted it was surrounded by troops, and no one was allowed to enter or leave. The village and all that it contained was destroyed by fire. With this route open continually it is evident that fresh importation must be expected sooner or later.

Apparently a new plague focus, independent of that in Yunnan and Hong Kong, has been recently discovered in Manchuria. The plague seems

to have existed in this province for more than ten years under the name of Tarabagan plague, and is believed to be spread by a rodent, the *Arctomys cobuc*, which is subject to a hemorrhagic pneumonia. The presence of such an independent endemic focus in Manchuria indicates the possibility of the spread of the disease by caravan to Lake Baikal, and thence by the Siberian railroad to Russia. Indeed, the epidemic of pneumonic type which began July, 1899, at Kolobovka, in Astrakhan, while it may have been imported from Persia, might also owe its origin to the Mongolian focus.

Russia, however, is not the only country endangered by the overland transmission of the disease. There are commercial highways which lead from Northwestern India through Baluchistan and Persia to the Caucasus, and through Turkey to Constantinople. Grave danger threatens from this source, and more especially from the cities along the Persian Gulf. Two important cities here are already infected, namely, Bushire, in Persia, and Bassorah on the Tigris, in Turkey. It would appear as if Turkey and Persia would escape with difficulty from a visitation of this dread disease.

Such, then, is the geographical distribution of the present outbreak of the plague. This, an apparently extinct disease, has suddenly reappeared and given evidence of its power to spread death and desolation. Fortunately, however, modern sanitary precautions are quite able to restrict its progress, provided they be applied at the proper time and place. Filth and overcrowding, protracted wars and famine, have been the powerful allies of the plague in the past. Through their aid this disease has made a deep impression upon the pages of history. It may not be out of place, therefore, to turn from the present outbreak of the disease and trace its grewsome past.

In ancient writings references are found which would seem to indicate the existence of the plague at a very early date. The Bible contains several such references (Deuteronomy, Chapter 28, paragraph 27. Samuel I, Chapter 5, paragraphs 6, 9). The latter especially deals with the plague which attacked the Philistines after they took the ark. The rôle of rats in the dissemination of the disease is, as some believe, apparently referred to in the trespass offering of "five golden emerods and five golden mice." The return of the ark, together with this trespass offering, brought also the plague, "because they had looked into the ark of the Lord, even he smote of the people fifty thousand and threescore and ten men." Poussin's painting of this Philistine plague, exhibited in the Louvre, shows several dead rats on

the streets. It is evident that the susceptibility of the rat to the plague had been noticed even at this early date. The plague of boils visited upon the Egyptians as related in Exodus (Chapter 9, paragraphs 9 and 10) has also been taken to indicate the pest of today, but neither of these scriptural references can be said to be sufficiently definite.

The Attic plague, which ravaged the Peloponnesus 430 years before Christ, has been accurately described by an eye-witness, the historian Thucydides. His narration may be considered the earliest exact record of an epidemic. Like all the great epidemics of subsequent ages, it was ushered in by the overcrowding, the misery and the famine consequent upon prolonged wars. The combustible material was there, and all that necessary was the spark to begin the work of death and devastation. It is noteworthy that the origin of the pest was traced by Thucydides to Egypt or Ethiopia, from whence it spread gradually overland to Asia Minor and thence by boat to Athens. The nature of this first great historic epidemic is and will remain uncertain. There are those who consider the Attic pestilence as one of bubonic plague, but the fact that in the very careful description of the disease no mention is made of buboes and the statement that death occurred from the seventh to the ninth day would indicate that the disease was something else. Buboes are characteristic, it is true, of the plague, but it should be remembered that outbreaks of the pneumonic form, with little or no glandular enlargement are not uncommon. Death, however, in the case of plague is very common on the second or third day, and is less liable to occur in more protracted cases. These facts lead to the commonly accepted belief that the Attic pest was not the bubonic plague. It may have been typhus fever, possibly smallpox.

The great pestilence which devastated Rome and its dependencies in 166, Anno Domini, is known as the plague of Antoninus or of Galen. This prolonged epidemic was brought to Rome by the returning legions from Seleucia. It was not characterized by buboes, and it is very probable that it was largely smallpox. On the other hand, the plague of Saint Cyprian, which prevailed from 251 to 266 Anno Domini, may have been partly bubonic in nature, since, it prevailed during the fall and winter months and ceased during the hot summer. The disease was said to be communicated by means of clothing and by the look. It spread from Ethiopia to Egypt and thence through the known world.

Although the above early epidemics cannot be identified with the bubonic plague, there is nevertheless excellent evidence of the existence of this disease in remote antiquity. The first undoubted testimony on this point is that furnished by Rufus of Ephesus, who lived in the first century of the Christian era. The writings of this author are no longer extant, but they are quoted by Oribasius, the physician and friend of Julian the Apostate, who lived in the fourth century. The writings of Oribasius were discovered in the Vatican Library and were published early in this century by Cardinal Mai. In the forty-fourth "Book of Oribasius" occurs the extract taken from Rufus of Ephesus, from which it appears that "the so-called pestilential buboes are all fatal and have a very acute course, especially when observed in Libya, Egypt and in Syria. Dionysius mentions it. Dioscorides and Posidonius have described it at length in their treatise upon the plague which prevailed during their time in Libya." The description which then follows of the buboes and of the disease is an exact counterpart of the present plague. The writings of the authors quoted by Rufus are no longer extant, but one thing is certain, and that is that the Dionysius referred to lived not later than 300 years before Christ. The other two physicians lived in Alexandria contemporaneous with the birth of Christ. It may, therefore, be considered as an established fact that the plague existed in Egypt, Libya and Syria as early as 300 years before Christ. This is of especial interest in view of the recent discovery by Koch of an endemic plague focus in British Uganda and German Kisiba, at the headwaters of the Nile. Whether it ever invaded European territory prior to the sixth century is unknown.

The great plague of Justinian which broke out in 542, Anno Domini, appeared first in Egypt, and from thence it spread east and west throughout the known world and persisted for more than a half century. So unknown was the plague in Europe at that time that the physicians of Constantinople considered it a new disease. Procopius, who was an eye-witness of the plague at Constantinople, states that the daily mortality in that city was at times over 10,000.

The pandemic of Justinian resulted in the distribution of the plague for the first time throughout the length and breadth of known Europe. From that time on the early chroniclers make repeated mention of devastating plagues consequent upon the miseries of war and famine. The descriptions of these pestilences are, as a rule, insufficient to identify them with the bubonic

plague. Typhus, scurvy, smallpox and other diseases undoubtedly alternated in the work of destruction. Of the scores of epidemics thus recorded during the eight centuries following this first visitation few, indeed, can be identified to a certainty with the bubonic plague, and yet there can be no doubt but that this disease occupied no second rank during the dreary darkness of the middle ages. This era in history may be said to have been ushered in by the Justinian plague, and it was closed by an even more disastrous outbreak of this same disease. All the ravages and slaughter consequent upon the great historic battles, when taken together, pale into insignificance on comparison with that dread visitation of the fourteenth century, the 'black death'.

It is noteworthy that this great historic epidemic did not originate in Egypt, as did many of its predecessors. Without exception the contemporaneous writers ascribe its origin to Cathay, or the China of today. This fact is of interest when it is borne in mind that at the present time we know of the existence of two endemic foci in China, besides that of Gurhwal in India, of Beni Cheir in Arabia and of Uganda and Kisiba in Africa. Whatever may have been its source, the fact is that it advanced from the Orient along the three principal routes of travel. One of these led from the Persian Gulf through Bassorah and Bagdad along the Euphrates, across Arabia to Egypt and Northern Africa. Another route passed from India through Afghanistan, and skirting the southern borders of the Caspian and Black Seas, eventually reached Asia Minor. A third route from Turkestan and China led around the northern shore of the Caspian Sea to Crimea, and thence to Constantinople. It was along these several routes that the plague advanced and spread over most of Western Asia and Northern Africa.

The European black death, however, can be traced with accuracy to the Crimean peninsula. Gaffa, a town in Crimea, now known as Theodosia, had been founded and fortified by the Genoese. It, as well as other cities along the Black Sea, was largely populated by Italians. One of these, Gabriel de Mussis, a lawyer in Gaffa, has left a faithful account of his experience and share in the introduction of the plague into Europe. In 1346 in the Orient numberless Tartars and Saracens were attacked with an unknown disease and sudden death. In the city of Tanais, through some excess, a racial struggle ensued between the Tartars and the Italian merchants. The latter eventually escaped and took refuge in Gaffa, which in time was besieged by

the Tartars. During the siege, which lasted three years, the Tartar hordes were attacked by the plague, which daily carried off many thousands. The besiegers, despairing of reducing the city by direct attack, attempted to do so in another way. By means of their engines of war they projected the dead bodies into the beleaguered city, which, as a result, soon became infected. The Christian defenders took to their ships, and abandoning Caffa, sailed westward, touching at Constantinople, Greece, Italy and France.

Wherever the infected vessels touched they left the plague. Constantinople thus became infected early in 1347. During the summer Greece, Sardinia, Corsica and parts of the Italian coast developed the disease. In the fall it reached Marseilles. The following year it spread inland into Italy, France, Spain, and even into England. In another year or two it spread over Germany, Russia, and crossed to the Scandinavian peninsula. Within four years it had completed the circuit of Europe, spreading untold death and misery. No greater catastrophe has been recorded in the history of the world.

The rapidity with which the disease spread among the fugitives from Caffa, and in the cities visited by their ships, is despairingly narrated by De Mussis, who, returning in one of the ships to Genoa, says: "After landing we entered our homes. Inasmuch as a grave disease had befallen us, and of the thousands that journeyed with us scarcely ten remained, the relatives, friends and neighbors hastened to greet us. Woe to us who brought with us the darts of death, who scattered the deadly poison through the breath of our words." According to this writer 40,000 died in Genoa, leaving scarcely a seventh of the original population. Venice was said to have lost 100,000, Naples 60,000, Sienna 70,000, Florence 100,000. All told, Italy lost half of its population.

Of the contemporaneous writers none has printed the horrors of the plague more vividly than does Boccaccio in his introduction to the 'Decameron.'

"What magnificent dwellings, what notable palaces were then depopulated to the last person! What families extinct! What riches and vast possessions left, and no known heir to inherit! What numbers of both sexes in the prime and vigor of youth, whom in the morning either Galen, Hippocrates, or Æsculapius himself but would have declared in perfect

health, after dining with their friends here have supped with their departed friends in the other world!”

From Marseilles the plague spread through Provence with disastrous results. In some monasteries not even a single survivor was left. In one of these Petrarch's brother buried thirty-four of his companions. At Avignon, the seat of the Pope, 1,800 deaths occurred in three days. In Paris more than fifty thousand died of the plague.

In England the black death appeared in August, 1348, and continued till the autumn of 1349, when it disappeared. London, which at that time probably had a population of 45,000, had a mortality of about 20,000. No exact statement can be made of the relative mortality in England, although many undoubtedly extravagant guesses are recorded by contemporaneous writers.

It is estimated that the population of Europe previous to the outbreak of the black death was about one hundred and five millions. One quarter of the population, or about twenty-five millions, are said to have died of the plague. This may be but a mere estimate, it may be grossly inaccurate, but it nevertheless indicates the deadly character of the pestilence. According to a report made to Pope Clement VI, the total mortality for the known world was placed at forty-three millions. One-half the population of Italy succumbed. The Order of Minorites in Italy lost 300,000 members. The Order of Capuchins in Germany lost 126,000 members, while the total of deaths in Germany was placed at 1,200,000.

The invasion of Europe by the black death was sudden and rapid. The seeds of the disease, once planted on European soil, persisted, as might be expected, for no little time. Although the great epidemic was said to have lasted till 1360, it must not be inferred that it then ceased altogether. Diverse localities retained the infection, and, as a result, new outbreaks, though to a less extent, continued to outcrop during the following years. From that time on every decade or two witnessed more or less pronounced outbreaks of the disease in France, England and Italy. The chroniclers of those local outbreaks during the latter half of the fourteenth and during the entire fifteenth century did not always make it clear that the pestilence described was the real plague. It was but natural to include typhus and other diseases under the dreaded term of pest. Nevertheless, the frequency of these

outbreaks indicates the persistence and the wide dissemination of the plague during those years.

During the sixteenth century the plague apparently began to show a decrease in its frequency, although during this period, as before, other epidemic diseases were mistaken for it. Germany, Holland, certain cities in France, and especially in Italy were scourged by the plague during this century. The noteworthy outbreak in Italy in 1575–77 was due to fresh importation from the Orient. The disease spread throughout Italy, and the devastation it caused was not inferior to that of the great plague two centuries before. For example, in 1576 in Venice 70,000 died of the disease.

During the seventeenth century the plague asserted itself with great severity. Following a famine, it prevailed in Russia in 1601–1603, and some idea of its destructiveness may be gained when it is stated that in Moscow alone 127,000 lives were taken. During the following decade even greater epidemics prevailed in Western Europe. France and England were invaded, and in Switzerland it even penetrated to the highest Alps. Basel in 1609–1611 had 4,000 deaths, while London in 1603 yielded 33,000.

The terrible epidemic which ravaged Northern Italy in 1629–1631 deserves more than a passing notice. During those years more than a million died of the disease. Scarcely a town in Northern Italy escaped. The city which, perhaps, suffered the most was Milan, where, in 1630, the deaths from all diseases are said to have amounted to 186,000. The Milan outbreak has been graphically described by Manzoni, in his celebrated ‘*I Promessi Sposi*.’ Unrecognized, the disease entered Milan in October, 1629. The mild cases which were met with during the winter months lulled the fears of the people and encouraged the mass of physicians to deny the existence of the plague. But in April the disease began to assert itself in terrible earnest. The frenzied populace, blind to the contagiousness of the disease, were possessed with the strange hallucination that obtained during former plague epidemics in other Italian cities, that the pest spread because of poison scattered about by evil-minded persons. Suspicious strangers were, as a result, stoned in the streets, imprisoned and even put to death by legal process because of such fanatical beliefs. To offset the growing pestilence, the people demanded of the Archbishop that a solemn religious procession be held, and that the holy relics of Saint Charles be exposed. At first this was refused, but eventually it was granted. The procession bearing the

saintly body was solemnly held on the 11th of June. The fanatical security which these devotions engendered was rudely shattered when, a few days later, the disease burst forth with renewed activity among all classes in all parts of the city. Nevertheless, as Manzoni observes, the faith was such that none recognized that the procession itself was directly the cause of the new outburst of the disease by facilitating the spread of the contagion. Again the belief asserted itself that the 'untori,' or poisoners, mixed with the crowd and with their unguents and powders had infected as many as possible. From that day the fury of the contagion continued to grow to such an extent that scarcely a house remained exempt from the disease. The number of patients in the pesthouse rose from 2,000 to 12,000, and later reached 17,000. The daily mortality rose from 500 to 1,200, then 1,500, and is even said to have reached 3,500. Milan, before the epidemic, was said to have had a population of from 200,000 to 250,000. The loss by death has been variously estimated at from 140,000 to 186,000. All these deaths were not due to the plague. Thus, large numbers of children died as a result of starvation consequent upon the death of their parents from the plague.

The horrors attendant upon such a dreadful visitation can well be imagined. Scarcity of help in removing the dead and in taking care of the sick made itself felt, to say nothing of the lack of food. Enormous trenches, one after another, were filled with the bodies of the victims, carried thither by the hardened *monatti*, the counterpart of the Florentine *becchini*, so well portrayed by Lord Lytton in his 'Rienzi.' These bearers of the sick and dead were naturally recruited from the lowest criminal classes, and it can, therefore, cause but little wonder that an epidemic of the worst of crimes was associated with that of the plague.

In 1656 Italy was again invaded by the plague, and on that occasion Genoa lost 65,000 of its population by death. About the same time terrible epidemics of the disease ravaged Russia, Turkey and Hungary.

London, in 1665, suffered dreadfully from the plague. The disease appears to have been imported from Holland, where it was known to have existed for some time. The progress of the disease in London has been vividly portrayed by Defoe in the 'Journal of the Plague Year' and in the 'Due Preparations for the Plague.'

It is supposed that the pest had been imported in bales of goods from Smyrna into Holland in 1663. From thence it crossed over to London, where the first deaths were reported about the first of December in 1664. Toward the end of that month another death occurred in the same house, but during the following six weeks no new case developed. About the middle of February, however, a person died of the plague in another house. From that time only occasional cases of plague were reported, although the weekly mortality was rapidly rising and was greatly in excess of the usual rate. Thus, while the ordinary weekly mortality ranged from two hundred and forty to three hundred, this was gradually increased, so that in the third week in January it had risen to four hundred and seventy-four. After a slight remission, the mortality again rose, so that early in May plague cases were reported more frequently. It soon became evident that the plague, as in Milan in 1630, had slowly but surely gained a firm foothold. The increased mortality was undoubtedly due to unsuspected plague cases of either the pneumonic or the septicemic type.

During May, and especially during the hot weather in June, the disease continued to spread. At the same time, the panic-stricken people began to leave the city in large numbers. In July the condition was truly deplorable. To quote Defoe:

“London might well be said to be all in tears; the mourners did not go about the streets, indeed, for nobody put on black or made a formal dress of mourning for their nearest friends; but the voice of mourning was truly heard in the streets. The shrieks of women and children at the windows and doors of their houses, where their dearest relations were perhaps dying, or just dead, were so frequent to be heard as we passed in the streets, that it was enough to pierce the stoutest heart in the world to hear them. Tears and lamentations were seen almost in every house, especially in the first part of the visitation; for toward the latter end men’s hearts were hardened, and death was so always before their eyes, that they did not so much concern themselves for the loss of their friends, expecting that themselves should be summoned the next hour.”

London at this time had a population of nearly half a million. The deaths from the plague during 1665, as reported in the bills of mortality, are

68,596. By far the larger number of these occurred in August, September and October. The weekly mortality from the disease rose from a few cases in May to over 7,000 per week in September. It may, indeed, be close to the truth when Defoe states that 3,000 were said to have been buried in one night.

The great plague of London in 1665 was by no means the only visitation of that kind. From the time of the black death in 1348, London had a continuous record of plague infection. On an average it had an epidemic of plague every fifteen years. Some of these were fully as severe as that of 1665. Thus, in 1603, with a population of 250,000, there were over 33,000 reported deaths from the plague. In 1625, 41,000 died of pest out of a population of 320,000.

One of the most remarkable facts in connection with the great plague is this—that it was the last in England. The great fire of 1666 is supposed to have extinguished the plague, but this cannot be said to be true. The disease continued to a slight extent in 1666 and isolated cases were reported as late as 1679, but after that date it disappeared completely and from that time until this year England has been absolutely free from the plague. The sudden extinction of the plague in England after it had become domesticated, so to speak, for nearly three centuries, is indeed difficult to explain. Creighton sees an inhibiting influence in the growth of the practice of burial in coffins. But the absence of famine, together with the cessation of domestic wars and strife and the abeyance of want and misery, had not a little effect. As will presently be seen, the extinction of the plague in England was no more remarkable than its disappearance from Western Europe.

The history of plague in the seventeenth century does not close with the London epidemic. From 1675–1684 the disease ravaged Northern Africa, Turkey, and from thence invaded Austria and even reached Southern Germany. The Vienna outbreak of 1679 can be said to have been no less terrible than that of Milan or of London. The deaths from the plague in Vienna in that year have been variously estimated at from 70,000 to double that number.

From Vienna the plague reached Prague, where in 1861 it is said to have caused no less than 83,000 deaths. It is not to be wondered at that a

nation scourged by thirty years of relentless warfare, by religious persecution and finally tried thus severely by the plague should inscribe upon the equestrian statue of their patron saint the heart-rending appeal, 'Lord, grant that we do not perish.'

The close of the seventeenth century saw the disappearance of the plague from Western Europe. In Eastern Europe, however, the disease continued to exist even during the eighteenth century. Nevertheless, a change had taken place for the better, and as the years went on the retrogression of the plague became more and more distinct.

During the first two decades of the eighteenth century the plague was widely distributed in Eastern Europe. It was present especially in Constantinople and in the Danubian provinces. From the latter it extended to Russia (Ukraine), and from thence to Poland. The disastrous invasion of Russia by Charles XII. of Sweden, ending in his defeat at Poltawa in 1709, led to its further dissemination to Silesia, Eastern Prussia, the Baltic provinces and seaports, and even to Scandinavia. It was during this epidemic that Dantzic, in 1709, lost 33,000, and Stockholm 40,000 by the plague. During the years 1709 and 1710 the plague mortality in the Baltic provinces exceeded 300,000. Three years later, in 1713, the plague spread up the Danube and reached Vienna, Prague and even Bavaria.

During these two decades Western Europe was entirely free from the dread disease. In 1720 the disease suddenly developed in Marseilles and extended from thence to neighboring towns and the country districts of Provence. Terrible as was this visitation it is of interest, inasmuch as it was the last occurrence of the plague on French soil, and the last in Western Europe until the recent outbreak in Portugal.

The plague was said to have been imported into Marseilles by a merchant vessel, the 'Grand Saint Antoine', from Syria. On its way to Marseilles several deaths occurred on shipboard, but the cause was overlooked. On the 25th of May, 1720, two days after the arrival of the vessel, another death occurred among the crew. The disease was still not believed to be the plague, and although quarantine was instituted, new cases appeared among the crew and the dock laborers employed in unloading the vessel, and it was not until the disease reached the city that its true nature was recognized. The germs of the disease had then been scattered broadcast.

Unsanitary a city as Marseilles is to-day, it must have been vastly more so in 1720. The result of the addition of plague germs to the want, misery and filthy condition was at once evident. During August the mortality averaged four and even five hundred per day. In September the daily mortality rose to 1,000. So great was the terror of the populace that it became impossible to secure bearers of the dead, to obtain nurses and attendants. The dead were left in heaps upon the streets, so that it became necessary to transfer to the city 700 galley slaves, who were required to remove the bodies. These same galley slaves were even pressed into service as nurses. The diseased were abandoned by friends and relatives, and under such conditions it need not be wondered at that they received little or no attention from others. Food and water were denied to the unfortunates, and when food was administered to the pesthouses it was thrown into the windows by machinery.

The disease continued in Marseilles until December, 1721, but isolated cases persisted until April, 1722. During the fifteen months of its duration it carried off 40,000 of the population. According to Defoe, there died of the plague in Marseilles and within a league of its walls 60,000.

From Marseilles the plague reached Aix, and in the winter of 1720 and 1721 it carried off 18,000 of its people. It also reached Arles, where, in 1721, out of a population of 23,000, 10,000 died (forty-five per cent). The same year, in Toulon, which had a population of 26,000, the plague attacked 20,000 of the population, and of these 13,000, or about one-half of the original population, died.

The country districts about Marseilles were likewise invaded. Out of a population of 248,000, there died of the plague 88,000, or fully thirty-five per cent.

It is evident from this description that the plague of 1720 was in nowise inferior to that of 1348. Fortunately, the disease did not spread beyond Provence. It is noteworthy that in many instances, in Marseilles, people secluded themselves in their houses, avoiding all communication with the outer world, and in this way escaped. Similar isolation of cloisters, insane asylums, likewise resulted in freedom from the disease which stalked so freely throughout the stricken city. It was experience of this kind in isolation of the healthy which led Defoe to write his 'Due Preparations for the Plague.'

Toward the middle of the century the plague reasserted itself in the Danubian provinces, the constant battleground between the Turks and Russians and Austrians. In 1738 it not only prevailed in Russia but also invaded Hungary. Of more importance than this occurrence is the outbreak of the plague in 1743 in Sicily. The last epidemic of plague had occurred in Messina in 1624. After a lapse of one hundred and twenty years, it reappeared with terrible results. In Messina, as in Marseilles and in London, the first cases were not recognized as plague cases and, as a result, the infection spread until, like a veritable explosion, the disease developed all over the city. The plague, with its attendant misery of lack of food, and even of water, was in vain combated by religious processions. The plague corpses were in heaps in the streets, as in Marseilles, and cremation was resorted to in order to effect their removal. That year 30,000 died of plague in the city of Messina. With the exception of a slight epidemic at Noja in 1815, this outbreak in Messina in 1743 was the last one to appear in Italy.

In 1755, the plague was introduced into Transylvania by an Armenian merchant from the Black Sea. Before it was extinguished, 4,300 deaths were recorded.

Next to that of Marseilles and of Messina, the most noteworthy outbreak of plague was that which occurred in 1771 in Moscow. The disease was introduced by troops returning from the Danubian provinces. As so often has been the history of plague, the first cases were not recognized, and the existence of pest was denied. When the plague was demonstrated to be present, it is said by Haeser that three-fourths of the populace deserted the city. The disease began early in March and increased during the early summer months. In August over 7,000 deaths resulted, while in September the records show that 21,000 died. In October the plague decreased, but still 17,000 deaths attested to its fearful power. Early in January it became extinct, after a duration of ten months, and after having caused the death of more than 52,000 people.

Toward the close of the eighteenth century, at the time of the Napoleonic invasion of Egypt and Syria, the French armies came into contact with the plague. Bonaparte's visit to the pest-stricken soldiers at Jaffa has been perpetuated in the historic canvas which is to be seen at Versailles.

During the nineteenth century the plague ravaged Northern Africa on diverse occasions. Constantinople was invaded in 1802, 1803, 1808. It was also present to a slight extent in the Caucasus and in Astrakhan. A notable plague epidemic appeared in Egypt in 1812, and soon spread through Turkey and Southern Russia. Constantinople and Odessa were severely scourged. In Odessa out of a population of 28,000 there died 12,000.

It is a noteworthy fact that the Napoleonic wars, with all their incident hardships and misery, did not develop or spread the plague in Europe. The outbreaks of the disease were limited during this period to Africa and to Turkey, Bosnia, Roumania, Dalmatia and to Southern Russia. Two exceptions, however, are to be noted. In 1812 the Island of Malta was infected and more than 6,000 of its people yielded to the disease. The epidemic of 1815 at Noja, in Apulia, was the first recurrence of the plague on Italian soil since 1743, and thus far it has been the last.

The Balkan Peninsula and Southern Russia were visited from time to time by the plague up to about 1841. For nearly forty years Europe was wholly free from the disease, which, however, continued its existence in Northern Africa, in Mesopotamia and in India. The Russo-Turkish war of 1878 brought the Russian troops into contact with the disease in the Caucasus, and the epidemic at Vetlianka on the lower Volga was unquestionably introduced by such returning soldiers.

Such, then, has been the history of the bubonic plague. No other epidemic disease can be traced authentically as far back as the 'Black Death.' The characteristic symptoms, the rapid death, the excessive mortality are all features which have been noted through more than twenty centuries. The plague bacillus discovered in 1894 by Yersin, judged by its effect, is neither more nor less virulent than its early progenitors. It has often died out in a given locality or country, it has even been forced back to its original ancestral home, but still the same type, the same species has perpetuated itself unchanged. If the plague on its present world-wide journey does not cause such terrible outbreaks as it has in the past, it will be not because the germ has been altered by time, but because man has changed in so far as he has slowly learned and profited by the lessons of previous epidemics.

GASOLINE AUTOMOBILES.

BY WILLIAM BAXTER, JR.

TO understand the operation of a gasoline vehicle it is necessary to be somewhat familiar with the principle on which gasoline motors act. Briefly stated, it is as follows: The gasoline is converted into a vapor, and in this state is mixed with a sufficient amount of air to cause it to ignite when heated to a proper temperature. This mixture of air and vapor is admitted into a cylinder in which a piston moves freely, this part being substantially the same as in a steam engine. By means of an electric spark or a hot tube, the mixture is ignited, burning so violently as to expand the products of the combustion with such rapidity as virtually to become an explosion. The force of this explosion pushes the piston to the further end of the cylinder, and by means of a connecting rod and a crank this movement imparts a rotary motion to a shaft.

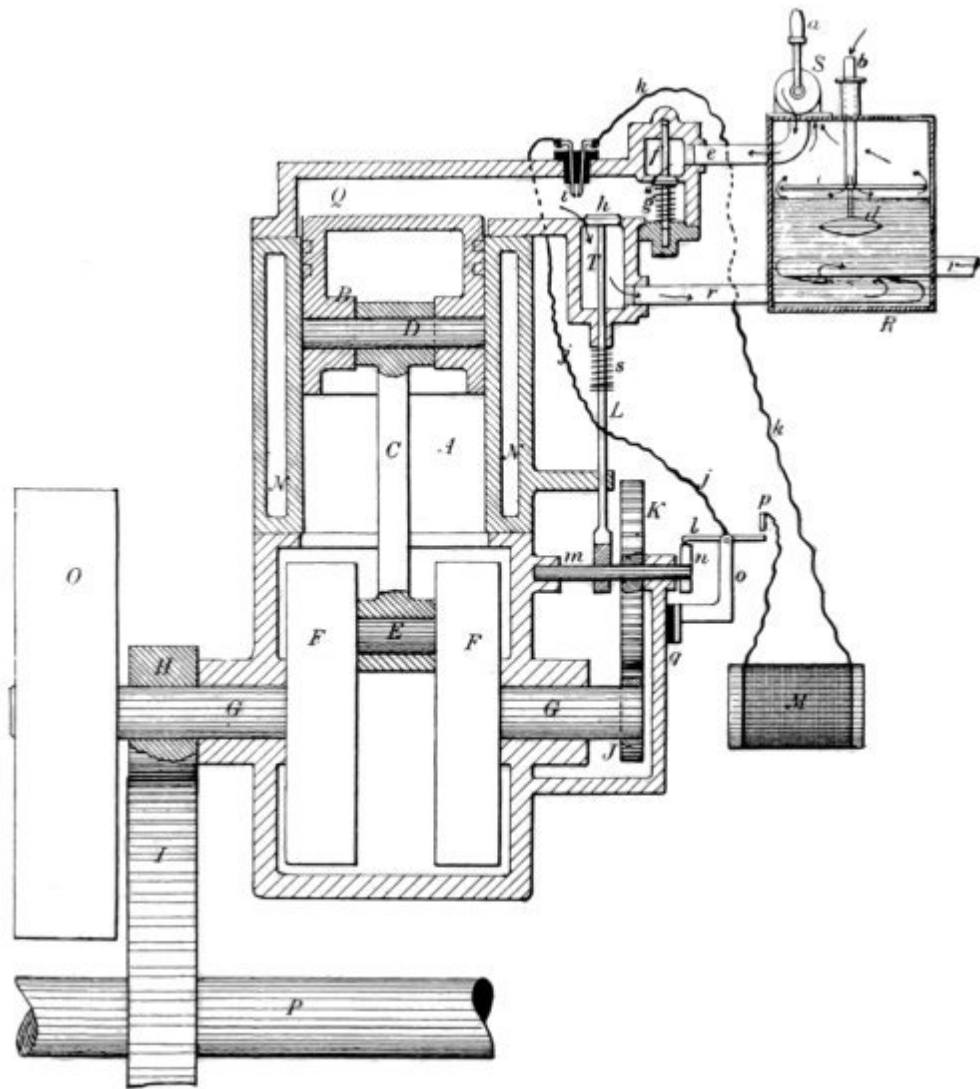


FIG. 1. GASOLINE MOTOR.

The entire operation is made perfectly clear by the aid of [Fig. 1](#), which is a simple diagram of a single cylinder motor. The chamber *R* contains the gasoline. Air enters this chamber through tube *b*, as indicated by the arrow, and passes out between the plate *c* and the surface of the gasoline. The float *d* keeps the plate *c* in the proper position regardless of the amount of liquid in the reservoir. The heated gases exhausted from the cylinder pass through the pipe *r*, and thus heat the gasoline so that it vaporizes freely and the air passing under *c* becomes charged with the necessary proportion of vapor. The mixed air and vapor enter a valve chamber *S*, from which the flow into

pipe *e* is regulated by the movement of handle *a*. In this chamber there is another valve, operated by an independent handle, and by means of this more air can be admitted into the mixture when desired. Through the pipe *e* and the valve *f* the vapor enters chamber *Q*, which connects with the top of the cylinder. Suppose the shaft *G* is rotating, then the piston will be drawn down from the position in which it is shown and thus a vacuum will tend to form in chamber *Q*. This action will cause the valve *f* to open and the mixture of air and vapor will flow into *Q* until the piston reaches its lowest position and begins to ascend. At this instant the valve *f* will close, and then the upward movement of the piston will compress the mixture in the chamber *Q*. When the piston reaches the upper position, after completing the down and up strokes, the lever *l* and the contact point *p* will come together, and an electric current developed in the induction coil *M* will pass through the wires *j* and *k* and produce a spark at *i* between the ends of the metallic terminals passing through the plug of insulating material, which is shown in dark shading. This spark will cause the mixed air and vapor to ignite, producing an explosion that will force the piston down for the second time. On the second upward movement of the piston the gases produced by the combustion of the vapor will be forced out through the valve *h* into the chamber *T* and the pipe *r*. The valve *h* and the lever *l* are operated by cams mounted on the shaft *m*, and they are so set that the spark at *i* occurs when the chamber *Q* is full of the explosive mixture and the piston is at the top of the cylinder. The valve opens when the piston begins to move upward after the explosion has forced it to the bottom position.

As will be seen, the piston must move down to draw in a supply of the explosive mixture; it then moves upward to compress it, and on the second down stroke it is pushed by the force of the explosion. From this action it can be clearly realized that the power developed by the motor comes from the force exerted by explosions at every alternate revolution of the shaft. On that account the cams that move the valve *h* and the lever *l* are placed on a separate shaft, which is geared to the main shaft in the ratio of two to one; that is, the wheel *K* is twice the diameter of the wheel *J*. As the force of the piston acts on the shaft only once in every two revolutions it is necessary to provide a heavy fly wheel *O*, which will store up enough momentum to continue the rotation of the motor through the ineffective revolution. Before the motor can put forth an effort it is necessary for the piston to move downward so as to draw in a supply of explosive gases and then to move up

so as to compress them and produce an explosion; therefore, the motor will not start of its own accord, but must be set in motion. In the act of starting the wheel *O* is turned by hand.

The combustion of the gasoline vapor within the chamber *Q* and the upper end of the cylinder develops a large amount of heat, and unless means are provided for dissipating it the temperature will soon rise to a point that will interfere with the proper action of the motor. Two ways are employed to carry off the heat. One is by surrounding the cylinder with a water jacket, as shown in the diagram at *NN*; and the other is to provide the exterior of the cylinder with numerous thin ribs so as to increase the surface exposed to the air and thus increase the radiation.

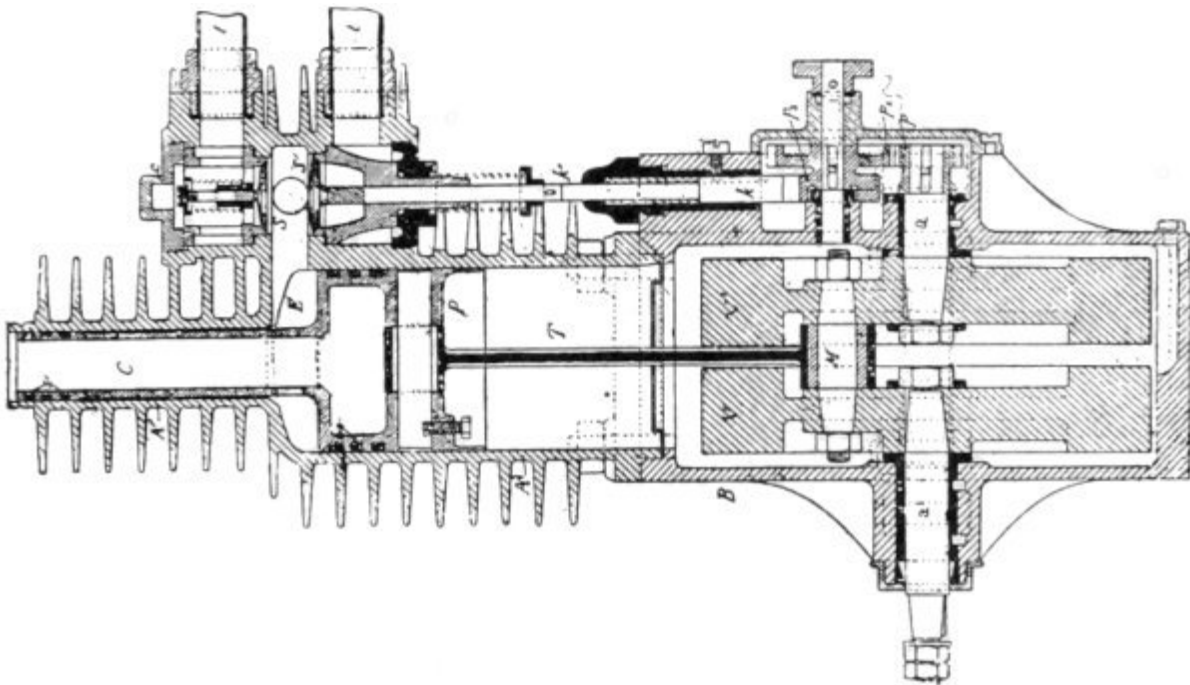


FIG. 2. PETROLEUM SPIRIT MOTOR.

The electric spark is a very effective igniter for the explosive mixture, and, by properly setting cam *n* the explosion can be made to take place just at the position of the piston that may be found the most desirable; but the points at *i* are liable to get out of order, and the battery that actuates the induction coil *M* and the coil itself can become a source of more or less

trouble, and on that account the igniting is effected in some motors by means of a hot tube. When this is used the cam n , the lever l and the electrical parts of the apparatus are not required. In their stead a tube is placed on the upper side of the chamber Q and this tube is maintained at a red heat by means of a flame impinging against its outer surface. When the explosive mixture is compressed it rises in the interior of the hot tube, and when it reaches the portion that is hot enough to produce combustion an explosion takes place. By many engineers this arrangement is regarded as superior to the electric spark on account of its simplicity.

Gasoline motors are made with one, two or more cylinders, but in each cylinder the action that takes place is that described above. The actual construction of a motor is not so simple as might be assumed from the appearance of [Fig. 1](#); many details are required which are not here shown. A more perfect idea of the actual construction of a gasoline motor can be had from [Fig. 2](#), which is a working drawing of a recent European invention. In this design it will be noticed that the cylinder is cooled by radiation into the surrounding air, the exterior surface being increased by numerous circular ribs and also by extending a hollow trunk from the upper side of the piston, so as not only to increase the radiating surface, but also to allow the hot air to escape from the chamber T in which the crank discs revolve. In this drawing E is the explosion chamber, corresponding to Q in [Fig. 1](#), and the valve s is the counterpart of f , while s' corresponds to the valve h . The upper pipe t is the pipe e of [Fig. 1](#) and the lower pipe t' is the pipe r of the same figure. Although the crank discs, connecting rods and other details are different in shape, it will readily be seen that their relation to each other is the same.

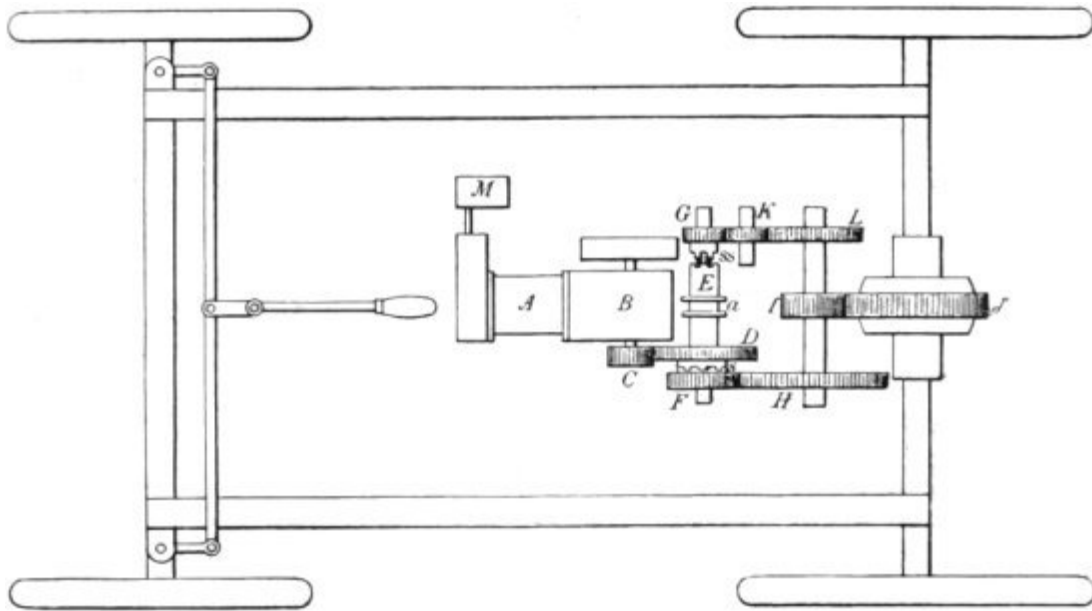


FIG. 3. REVERSING MECHANISM.

Since a gasoline motor cannot start of its own accord, it is necessary in vehicles in which they are used so to arrange the driving gear that the motor may be kept in motion all the time and always in the same direction, hence, to reverse the direction of the carriage, reversing mechanism, independent of the motor, must be provided. The most simple mechanism for a gasoline vehicle employing spur gearing exclusively is shown in diagrammatic form in [Fig. 3](#). In this figure *A* represents the cylinder of the motor, *B* the crank disc chamber and *M* the vaporizing receptacle, which is generally called the carburetor. The pinion *C*, on the end of the motor shaft, meshes into a gear *D* which is mounted upon a sleeve *E* which revolves freely round shaft *G*. This sleeve has its ends formed so as to engage with the gears mounted upon shaft *G*, and by means of a lever, which is not shown, but which works in groove *a*, the clutch either *s* or *ss* can be thrown into engagement with its corresponding gear. If *s* is thrown into gear, as shown in the drawing, the wheel *F* will turn *H* and the pinion *I* will rotate the gear *J* which is mounted upon the axle of the carriage. If the clutch *ss* is thrown into engagement, the gear *G* will turn *K* and this wheel will turn *L*; but, as can be clearly seen, the direction in which *L* will revolve will be opposite to its motion when driven through *F* and *H*, therefore, if when *F* drives the carriage runs forward, when *G* drives it will run backward, and when *E* is moved to the central

position, so that neither *s* nor *ss* engages with their respective gears, the vehicle will stand still, but the motor will continue to revolve.

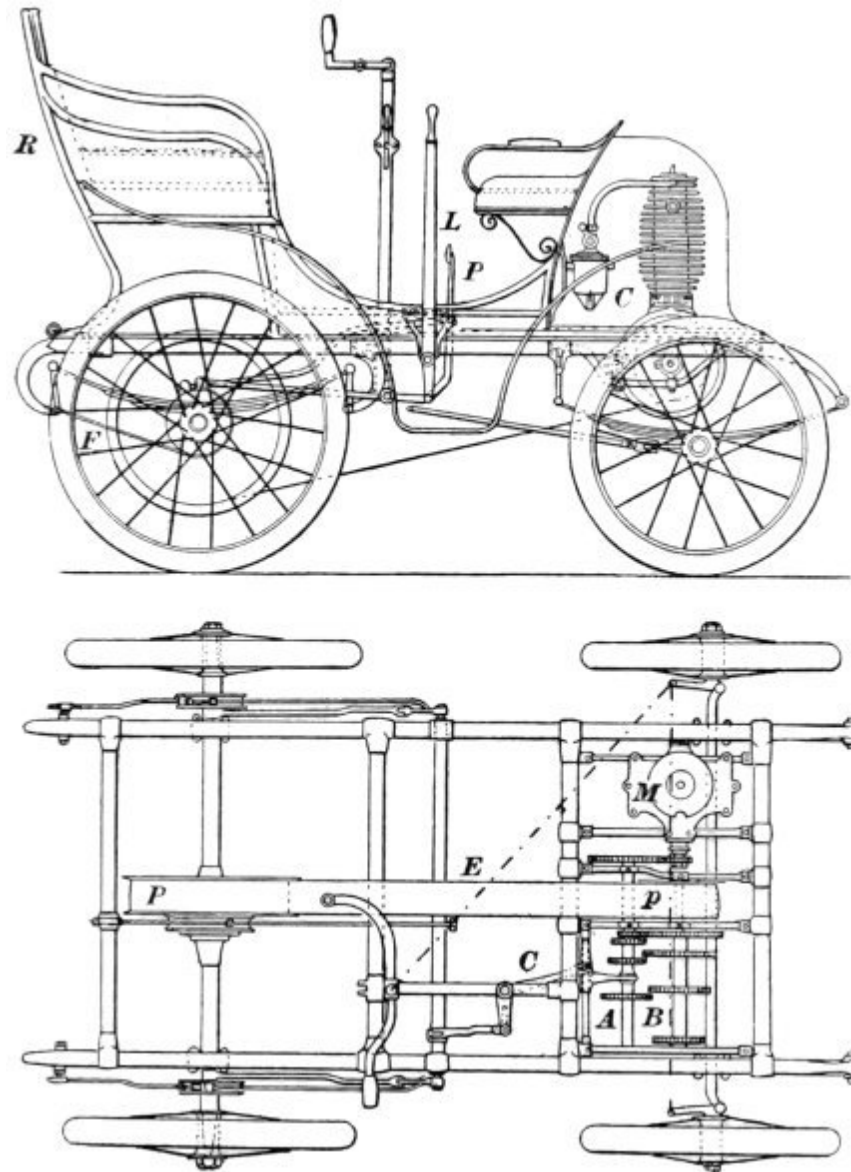


FIG. 4. PLAN AND ELEVATION OF UNDERBERG MOTOR VOITURETTE.

This diagrammatic arrangement is more simple than the gearing actually used and is not as complete in action as many of the devices, as it only provides means whereby the direction of rotation of the axle may be

changed, while in many carriages the gearing also varies the ratio between the speed of the motor and the driving wheels. It is also quite common to combine in the train of gearing spur gears and sprocket wheels, and in some instances even belts. [Fig. 4](#) illustrates a French gasoline automobile made by Underberg, of Nantes. The first figure is a side view, and the second is a plan of the truck and driving mechanism.

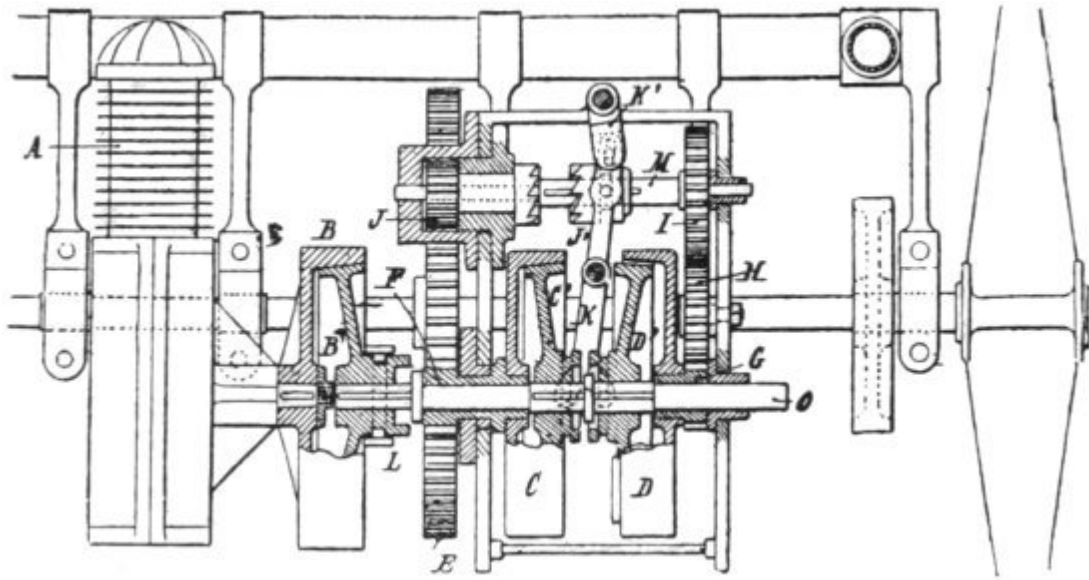


FIG. 5. CHERRIER TWO-SPEED GEAR.

The motor, which is of the single cylinder type, cooled by radiation into the air, is located at *N*. The pinion on the end of the motor shaft engages with the wheel on the end of shaft *A*. This shaft carries four gears, which can be moved by means of lever *C*, so as to engage with corresponding gears on shaft *B*, thus providing four different speeds. The motion of *B* is transmitted to the rear axle by means of a belt that runs over the pulleys *p* and *P*, the latter being carried by a differential gear, so as to run the two driving wheels at proper velocities. The circular ribs surrounding the motor cylinder are well shown in the figure, in which the carburetor of *C* is also seen. The housing for the motor is open at the sides so as to give air currents free access. In [Fig. 4](#) the speed changing gears are shown, the reversing train being omitted; but if it were also drawn in, the diagram would be far more elaborate than [Fig. 3](#).

Another form of variable speed gear is shown in [Fig. 5](#). This provides for two speeds. The large wheel *E* is on the carriage axle, and it is driven either by a pinion *F*, or by *J*. Upon the shaft *O* there are two friction clutches *C D*, and when *C* acts the pinion *F* drives *E*, and when *D* acts the pinion *G* drives *H*, which in turn drives *I*, and this wheel is mounted on the same shaft as *J*.

Some of the best-known makers of gasoline vehicles do not employ variable gears and depend for changes in the speed wholly upon variation in the velocity of the motor. The De Dion carriages are made in this way, the gearing being substantially as illustrated in [Fig. 3](#).

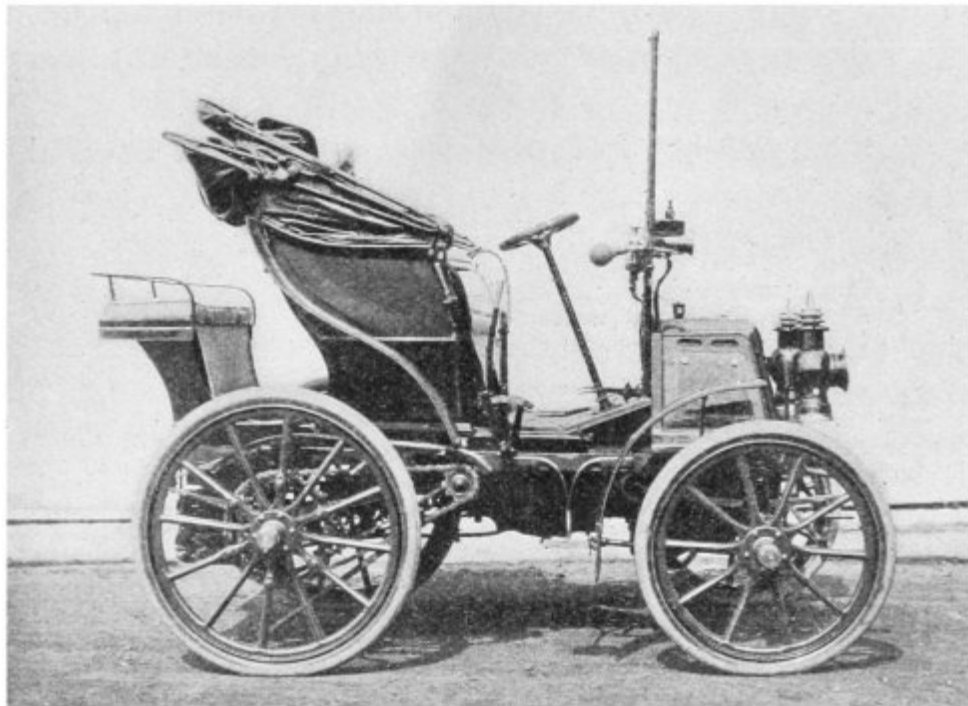


FIG. 6. PANHARD & LEVASSOR VEHICLE.

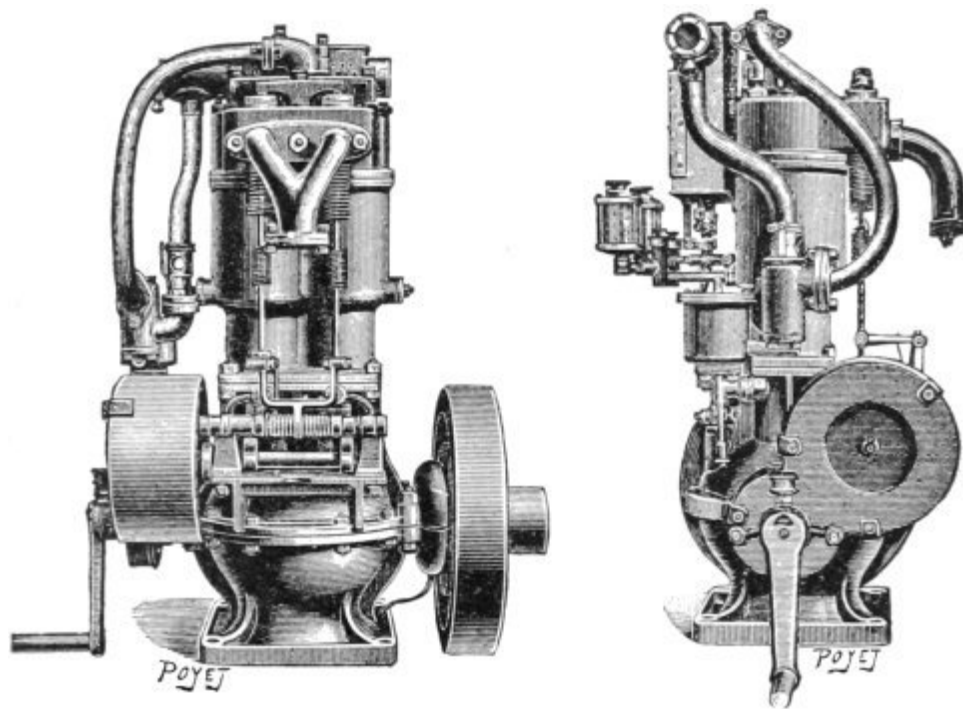


FIG. 7. MOTOR OF VEHICLE.

[Fig. 6](#) shows a gasoline vehicle made by Panhard & Levassor, who are perhaps the best known French manufacturers of automobiles, as their vehicles have been the winners in all the notable races held within the past few years. The motor they use is shown in [Fig. 7](#), and, as can be readily seen, is of the two-cylinder type, cooled by a water jacket, just as in [Fig. 1](#). The explosion is produced by means of a hot tube, as explained in connection with the last-named figure. This motor is placed under the body of the vehicle, and is connected with the rear axle by means of a train of gearing which terminates in sprocket wheels and chains that connect with driving wheels, each one being operated by a separate chain. In [Fig. 6](#) the sprocket wheel and chain are well defined, and forward of these can be seen the outline of the casing enclosing the gearing.

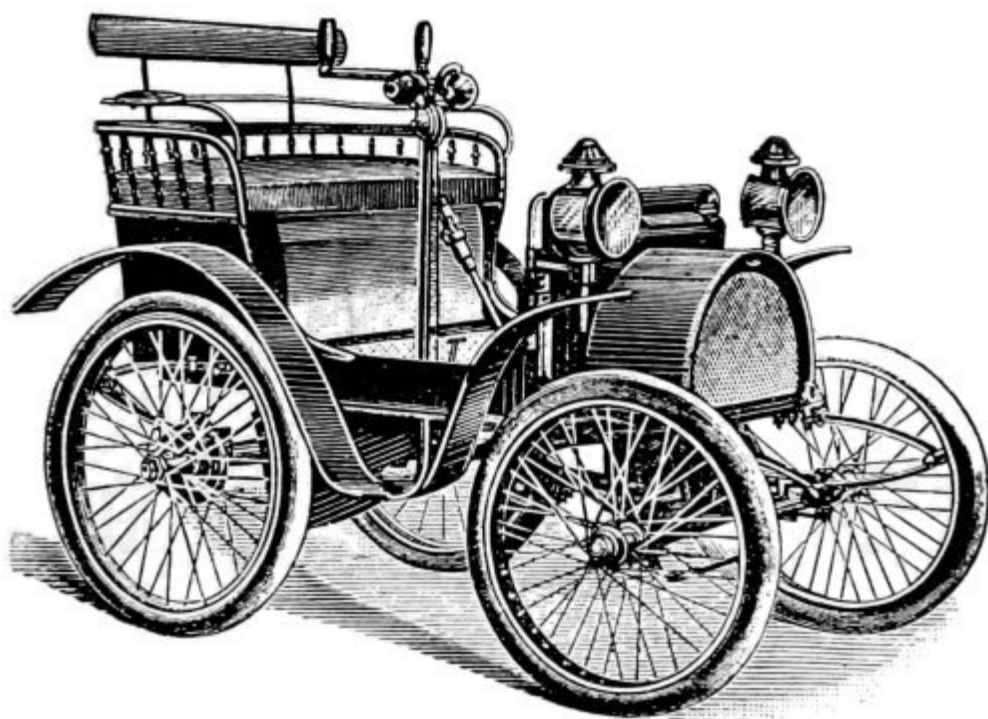


FIG. 8. GENERAL VIEW OF RENAULT VOITURETTE.

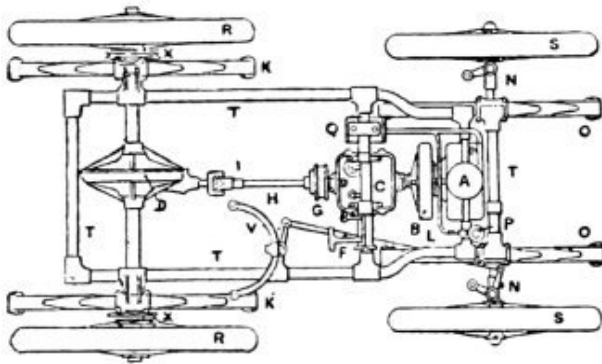


FIG. 9. PLAN OF THE TRUCK.

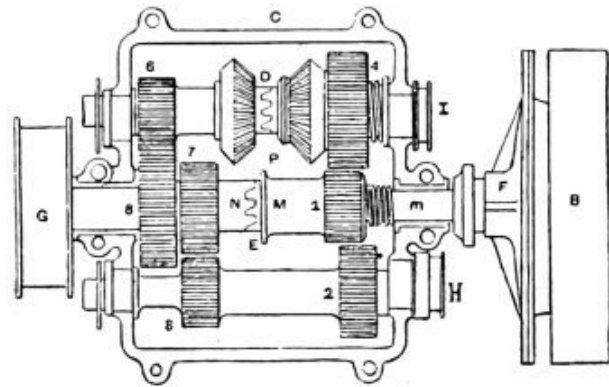


FIG. 10. VARIABLE SPEED GEAR.

[Fig. 8](#) shows another European design, in which a variable-speed gear is used. The plan of the truck, showing the general arrangement of the mechanism, is presented in [Fig. 9](#), and the details of the variable-speed gear are shown in [Fig. 10](#). The motor is located at A, and through a friction clutch B, and the variable speed gear C, it rotates the shaft H, which runs lengthwise of the vehicle. Motion is imparted to the hind axle by means of bevel gears contained within the casing D. The large bevel gear on the axle is of the differential type, so as to drive the wheels R R at the proper velocities.

When a high speed is desired, the variable speed gear, [Fig. 10](#), is set so that shaft M drives N direct, the clutch at E being moved so as to interlock. N is the end of shaft H, so that with this connection the bevel pinion, which meshes into the axle gear at D, revolves at the same velocity as the motor

shaft. By moving the handle *V*, [Fig. 9](#), to the right, an intermediate speed is obtained, and by moving it to the left, the carriage is run at the lowest velocity. When the handle *V* is turned to the right, the ends *M* and *N*, which form the clutch *E*, [Fig. 10](#), are separated, and at the same time the lower shaft *H* is moved toward *M*, so as to cause gear 1 to mesh into gear 2, and also 3 into 7. By this means the end *N* of the axle-driving shaft is rotated through the train of gears 1, 2, 3 and 7. If the handle *V* is turned to the left, the shaft *I* is moved toward *M*, so as to cause gear 1 to mesh into gear 4, and gear 6 into 8, the latter being secured to end *N* of the axle-driving shaft. The speeds obtained by these changes are in the ratio of nearly 1, 2 and 4.

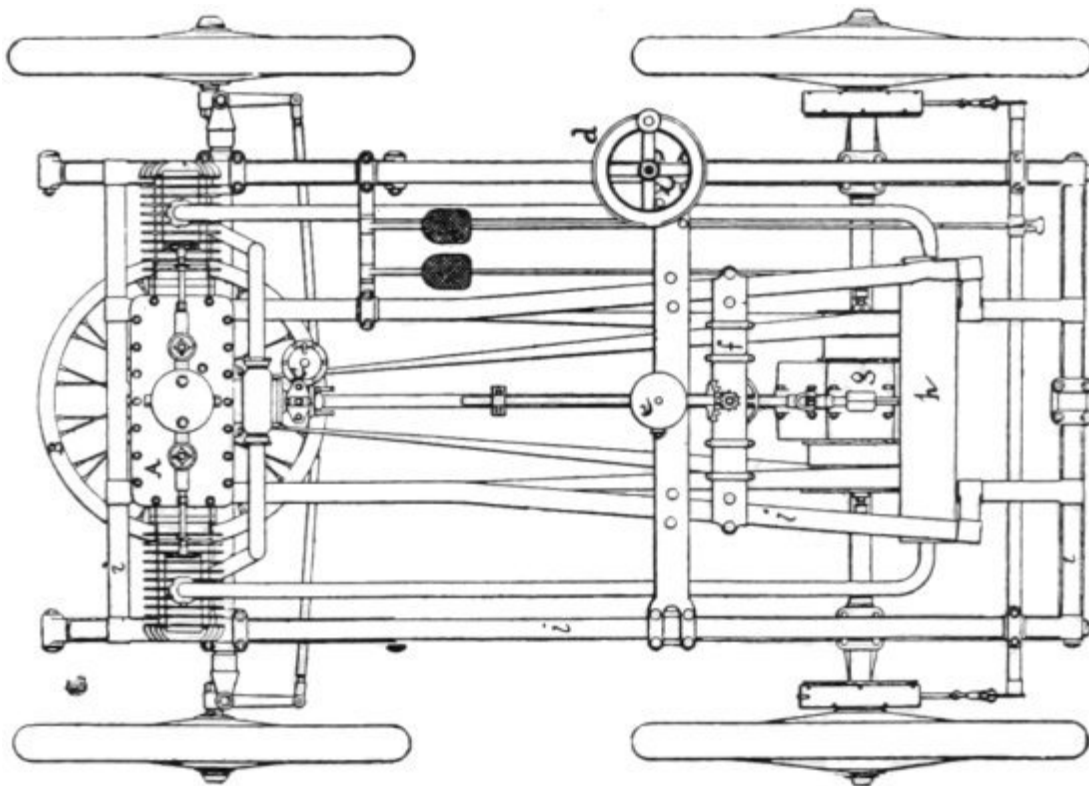


FIG. 11. PLAN OF THE TÜRĞAN-FOY VOITURETTE.

[Fig. 11](#) shows the plan of a light French carriage, which is equipped with a double cylinder motor, set in a horizontal position above the front axle, and arranged to impart motion to the hind axle by means of belts. The motor, which is located at *A*, turns a vertical shaft, and this, through spur

gears, rotates a horizontal fly wheel, *B*. Two pulleys are mounted upon the motor shaft, and from these belts run to tight and loose pulleys on a countershaft, *S*. From the latter the rear axle is driven through two sets of spur gearing, which give two different speeds. By means of the belts, two other speeds are obtained, thus giving, in all, four different velocities. To stop and start, the belts are shifted from the tight to the loose pulleys by a belt-shifter, *f*. At *h*, a muffling chamber is located, into which the motor exhausts, so as to reduce the noise.

The elevation and plan of one of the celebrated French racing-machines, the Vallée car, is shown in [Fig. 12](#). The motor of this machine is of sixteen horse-power capacity, has four cylinders, and is connected so as to impart motion to the hind axle by means of a single wide belt, which is marked *G* in both the line drawings. The driving-pulley on the motor shaft is located at *H*, and the axle pulley at *H'*. Within the latter there is a train of gears for reversing the direction of rotation of the axle, and also for obtaining the differential velocities of the two driving wheels. There is no mechanism for variable speed, this being obtained wholly by changes in the velocity of the motor. The motor speed can be made to vary through a wide range by using four cylinders, with which it is possible to reduce the velocity so low that it would be likely to bring the machine to a standstill if provided with one, or even two, cylinders. The change in the motor velocity is obtained in part by the action of a governor located in a chamber at *A*, and in part by the action of the electric ignition device which is arranged so that the time when the spark is produced can be varied. The rear axle is so held that it can be moved through a short distance, horizontally, by manipulating the lever *D*, and in this way the belt *G* can be made tight or loose, thus affording another means for varying the speed. A brake is provided which presses against the inner side of the axle pulley, *H*. This brake is used ordinarily, but in the case of an emergency another brake can be operated which presses against the outside of the wheel in the space between the two sides of the belt. It is claimed for this vehicle that by the elimination of mechanical speed-changing devices, a great deal of weight is saved, and that this is more than enough to compensate for the extra weight of the motor, arising from the use of four cylinders. In most gasoline carriages it is necessary to provide a slow-speed gear for hill-climbing, as the motor cannot put forth a sufficient effort to ascend a steep grade at the normal

velocity. With this racing-machine such a gear is not required owing to the enormous power of the motor.

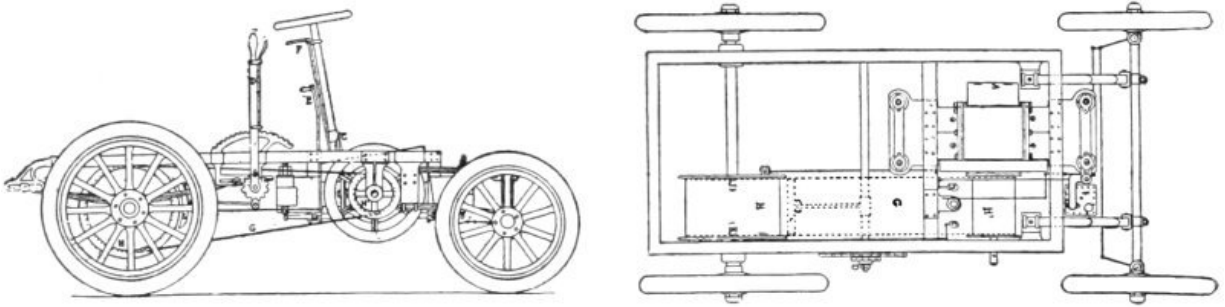


FIG. 12. ELEVATION AND PLAN OF VALLEE CAR.

There are quite a number of gasoline automobiles manufactured in this country, and, as in the case of the steam and the electric carriages, they compare most favorably with the best European products, in so far as the artistic effect is concerned. That such is the case can be realized at once by an examination of [Figs. 13](#) and [14](#). We regret our inability to illustrate the mechanism of these vehicles, but the truth is, that the manufacturers appear to be unwilling to make public the details of their designs. In the phaeton shown in [Fig. 13](#), a single-cylinder motor is used, and it is so arranged that it can run at different velocities, so that no variable speed mechanism is required, except a single train of gears, which is thrown into action when running uphill. The motor itself can be run at any velocity from 200 to 800 revolutions per minute, thus giving a speed variation of four to one. A carriage of this make competed in the last international automobile race from Paris to Lyons, France, and although it failed to come in first, it made a remarkable showing, which might have been considerably improved if it had not been for an accident which compelled it to retire from the contest.

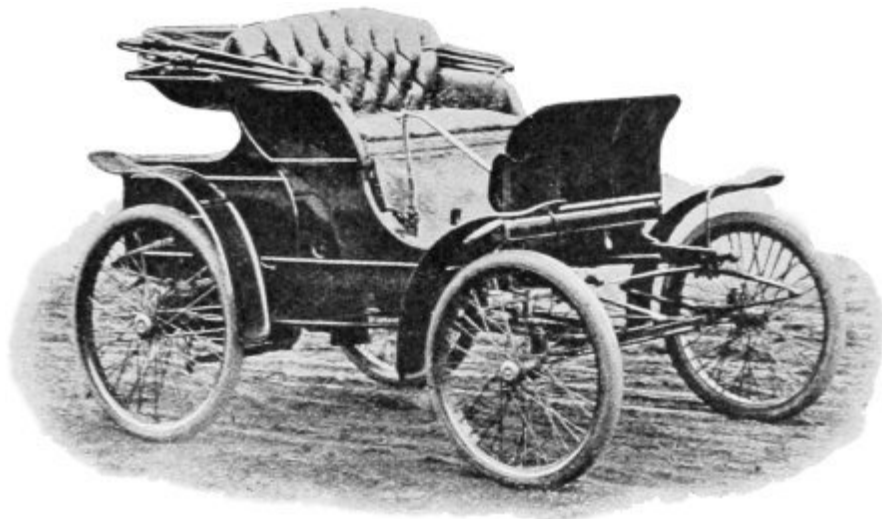


FIG. 13. WINTON PHAETON.

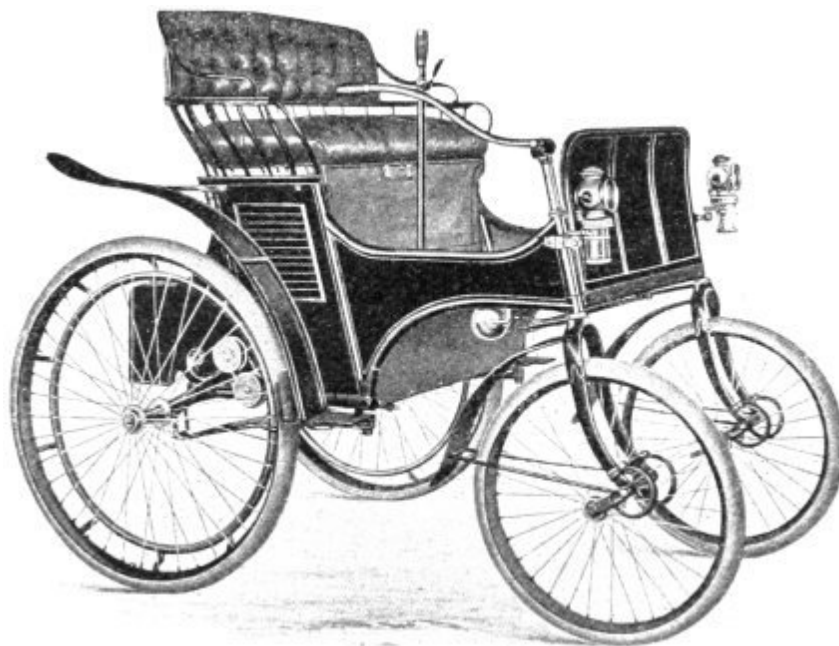


FIG. 14. OAKMAN VEHICLE.

The vehicle shown in [Fig. 14](#) is of small size and light construction, although amply strong for the purpose for which it is intended. The power of the motor, which is located under the seat, is transmitted through friction

wheels. In looking at the illustration it will be noticed that the hind wheels have a circular rim attached to the inner side, and of a diameter somewhat smaller than the wheel itself. Two small friction wheels are placed so that either one may be pressed against the inner surface of this rim. The shape of the rim, as well as that of the small wheels, is such that they hug each other firmly, so that the rim is carried around in a direction which corresponds with the direction of rotation of the friction wheel. In operating the carriage the motor is set in motion, and then one or the other of the two friction wheels is pressed against the rim on the driving wheel, according to whether it is desired to run forward or backward. While this arrangement might not operate with entire success if applied to a heavy vehicle, it appears to be all that could be desired for a light carriage.

Three-wheel vehicles have been used, but there is a difference of opinion as to their value, as the construction has disadvantages as well as advantages. It is evident that such a vehicle can be steered with greater ease than one running on four wheels, but on country roads, where the wagon wheels roll down a smooth surface, and leave the space between in a rough condition, it is equally evident that the third wheel, in passing over this uneven surface, would jolt the vehicle to a considerable extent. On a smooth pavement the three-wheel vehicle will run fully as well as the four-wheel; but, on the other hand, on such a pavement the latter can be steered with as little effort as the former, so that the question of superiority of design is one that probably depends upon individual taste.

From the descriptions of automobiles given in this and the two preceding articles, it will be seen that although many of them are used, especially in France, they are not entirely free from objectionable features. The electrical vehicles are provided with the most simple and durable machinery, and, being noiseless, odorless and free from smoke, are all that could be desired in so far as their operation is concerned; but they are heavy and can only be used in places where the batteries can be recharged. The steam vehicles are light, have simple mechanism and can run anywhere; but they exhaust steam into the air, which is clearly visible in cold or wet weather, and the heat from the engine and boiler is an objection, at least in summer time. The gasoline vehicles run well, but are noisy, and the odor of the gasoline is disagreeable as well.

SOME SCIENTIFIC PRINCIPLES OF WARFARE.

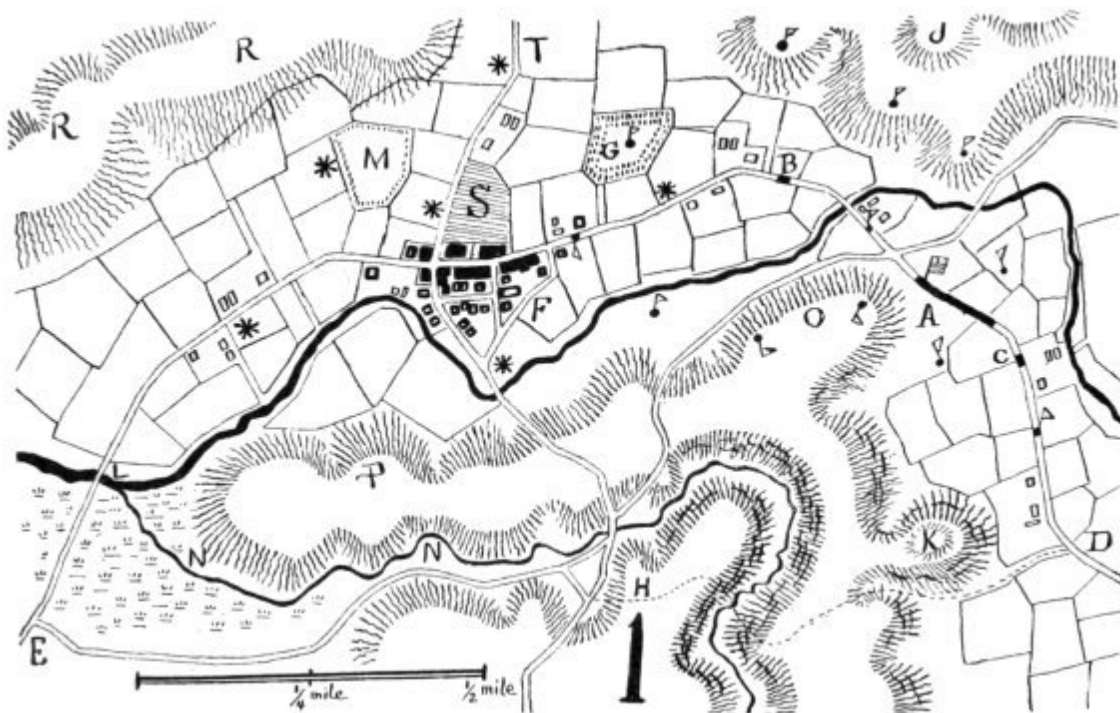
BY WILLIAM J. ROE.

AS in boxing, fencing, saber and bayonet exercises, there are comparatively few postures, guards, thrusts and strokes, so in warfare, whether the numbers be large or small, the arms most modern or ancient, there are just a few principles to whose steady adherence and skilful manipulation all success is due. In order that these may become apparent without irksome study of military details, let us imagine a command of say a thousand men, fairly well drilled, of good ordinary intelligence and engaged in a cause worthy of being fought for. We have been in camp for some time, but an order has now come to join the main army. This is a long distance off, the railway communications have been broken, and the intervening country, though possessed of good roads, is more or less in the hands of the enemy.

Our scouts have kept us informed as to the condition of the country for several miles around; our first day's march is, therefore, not hampered with any especial dread of surprise. We move quickly and at ease. Safe as everything appears to be, the commander relaxes none of the needful precautions; at least fifty men, under command of an experienced officer, are sent quite far to the front, the distance varying with the nature of the country—the farther, the more broken it may be. The best roads are followed; the men are allowed to march at ease, though always preserving their company organization, while the officers are always more or less on the alert. There is a small rear guard, but it is upon the advance that the main responsibility falls. Of the fifty thrown forward, about half will remain together; the rest are scattered; some far to the front along the highway; others on either side of the route, riding up the hills on either hand, making sure that no deep gorge, dense growth of forest or thicket, nor even a field

of grain conceals an enemy. It is upon the alertness of those vedettes on front and flanks that the safety of the force in great measure depends. History records many relaxations of this principle of precaution, and for lack of it sudden ambushes and deplorable disasters. It was thus, in spite of Washington's repeated warnings, that Braddock fell into a cunning ambush, and thus (not to multiply examples) that Custer and his command were massacred to a man among the high Rockies.

On the annexed map the men may be located at 'A' marching from 'D' in the direction of the village, 'F'. The advance is at 'B', the rear guard at 'C'. The commander rides with the main column, near the front. The black dots, with pennons, indicate the general position of the vedettes at this point, though, of course, they are continually advancing. The commander has noted on his map a foot path, beginning at 'D', leading over the rugged hills. By taking this path a considerable distance could be saved; but it is quite impracticable for the wagons, and the troops, therefore, continue along the high road. The valley is gently undulating, with a gradual slope from the low hills towards the stream.



The projecting hills near the head of the column form an especially dangerous point. What easier than for an enemy to plant batteries here on either side of the road. A sudden, heavy fire would throw a negligent force at once into disorder; a situation to be taken instant advantage of by a vigorous adversary; a charge of horse concealed behind the hill at 'O', and nothing might be left except flight, with great loss of life, and surrender with loss—if not of honor, at least of reputation as a safe leader.

Happily, we shall avoid both alternatives. Our scouts have explored most thoroughly every possible vantage ground. They have not been content with any mere glances; their instructions are to take nothing for granted. That field, marked 'G', looks innocent enough, but the tall, thick rye or corn may cover a skilfully placed battery. The plot marked 'M' may be simply a vineyard; but it does no harm to inquire. The inhabitants of the country are friendly, and, therefore, the chances are not favorable to this sort of surprise; but in war it is often not the likely, but the unexpected that happens; the commander who knows his business guards against the remote possibility.

Though we have imagined a force of a thousand, it must not be lost sight of that the same kind of precautions should be employed for very much larger numbers; indeed, you need only alter the scale of the map, imagine additional roads, a railway line or two, increase to thousands, if necessary, the fifty of our vanguard, and the result is but an application of the very first principle of warfare: Eternal vigilance is the price of safety as well as of liberty.

The troops have been in camp for some time; their condition is excellent for a long march. As the corn and rye are not yet gathered, the time is early summer. The roads are in prime condition. They set out by sunrise and halted for perhaps two hours at noon. It is by thus sparing his troops during the heat of the day that the colonel will have a body of men fresh enough at nightfall to march, if necessary, all night. But no such urgency exists; it is nearing sunset, and preparations are now being made to encamp. By his map the colonel has informed himself in the matter of distances, and has decided that they shall pitch their tents somewhere in the vicinity of the village ('F'). The scouts report an eligible location for camp at 'S', and this is finally chosen. It has several advantages, being comparatively level, and yet upon high ground, and has in close proximity several wells of good water. The train containing provisions and

ammunition is parked in the safest locality, the horses picketed, and the guns—perhaps two or three field pieces and machine guns—placed where they can be most easily handled.

By all means, give the men as good a supper as the neighborhood affords. It will be wise not to encroach upon the rations, but rather draw supplies from the village; there are, no doubt, purveyors of one sort or another to be found ready enough to supply us, the more so that they will be amply paid.

Refreshed by their supper the men are ready to turn in at tattoo; by the time ‘taps’ have sounded most are soundly sleeping. But some are awake; if doing their full duty, wider awake than ever they are likely to be in times of peace. The same attention to the bodily comfort of his men which impelled the colonel to give them a long rest at midday and a comfortable meal, applies with increased force to those detailed early in the morning for the night’s guard; during the march these have been spared as far as possible, even being allowed a lift now and then in an ambulance. Such privileges are not granted by a commander who knows his craft as a concession to the laziness, but rather as a preparation for the effectiveness of his men. This is a principle of action, and may apply to business as well as war, that the strong head never withholds reasonable and proper indulgence; the better, it may be said, to enforce at needful times reasonable and proper exertions.

As soon as the camp is established the guards are posted. If great precautions were needed during the day, much more are they by night. If fifty were sufficient on the march we need a hundred during the hours of darkness. In the case of a large army an elaborate system of night guards is necessary: First, ‘advanced guards’, occupying strong positions at some distance from the main body; beyond these are the picket guards; further still towards the front what are called ‘grand guards’, from which are thrown forward the outposts, to which the line of sentinels is directly attached. In case of alarm, the sentinels fall back upon the outposts; these upon the grand guards; they, in turn, if necessary, upon the pickets; the necessities of the case and the strength of the enemy’s demonstration determining the movements of the defense, even perhaps to the ‘long roll’ and rousing of the entire army.

In our case, no such elaborate system is possible; we content ourselves with outposts and the line of sentinels, all that will be needed, if vigilant, to guard against surprise. The colonel, attended by the officer in command of the guard, will select the sites for outposts. These, five in number, are marked by stars upon the map. The direction from which an attack is most probable is from the ridge ('R', 'R').

The men are usually on the sentinel line for two hours at a time, with opportunity for four hours' sleep; that is, with shifts, or, as they are called, 'reliefs' of three parties, two hours on and four off. This is not, however, invariable, it being sometimes wiser to relieve the men oftener or not so often, this being regulated by circumstances—the state of weather, distance of posts apart, fatigue of the men, etc., etc. The sentinels will be posted on clear nights generally upon high ground; in bad or foggy weather the foot of the slope is preferable. The officer will see that no obstacle prevents the sentinel from retreating upon his outpost if attacked. The men will be directed to take advantage of any cover that offers, always to keep in easy touch with one another and watchful, never to raise a false alarm, but quickly and decidedly a real one, and while not failing to discover the meaning of anything unusual in their front, never to expose themselves from mere bravado.

What measures shall be taken in case of an attack in force must, of course, depend entirely upon circumstances. A night attack, intended merely as an annoyance, or 'feeler', or at most to stampede some of the cattle, or to gather information as to strength, resources, etc., is quite a different affair from one planned for the purpose of complete victory, either the destruction, dispersal or capture of the command.

A mere night foray is generally executed by comparatively few. The opposing chief may be desirous of getting information concerning the force that his scouts have reported is advancing down the valley. A little expedition like ours sometimes serves as a disguise for a momentous strategical movement. The chief determines to find out all he can as to our purpose. He has found us vigilant by day; he resolves to try what the night may disclose. This sort of surprise is apt to produce better results than the project of some dashing subaltern, anxious for the bauble reputation.

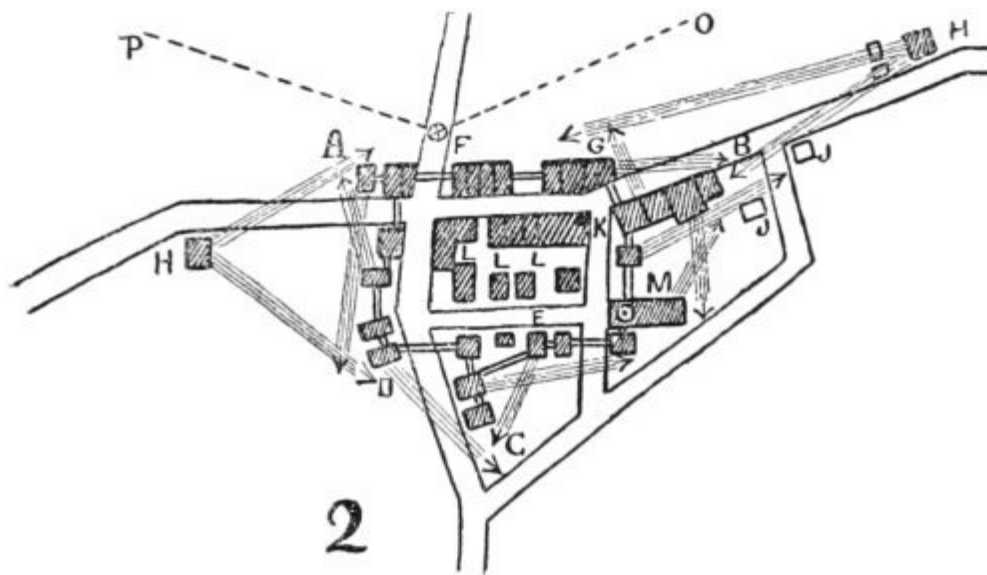
For such an attack an hour near midnight is usually selected, that the information may be gathered or the mischief done and a retreat effected under cover of darkness. A dark, wet, blustering, or—if the time be winter—an especially cold night is chosen. The degree of success to be attained depends naturally upon the element of surprise. Unless this be complete the attacking party will find their attempt usually quite futile.

The other sort of attack—that which has for its object the capture of the position—is usually planned to take place during the extreme darkness just preceding daybreak. The enemy has perhaps crawled on hands and knees up the slopes towards the line of sentinels. The van of this force is composed entirely of picked men, officered by the coolest heads. Signals are agreed upon, exact times for action arranged, and everything calculated to a nicety to insure that suddenness which is the very soul of success.

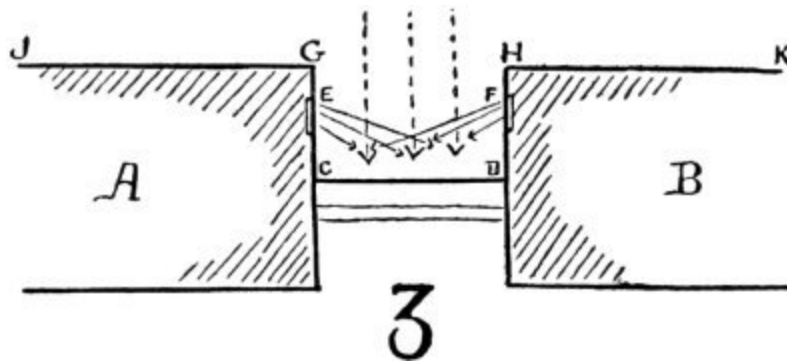
It is in the planning of such an expedition that true qualities of generalship are shown. It is the fashion rather to decry the military merits of Washington; yet I know of few events in history that show more sagacity than the swift crossing of the wintry Delaware and the surprise of Trenton. It was sagacious chiefly for the accurate comprehension of the probabilities. Washington knew the convivial habits of Rahl's Hessians, especially at Christmas-tide; he reckoned upon finding them in the midst of carousals, and the result proved the value of his forethought.

Under ordinary circumstances, on the march, to quarter a command inside four walls is never advisable. The men are not as readily under the eye of their officers; in case of surprise they cannot be called into the ranks as quickly; discipline insensibly relaxes, and the machine (for an armed force ought to be that, however intelligent its units) fails to respond instantaneously to the word of the chief. In case of a serious attack, however, the village may serve a most important purpose. Should the houses be substantial ones of stone or brick, each may become a most efficient, if temporary fortification. One consideration which might have prevented its occupation has now no longer any weight. Apart from any natural feeling of good will for our fellow citizens, how unwise it would be to unnecessarily exasperate them. But now in the face of the enemy, it will be surprising if any soul is churl enough to grudge a patriotic hospitality. Most of the denizens will, indeed, make haste to hide their precious persons in the cellar, but will seldom grumble at the necessity.

With the utmost celerity the baggage and horses are moved to the most sheltered spot; the guns, under strong guards, posted where they may be best utilized; some of the men, previously detailed for just such an emergency, are engaged in throwing up earthworks, piling logs, stones, anything that can be utilized for barricades. The officer charged with that duty, if possible a skilled engineer, goes quickly from place to place, hurriedly indicating the lines of defense; these connecting the several buildings in such a manner as to enclose the entire command within lines of quite formidable intrenchments. All this time the troops, having taken possession of the houses, have poured an uninterrupted fire upon the assailants, obliging them to retire, or at least giving the diggers—or sappers, as they are called—time to complete their labor of defense. Surrounded by a force sufficiently large to make resistance in the field quite hopeless, we are at least in position to protract the struggle, and one capable of defense, except against an assault in overwhelming numbers, or against heavy artillery. The latter they are not provided with, or the measures we are taking might all go in the end for nothing. Several assaults are attempted during the day, but are easily repulsed with no small loss. The enemy at last withdraws, and we now see that he is busy throwing up intrenchments. Meanwhile, we have not been idle. To facilitate communication, and to enable us to concentrate our forces under cover, passageways have been constructed between the various buildings, inner partitions preventing free access from room to room within the houses have been broken through, and the débris, together with beds broken up, mattresses and ‘any old thing’ capable of arguing with a bullet, piled in the window embrasures, leaving loopholes here and there, as occasion offers, while galleries may be constructed with loopholes in the floor to fire downwards.



One of the most important matters to be attended to is the securing of as many good positions as possible, from which fire may be concentrated upon exposed points. In a regular siege the points of attack selected will always be those most exposed, on account of their projecting beyond the line of defense. In the case of a village like this resisting an attempt at capture the principles are identical; it will certainly be the points that project that will be danger spots and which will therefore require especial attention.

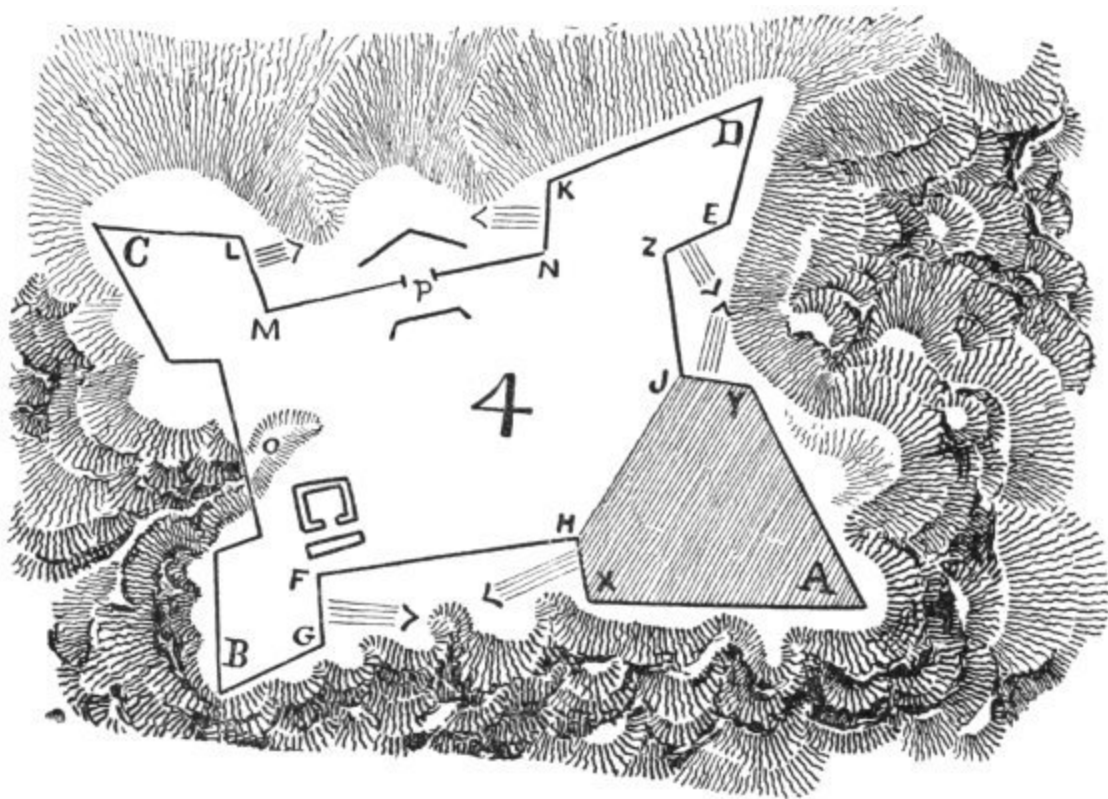


You observe on the enlarged map of the village that there are double lines between the outer buildings; these are the improvised intrenchments. Notice that they have not been constructed flush with the face of the outer walls in any instance; but always considerably retired. The object of this arrangement is more effectually to defend the barricades. In the annexed sketch (No. 3) 'A' and 'B' represent the two adjacent buildings and the lines 'CD' the breastwork. In the buildings are windows—'E' and 'F'—from which a heavy fire can be concentrated upon the assailants, as may be seen from the direction of the arrow heads. On the outer line are several projecting, and, therefore, especially exposed points; such as those at 'A', 'B' and 'C'. The arrow heads show the direction of protective fire. As additional protection, it might be wise to hold the two buildings ('H', 'H') outside the village. If not held, they ought, if possible to be destroyed, as also those marked 'JJ', not included in the defensive lines, as they offer excellent cover for the enemy. The utmost care should be taken to provide a safe magazine for the ammunition and to cover well the place selected for a hospital. The wagons and horses would be best protected in the space marked 'LLL'.

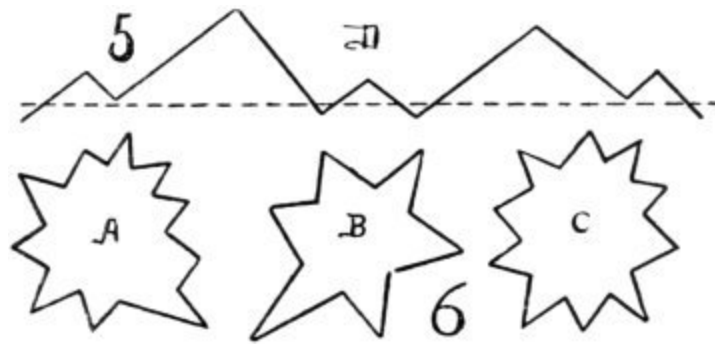
Should our defense prove too obstinate for direct assault, it may be that the enemy will construct regular intrenchments from which to dig a trench deep enough to protect, and large enough to hold a body of troops, thus enabling them to approach sufficiently near to assail some weak point, without too great risk. The modern repeating rifle, dangerous at a thousand yards, and fatal at a hundred, has given the defense so great preponderance

that it requires quick work indeed to capture a stronghold. Observe the broken lines 'OF and 'PF'; these show the direction of possible trenches dug by the enemy. But 'OF' would be raked by the fire from the outlying house, 'H'; the other is, therefore, the only feasible mode of approach.

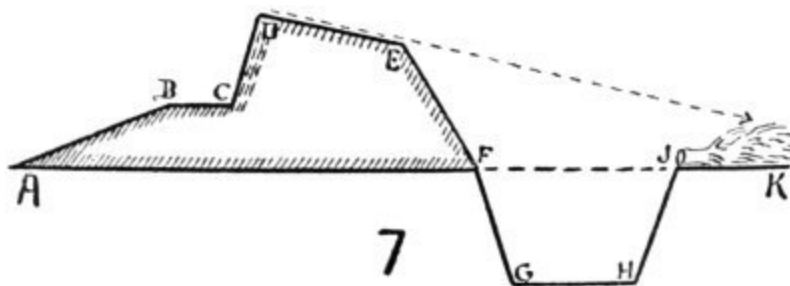
The principle of defense, shown by the direction of the arrow heads in the case of the beleaguered village, is applicable to all conditions where ramparts are used. Suppose the command whose fortunes we have followed had been attacked while on the march at the point 'A' on [Map 1](#). The opposing force was manifestly too strong for resistance in the field; they retreat to the rocky eminence 'K' and there proceed to fortify the position. A glance at [Diagram 4](#) will show what they will try at least to accomplish. In military language that shaded portion of the work to be constructed is called a bastion; it consists of two faces ('AX' and 'AY'), and the two flanks ('JY' and 'HX'). The faces of this bastion are defended (as the arrow heads indicate) by the flanks of adjacent bastions; that is, the face 'AY' is swept by a raking fire from 'ZE', and the face 'AX' from 'FG'. Reciprocally, 'HX' rakes the face 'BG', and 'JY' the face 'ED', and so on round the intrenchment.



All that has been said as to protecting the ammunition and stores will apply to this work as it did to the village. If a spring of water can be included, as at 'O', this will be found of incalculable advantage. Of all forms of defensive ramparts the straight line is the worst; if time does not permit a work with bastions, however irregular, an enclosure shaped somewhat like a star is serviceable (shown in [Diagram 6](#), Figs. 'A', 'B' and 'C'). Should an enclosed work be impracticable, the line should have its ends (or 'flanks') strongly guarded, and be broken up, as in [Diagram 5](#) 'D' into short straight lines nearly at right angles, to serve for mutual support. This principle of mutual support, however achieved, is called that of 'defensive relations', and is capable of adaptation to all kinds of defensive works, whether of a few men beleaguered in an improvised fortification, a considerable number in a scientifically constructed work—permanent or field fortification—a fortress with an entire army behind its ramparts, or a cordon of forts surrounding a great city.

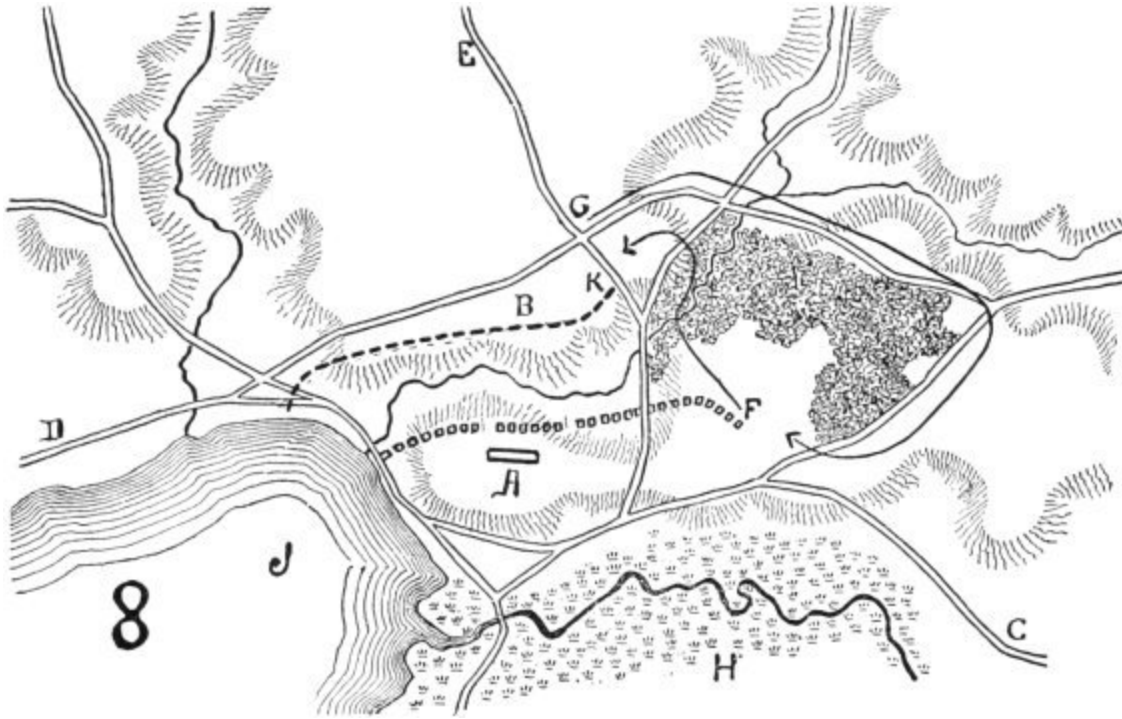


The ground plan of the work having been decided upon and staked out the men start in with pick and shovel, digging, if possible, a ditch, and throwing the material into the shape of the shaded portion of Diagram 7. The ditch, outside the fort, indicated by the figure 'FGHJ' serves the twofold purpose of getting material for the parapet 'ABCDEF', and for embarrassing an enemy in any attempt at assault. To further embarrass him every sort of obstacle that may be at hand should be put to use—trees, butts turned our way, boughs interlacing; stakes driven deep into the soil close together; barbed wires wound in and out; in short, every expedient that may delay his advance and keep him as long as possible exposed to our most effective fire.



The drawing (7) was made with no attempt at exactness of proportion, and simply to show the essentials; the slope 'EF' is made as steep as the nature of the soil will permit; 'DE' slopes enough to enable a soldier standing upon 'BC' to fire upon an enemy entangled among the obstacles at 'J', but never enough to weaken the mass of earth at and near 'D'.

Observe how common-sensible all these arrangements are; not one too many or too few; just the things that a practical man, if he could think as he felt, would do if suddenly called to command with an enemy advancing upon him. Unfortunately, perhaps, for the purposes of a patriotic and peaceful people, men are inclined, even though brave as courage itself, to get nervous or nerveless in the immediate presence of danger. This is the reason, rather than for any especial erudition involved in war's art, that we need trained soldiers—men trained to think mechanically and to act automatically amid the uproar of battle.



We have carefully, if briefly, considered the requirements of the first maxim of strategy—CAUTION—the need of it, and the practical methods of securing it; and also of the second maxim—DEFENSIVE RELATIONS—their necessity, and how to secure them. It now remains to consider the meaning of that phrase, ‘turning a position’, or ‘flanking’ an enemy, as to which of late we read so much in the daily press. The map (marked 8) gives an idea of a section of country where two armed bodies meet under conditions that permit one flank to be completely guarded from attack; these are the left

flank of the force 'A', and the right flank of 'B'. Both rest upon a lake or broad river. A steep precipice or deep morass, as at 'H', would serve as well. Suppose our force has advanced from the direction 'C', the enemy down the road from 'E' to 'G'. Soon they form opposed lines facing each other, the reserve somewhat to the rear and sheltered by some inequality of ground, the 'thin blue line', almost, but not quite, touching elbows, stretched along the crest of the ridge in front, taking advantage of every chance to protect themselves—trees, stone walls, ditches; kneeling, crawling, lying face down, eyes along the rifle barrel, finger on trigger, keen and murderous, but prudent, and parsimonious of life. The solid formations, such as went out of vogue with old-time weapons, would melt away before machine guns and Krag-Jørgensens like frost before an August sun. It seems as if all chivalry had departed; it has but changed its ways.

The object of 'flanking' a position is to so manage as to turn that attenuated line into a mass of men upon which to let loose with dire effect either the quick-firing guns or the sharp edges of our horsemen's sabers.

Notice those long, bent, black lines, bending like fish hooks. The arrow heads indicate the direction of a flanking attack; from 'F', through the woods, up the ravine, to fall upon the exposed end of the enemy's front at 'K'. Such would be our most feasible method of flanking; the foe might, however, have anticipated us, either by providing a bloody hospitality somewhere in that ravine, or by a flank movement of his own, as the bent black line shows, around the woods, to fall upon our right flank at 'F'. Such an operation, if successful for them, would be utterly disastrous to us.

Surprised by a sudden and unexpected attack upon the weakest point and unable to change front in time, men lose heart, forget discipline, huddle in masses, confused and disorganized, or fly like sheep, in either case food for firearms, gluttonous of such occasions. It requires sometimes but a very small force upon a flank to produce great results; the appearance upon the field, even at a distance, of Joseph E. Johnston's corps at the first Bull Run was sufficient to demoralize the whole Union army, and at the battle of Arcola, Bonaparte completely flanked the Austrians with a few flourishes of his trumpets.

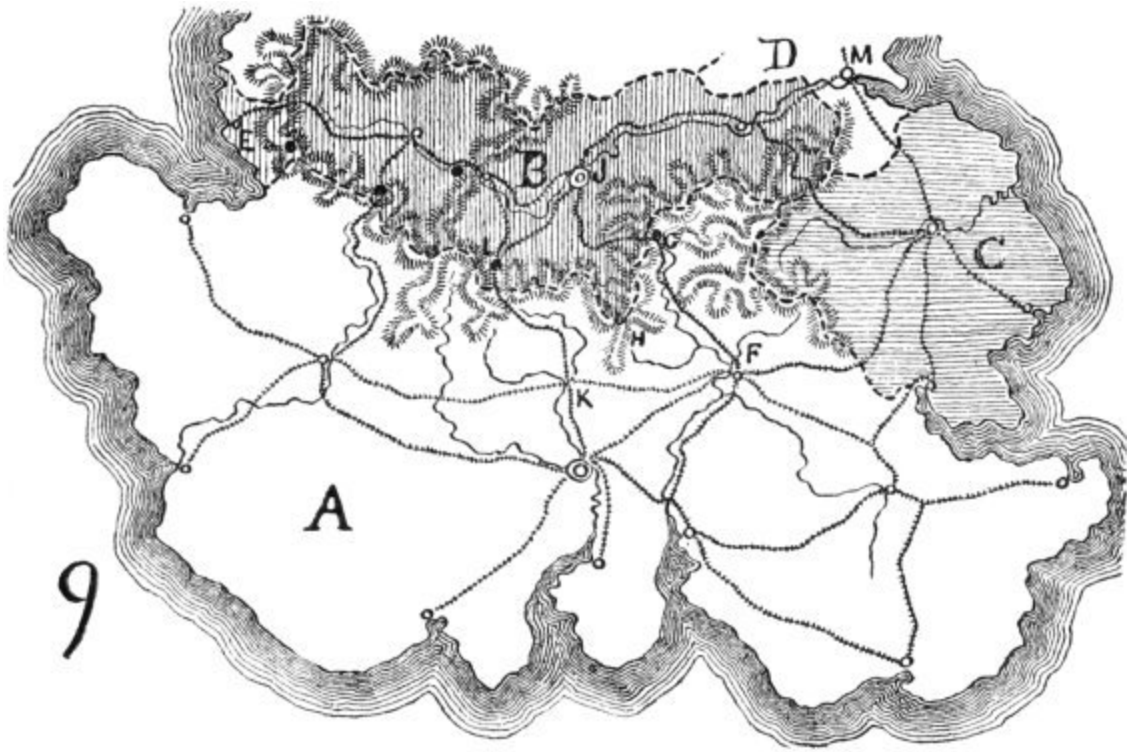
So we have for a third maxim of war the necessity of PROTECTED FLANKS. If we know or think that a Johnston lurks on either hand, we ought

to be sure of our Pattersons; if we apprehend an unfriendly visit from a Blucher, we should see to it that our Grouchy is trustworthy.

Let us now broaden our view of operations, that we may see how the principles established for a limited number of men on the march, in the field, or behind fortifications, may apply upon a larger scale. To this end a brief study of the map (9) will show four contiguous countries—‘A’, very populous, powerful and wealthy, having a navy capable of control of the high seas, and a large and efficient army; ‘C’ represents a country even more populous, but not aggressive, ‘D’ an insignificant power, while ‘B’ is a country considerable in extent, but largely mountainous, and sparsely inhabited by a rude but warlike people.

A cause of war comes up between ‘A’ and ‘B’. In ancient times the ruder nation would have been the aggressor, tempted by the wealth and invited by the enervated populace of the larger civilization. Now the conditions are likely to be reversed. However, war begins; the forces of ‘A’ move hastily towards the frontier, while his fleet blockades ‘B’s’ solitary seaport at the point ‘E’. The maxim of CAUTION now naturally expands; instead of information culled by a few daring riders from a narrow circuit, it should be made to embrace the widest area of country and the utmost latitude of information—the condition of the enemy as to armament, resources, position of forces, possible disaffection among the people—everything. In war no item comes amiss. The wealthier country will here have a manifest advantage; it can afford to hire spies, and can even (as England did during the Revolution) purchase the treason of some disaffected chief. Caution for the lesser country will—if good generalship prevails—take the shape of occupying and strengthening the natural strategic positions. These are nothing but flanks of a bastion on a large scale. Upon the map round black dots represent strategic positions along the frontier. They are points susceptible of thorough fortification which control the several passes in the mountain range between the two nations; also heads of valleys, where several meet, and from which attacks could be made at will in a number of directions. This entire frontier, which may be hundreds of miles broad, is mountainous, capable of being fortified at countless points, and having natural ‘defensive relations’ needing only the art of warcraft to render them almost impregnable. Modern murderous arms lend their services more readily to defense than to offense. It is even

possible that the country 'B', warned in due season of the purposes of her powerful rival, may have plotted out each rod of ground among those mountain passes, and that artillery service, once a matter of gunnery, has now become a matter of mathematics.



We now come to the fourth maxim of war; it is that of efficient SUPPLY. An army, as the saying is, moves on its belly. An invading force must ordinarily provide for all its needs from some safe place in the rear, called a 'base of operations'; it must also provide that the line of transit of its provisions and ammunition to the front shall not be liable to interference. Assuming that at 'F' is a strongly fortified city, the railway line or the adjacent rivers would furnish 'A' with a practical base; his line of advance would be in the direction 'FG', called the 'line of operations'; 'G', a fortified pass, the proximate, and 'J', the capital of 'B', the ultimate objective point of the campaign. But it will be noted with what facility a determined enemy could fall upon 'A's' communications from the point

‘H’, which would also be the case were the advance made from ‘K’ towards ‘L’.

Of course, in the end, the larger resources will prevail; but it may be that ‘A’, baffled and exasperated by a stubborn resistance, and finding that ‘B’ is being supplied through the neutral and insignificant country ‘D’, may finally conclude, “in the interests of a higher civilization,” to violate their territory, seize the port ‘M’, and thus, by a far-reaching and bold flank movement, gain entrance into ‘B’s’ country. Such devices are not unknown in the history of war. Such a course would be a distinct violation of the ‘law of nations’; but there would be apologies and ample indemnity to ‘D’, with which, doubtless, she would be satisfied.

In imagining such a campaign no account has been taken of the attitude of the country ‘C’, or of that of any foreign nation. In war these things must be reckoned with. Neutral nations are always liable, however disposed to maintain neutrality, to be touched at some sensitive point by one or the other of the contending parties.

MODERN MONGOLS.

BY F. L. OSWALD, M.D., A.M.

THE political supremacy of the Caucasian race was supposed to have been decided by the fall of Carthage, more than two thousand years ago, but was thrice afterwards imperiled by an encounter with a rival of long-unsuspected resources.

The Scythians of Strabo were probably not Tartars, but Slavs ('Sarmatians'), or, like their allies, the Getæ, Slavs, mingled with Teutons. Parthia, too, had a semi-Aryan population; but the campaign of Attila gave the champions of Europe a chance to measure their strength with that of a new foe, as shifty as the Semites, and of far greater staying-power. His Huns were undoubtedly Mongols, and came so near overpowering the inheritors of Roman strategy that at one time the fate of western civilization hung upon the issue of a single battle. The western coalition triumphed, yet its victory on the plains of Chalons (October, 451), was due to the numerical inferiority of their enemies as much as to the predominance of their own skill or valor. The very retreat of the vanquished chief established his claim to the prestige of a superlative tactician.

Again, in 1402, only the accidental quarrel of two Mongol conquerors saved Europe from the fate of its ravaged borders. Sultan Bajazet had vanquished all his western foes, and the union of his forces with those of Tamerlane would undoubtedly have sealed the doom of the Mediterranean coast lands, if not all of Christendom.

A hundred years later the generals of Solymán II. came very near retrieving the neglected chance. They vanquished Austrian, Hungarian and Italian armies, and in 1560 defeated the combined armadas of the Christian sea-power at Port Jerbeh—so completely, indeed, that the allies were eager to make peace by betraying each other.

And it would be a great mistake to ascribe these victories to a mere triumph of brute strength. That same Solymán, with all his fanaticism, was a patron of every secular science, and at a time when western princes had to sign their names by proxy, Mohammed Baber Khan, the conqueror of India, wrote essays in four different languages and published memoirs abounding with shrewd comments on social and ethical questions and problems of political economy. He was a poet, too, and liberal enough to compose a dirge in memory of a prince whom he had slain in single combat.

Ethnologically, there is, therefore, nothing abnormal in the outburst of intellectual vigor that has lifted Japan to the front rank of civilized nations. It is merely a revival, analogous to the dambreak of pent-up energies that followed the collapse of mediæval despotism. Instead of having to work out their salvation by tentative efforts, the Japanese, it is true, had the advantage of ready-made patterns, but that difference has perhaps been more than offset by achievements affecting the reforms of four centuries in as many decades, and by modifications which, in more than one instance, have improved upon Caucasian models.

“The organization of the Japanese transport system,” says a press dispatch from Taku, “was a revelation to western staff officers; bodies of troops, with their equipments of stores and camping outfit, were landed without a hitch, in quick succession, and moved to the front without a moment’s loss of time. No delay, no confusion, no blockades of wharf-boats and baggage carts; everything worked in smooth grooves and in evident conformity with a prearranged and oft-rehearsed plan.”

And in 1897, after the affront of the Russian intervention, the victorious islanders, compelled to forfeit half the rewards of their valor, proceeded to make the very best of the other half, and their provoked diplomats managed to preserve their dignity, as well as their complete presence of mind. The Japanese police enforces law and order without waging Blue-Law wars against harmless amusements; there are no associations for the prosecution of bathing youngsters, no anti-concert crusades, no suppression of outdoor sports on the day when ninety-nine of a hundred wage-earners find their only chance for leisure.

The ‘Yankees of the Orient’ have a code of honor without duellos, trade syndicates without ‘trusts’, giant cities and ghetto suburbs without

anarchists. Their labor riots are settled by a dispassionate court of appeal. Their schools, Professor Arnold informs us, are hampered by 'fads' and experiment committees, but not by boards of bigot trustees. In spite of Buddhist conventicles, the emergence of the educated classes from the shadows of religious feudalism is a complete emancipation. The Japanese 'Council of Finance' has adopted American custom-house methods and Belgian systems of graded taxation. There is, indeed, a good deal of eclecticism in the supposed surrender of indigenous institutions; foreign methods have been adopted only on the evidence of their efficiency, and always with a view to making them subservient to national purposes. The key to the distinctive characteristics of the North Mongols can be found in Sir Edwin Randall's definition of 'Perseverance combined with shiftiness.' The Asiatic Yankees can turn, dodge and deviate while keeping a pre-determined aim steadily in view, and it is by no means improbable that Mongol influences have impressed similar peculiarities on the character of the northeastern Slavs. Muscovy was a Tartar Khanate for a number of centuries, and Russian diplomats, since the days of Czarina Katherine, have accommodated themselves to emerging circumstances by crawling or strutting, without ever losing sight of the road to Constantinople.

In the shaggy Ainos of Yesso (probably the original home of our 'Shetland' ponies), that perseverance takes the form of mulish stubbornness. They strenuously object to foreign imports and stick to their sheepskin cloaks like Scotch Highlanders to their kilts, but in stress of famine seem now to take an interest in the harpoon-guns of their Russian neighbors, and now and then sell specimens of their poodle-faced youngsters to the agents of a transpacific museum.

Japan still produces athletes, as well as unrivaled acrobats, partly, no doubt, on account of bracing climatic influences, but partly, also, of a vice-resisting worship of physical prowess. About sixty years ago the slums of the large seaport towns were expurgated by a national revolt against the spread of the opium habit, and the consequent reform movement appears to have kept step with the Swedish crusade against the spread of the alcohol curse.

China may be forced into the arena of regeneration, but thus far seems to view the collapse of her ring-wall only as a blessing in a rather effective disguise. The policy of non-intercourse, indeed, had the sanction of a

physical necessity in the opinion of as shrewd a statesman as the vizier of the great Kooblai Khan, who conquered rebels from Mantchooria to Siam, but recognized the hopelessness of ordinary measures for protecting the peaceful toilers of the eastern provinces against the predatory hordes of the northwest. A standard army of home-guards, he argued, would have to be composed either of natives who could not fight, or of foreign auxiliaries who might revolt; so, all things considered, it was deemed best to bar a foe that could not be beaten. Strategically, the plan succeeded, stone walls being then so inexpugnable to spear-armed besiegers that the proprietors of a stone-built robber castle could defy the wrath of the public for a series of generations. The Tartar marauders were kept at bay, but so were trading caravans and traveling philosophers; the disadvantages of all obstacles to free competition began to assert themselves. The nation, as it were, sickened in a marasmus of intellectual inbreeding. Protected incompetence propagated its species; monopolies flourished. The survival of the fittest no longer favored the brave; cowards and weaklings could find refuge under the telamonian shield of the big wall.

Within the last hundred years that process of degeneration has been hastened by two incidental afflictions—spring floods and summer droughts. The rapid increase of population has driven home-seekers into the highlands of the far west, and the destruction of land-protecting forests avenged itself in the usual manner. Every heavy snowfall in the mountains became a menace to the settlers of the lowlands; a sudden thaw was always apt to turn brooks into rivers and rivers into raging seas. The summers, at the same time, became warmer and drier. Famines, such as only India had seen before, crowded the cities with refugees. Charitable institutions were managed by agents of a paternal government, and paupers were rarely suffered to perish in wayside ditches, but hundreds of thousands were huddled together in parish suburbs and fed on minimum rations of the cheapest available food.

It was then that the masses were forced to apostatize from the dietetic tenets of Buddhism; abstinence from animal food became impossible; sanitary scruples had to be disregarded; whole settlements of famine victims were compelled to subsist exclusively on offal.

Millions of mechanics had to fight to struggle for existence by reducing their wants. The prices of food had doubled, and in order to pay the cost of

one daily meal all luxuries had to be relinquished. Sleep and oblivion of misery became the only alternatives of hopeless toil, and those who could save a few *taels* yielded to the temptation of supplementing those blessings by means of chemical anodynes. Opium-smoking became a national vice.

The 'opium war' did not rivet the yoke of that curse. It merely clinched the grip of a British trading company. The Chinese government had attempted to cancel their franchise, but only with a view to diverting its profits into the pockets of their own speculators. The total suppression of the traffic would have been not only difficult, but practically impossible. We might as well try to prohibit tobacco in North America.

Yet the results of these coöperating factors of degeneracy have stopped short of the extremes that might have been expected in a land of earth-despisers. Buddhism in its orthodox Chinese form is radically pessimistic. It inculcates a belief in the worthlessness of all terrestrial blessings, and considers life a disease, with no cure but death. And not death by suicide, either; the victims of misery must drain life's cup to the dregs, to cure the very love of existence, and thus prevent the risk of re-birth.

The value of health and wealth is thus depreciated in a manner that might tend to aggravate the recklessness of life-weariness; yet the South Mongol is conservative, even in his vices. An inalienable instinct of thrift makes him shrink from senseless excesses. Tavern brawls are less frequent in Canton than in Edinburgh; the toppers of the Flowery Kingdom get less efflorescent than ours, their love-crazed swains less extravagant. Absolute imbecility, as a consequence of poison habits, is a rare phenomenon in Mongoldom; nine out of ten sots remain self-supporting; the heritage of industrial habits is hardly ever lost altogether.

Nor should we forget to distinguish the primitive rustics of the inland provinces from the vice-worn population of the coast plains. Degeneration has not left its marks far above tide-water, and has hardly begun to affect the natives of the highlands, the Yunan hunting tribes, for instance, who, though South Mongols, have renounced the tenets of Buddha and adopted those of militant Mohammed.

Their chieftains welcomed war for its own sake, while the lowland conscripts were in the predicament of desert dwellers, caught in the flood of a sudden cloudburst. Thousands at first succumbed almost without a

struggle; the levies drilled to oppose the Japanese invasion stood to be slaughtered like sheep, being, moreover, morally handicapped by a misgiving that the war with the champions of the north had been wantonly provoked.

Discipline has begun to break the spell of that apathy, but the desperate valor that surprised the veterans of the allies at Taku and Yangtsun had a very different significance. Fury supplied the defects of military training; the listless life-renouncers had at last been goaded into a frenzy of nationalistic resentment. It was the same delirium of retributive wrath that rallied a million Frenchmen around the standards of the invaded Republic, and hurled a horde of Russian volunteers into the bullet-storm of Borodino.

‘A united nation of fifteen millions is not vincible’, wrote Jean Jacques Rousseau, in reply to an appeal of the Polish patriots. South Mongols were supposed to be hardly worth an expedition of Caucasian regulars, but even a world coalition might find use for intrenchments if the vendetta rage of a war for national existence should arouse a land of 385,000,000 inhabitants.

Whether that storm will purify the social atmosphere of the vast empire or subside into the calm of exhaustion, is a different question. It would even be premature to accept the appearance of a few able leaders as a propitious omen of regeneration. In a land ten times the size of France the crisis of a fearful peril will always evolve a Carnot, a Danton and a Dumouriez, if not a storm-compelling Bonaparte.

The days of the West Mongol Empire, the dominion of the turbaned Turk, are undoubtedly numbered, but not as a result of national decrepitude. The successor of Sultan Bajazet will succumb, not as a ‘sick man’, but as a cripple; an invalid worn out in a fight against hopeless odds. Within the last hundred years the stadtholders of the Prophet had to defend their throne against Russian, Austrian, Greek, French and British attacks, and more than once against a West-European alliance, backed by African and Asiatic insurgents. Within that period 3,000,000 Mongol Mussulmans have perished on the battlefield, a million for every generation of an impoverished and not specially reproductive race. Their empire will collapse, but its defenders are still the hardiest soldiers of Europe, the most unconquerable by hardships, wounds and hunger. The burden-carriers of Constantinople are still the stoutest men of our latter-day world. We might

as well impeach the degeneracy of the Circassian highlanders, who resisted the power of the Russian monarchy for sixty-five years, and in their last stronghold stood at bay with drawn hunting knives—after blunting their sabres and exhausting a stock of ammunition purchased by the sacrifice of their herds and harvests. For these heroic mountaineers, too, were Mongols, kinsmen of the martial Turkomans and chivalrous Magyars. The Turanian race—a synonym of the Pan-Mongolians—comprises as many different types as the Aryans and Semites taken together.

In 1863 some twenty clans of the vanquished highlanders left the Caucasus *en masse* to settle in the mountains of the Turkish province of Adrianople. They will share the fate of their protectors, and may soon be obliged to follow their flight across the Hellespont.

But the final expulsion of the West Mongols will, after all, mean only that the Caucasians have recovered lost ground, and freed at least Europe from an intrusive tribe of their most persistent and most formidable rivals.

RELIGIOUS BELIEFS OF THE CENTRAL ESKIMO.^B

BY PROFESSOR FRANZ BOAS.

^B A description of the religious beliefs of the Central Eskimo, based upon observations made by the writer, was published in the Sixth Annual Report of the Bureau of Ethnology. The following account embodies observations which Capt. James S. Mutch, of Peterhead, Scotland, following a suggestion of the writer, had the kindness to make. The material for this study was collected by Capt. Mutch during a long-continued stay in Cumberland Sound.

THE Eskimo who inhabit the coasts of Arctic America subsist mainly by the chase of sea-mammals, such as seals of various kinds, walruses and whales. Whenever this source of supply is curtailed, want and famine set in. The huts are cold and dark—for heat and light are obtained by burning the blubber of seals and whales—and soon the people succumb to hunger and to the terrors of the rigorous climate. For this reason the native does everything in his power to gain the good-will of the sea-mammals and to insure success in hunting. All his thoughts are bent upon treating them in such a manner that they may allow themselves to be caught. On this account they form one of the main subjects of his religious beliefs and customs. They play a most important part in his mythology, and a well-nigh endless series of observances regulates their treatment.

The mythological explanation of all the prevailing customs in regard to sea-mammals is contained in a tale which describes their origin:

“A girl named Avilayuk refused all her suitors, and for this reason she was also called ‘She who does not want to marry.’ There was a stone near the village where she lived. It was speckled white and red. The stone transformed itself into a dog and took the girl to wife.

“She had many children, some of whom became the ancestors of various fabulous tribes. The children made a great deal of noise, which annoyed Avilayuk’s father, so that he finally took them across the water to a small island. Every day the dog swam across to the old man’s hut to get meat for his family. His wife hung around his neck a pair of boots that were fastened to a string. The old man filled the boots with meat, and the dog took them back to the island.

“One day, while the dog was gone for meat, a man came to the island in his kayak^C and called the young woman. ‘Take your bag and come with me,’ he shouted. He had the appearance of a good-looking, tall man, and the woman was well pleased with him. She took her bag, went down to the kayak, and the man paddled away with her. After they had gone some distance, they came to a cake of floating ice. The man stepped out of the kayak on to the ice. Then she noticed that he was quite a small man, and that he appeared large only because he had been sitting on a high seat. Then she began to cry, while he laughed and said, ‘Oh, you have seen my seat, have you?’ [According to another version, he wore snow-goggles made of walrus-ivory, and he said, ‘Do you see my snow-goggles?’ and then laughed at her because she began to cry.] Then he went back into his kayak, and they proceeded on their journey.

^C The one-man hunting canoe of the Eskimo.

“Finally they came to a place where there were many people and many huts. He pointed out to her a certain hut made of the skins of yearling seals, and told her that it was his, and that she was to go there. They landed. The woman went up to the hut, while he attended to his kayak. Soon he joined her in the hut, and staid with her for three or four days before going out sealing again. Her new husband was a petrel.

“Meanwhile her father had left the dog, her former husband, at his house, and had gone to look for her on the island. When he did not find her, he returned home, and told the dog to wait for him, as he was going in search of his daughter. He set out in a large boat, traveled about for a long time, and visited many a place before he succeeded in finding her. Finally he came to the place where she lived. He saw many huts, and, without leaving his boat, he shouted and called his daughter to return home with him. She came down from her hut, and went aboard her father’s boat, where he hid her among some skins.

“They had not been gone long when they saw a man in a kayak following them. It was her new husband. Soon he overtook them, and when he came alongside he asked the young woman to show her hand, as he was very anxious to see at least part of her body, but she did not move. Then he asked her to show her mitten, but she did not respond to his request. In vain he tried in many ways to induce her to show herself; she kept in hiding. Then he began to cry, resting his head on his arms, that were crossed in front of the manhole of the kayak. Avilayuk’s father paddled on as fast as he could, and the man fell far behind. It was calm at that time and they continued on their way home. After some time they saw something coming from behind toward their boat. They could not clearly discern it. Sometimes it looked like a man in a kayak. Sometimes it looked like a petrel. It flew up and down, then skimmed over the water, and finally came up to their boat and went round and round it several times and then disappeared again. Suddenly ripples came up, the waters began to rise, and after a short time a gale was raging. The boat was quite a distance away from shore. The old man became afraid lest they might be drowned; and, fearing the revenge of his daughter’s husband, he threw her into the water. She held on to the gunwale; then the father took his hatchet and chopped off the first joints of her fingers. When they fell into the water they were transformed into whales, the nails becoming the whalebone. Still she clung to the boat; again he took his hatchet and chopped off the second joints of her fingers. They became transformed into ground seals. Still she clung to the boat; then he chopped off the last joints of her fingers, which became transformed into seals. Now she clung on to the boat with the stumps of her hands, and her father took his steering-oar and knocked out her left eye. She fell backward into the water and he paddled ashore.

“Then he filled with stones the boots in which the dog was accustomed to carry meat to his family, and only covered the top with meat. The dog started to swim across, but when he was halfway the heavy stones dragged him down. He began to sink and was drowned. A great noise was heard while he was drowning. The father took down his tent and went down to the beach at the time of low water. There he lay down and covered himself with the tent. The flood tide rose and covered him, and when the waters receded he had disappeared.”

This woman, the mother of the sea-mammals, may be considered the principal deity of the Central Eskimo. She has supreme sway over the destinies of mankind, and almost all the observances of these tribes are for the purpose of retaining her good-will or of propitiating her if she has been offended. Among the eastern tribes of this region she is called Sedna, while the tribes west of Hudson Bay call her Nuliayuk. She is believed to live in a lower world, in a house built of stone and whale-ribs. In accordance with the myth, she is said to have but one eye. She cannot walk, but slides along, one leg bent under, the other stretched forward. Her father lives with her in this house, and lies covered up with his tent. The dog watches the entrance, being stationed on the floor of the house.

The souls of seals, ground seals and whales are believed to proceed from her house. After one of these animals has been killed its soul stays with the body for three days. Then it goes back to Sedna's abode, to be sent forth again by her. If, during the three days that the soul stays with the body, any taboo or prescribed custom is violated, the violation becomes attached to the animal's soul. Although the latter strives to free itself of these attachments, which give it pain, it is unable to do so, and takes them down to Sedna. The attachments, in some manner that is not explained, make her hands sore, and she punishes the people who are the cause of her pains by sending to them sickness, bad weather and starvation. The object of the innumerable taboos that are in force after the killing of these sea animals is therefore to keep their souls free from attachments that would hurt their souls as well as Sedna.

The souls of the sea animals are endowed with greater powers than those of ordinary human beings. They can see the effect of the contact with a corpse, which causes objects touched by it to appear of a dark color; and they can see the effect of flowing blood, from which a vapor rises that

surrounds the bleeding person and is communicated to every one and every thing that comes in contact with such a person. This vapor and the dark color of death are exceedingly unpleasant to the souls of the sea animals, that will not come near a hunter thus affected. The hunter must therefore avoid contact with people who have touched a body, or with such as are bleeding. If any one who has touched a body or who is bleeding should allow others to come in contact with him he would cause them to become distasteful to the seals and therefore also to Sedna. For this reason the custom demands that every person must at once announce if he has touched a body or if he is bleeding. If he does not do so, he will bring ill luck to all the hunters.

These ideas have given rise to the belief that it is necessary to announce the transgression of any taboo. The transgressor of a custom is distasteful to Sedna and to the animals, and those who abide with him will become equally distasteful through contact with him. For this reason it has come to be an act required by custom and morals to confess any and every transgression of a taboo, in order to protect the community from the evil influences of contact with the evil-doer. The descriptions of Eskimo life given by many observers contain records of starvation which, according to the belief of the natives, was brought about by some one transgressing a law and not announcing what he had done.

I presume this importance of the confession of a transgression with a view to warning others to keep at a distance from the transgressor has gradually led to the idea that a transgression, or we might say a sin, can be atoned for by confession. This is one of the most remarkable religious beliefs of the Central Eskimo. There are innumerable tales of starvation brought about by the transgression of a taboo. In vain the hunters try to supply their families with food; gales and drifting snow make their endeavors fruitless. Finally the help of the angakok^D is invoked, and he discovers that the cause of the misfortune of the people is due to the transgression of a taboo. Then the guilty one is searched for. If he confesses, all is well, the weather moderates, and the seals will allow themselves to be caught; but if he obstinately maintains his innocence, his death alone will soothe the wrath of the offended deity.

D The medicine-man or shaman of the Eskimo.

While thus the reason appears clear why the taboos are rigorously enforced by public opinion, the origin of the taboos themselves is quite obscure. It is forbidden, after the death of a sea mammal or after the death of a person, to scrape the frost from the window, to shake the beds, or to disturb the shrubs under the bed, to remove oil-drippings from under the lamp, to scrape hair from skins, to cut snow for the purpose of melting it, to work on iron, wood, stone, or ivory. Women are, furthermore, forbidden to comb their hair, to wash their faces and to dry their boots and stockings.

A number of customs, however, may be explained by the endeavors of the natives to keep the sea mammals free from contaminating influences. All the clothing of a dead person, more particularly the tent in which he died, must be discarded; for if a hunter should wear clothing made of skins that had been in contact with the deceased, these would appear dark and the seal would avoid him. Neither would a seal allow itself to be taken into a hut darkened by a dead body, and all those who entered such a hut would appear dark to it and would be avoided.

While it is customary for a successful hunter to invite all the men of the village to eat of the seal that he has caught, they must not take any of the seal meat out of the hut, because it might come in contact with persons who are under taboo, and thus the hunter might incur the displeasure of the seal and of Sedna.

It is very remarkable that the walrus is not included in this series of regulations. It is explicitly stated that the walrus, the white whale and the narwhal are not subject to these laws, which affect only the sea animals that originated from Sedna's fingers. There is, however, a series of laws that forbid contact between walrus, seal and caribou. It is not quite clear in what mythical concept these customs originate. There is a tradition regarding the origin of walrus and caribou which is made to account for a dislike between these two animals. A woman created both these animals from parts of her clothing. She gave the walrus antlers and the caribou tusks. When man began to hunt them, the walrus upset the boats with his antlers and the caribou killed the hunter with his tusks. Therefore the woman called both

animals back and took the tusks from the caribou and gave them to the walrus. She took the antlers, kicked the caribou's forehead flat and put the antlers on to it. Ever since that time, it is said, walrus and caribou avoid each other, and the people must not bring their meat into contact. They are not allowed to eat caribou and walrus meat on the same day except after changing their clothing. The winter clothing which is made of caribou-skin must be entirely completed before the men will go to hunt walrus. As soon as the first walrus has been killed, a messenger goes from village to village and announces the news. All work on caribou-skins must cease immediately. When the caribou-hunting season begins, all the winter clothing, and the tent that has been in use during the walrus-hunting season, are buried, and not used again until the following walrus-hunting season. No walrus hide, or thongs made of such hide, must be taken inland, where is the abode of the caribou.

Similar laws, although not quite so stringent, hold good in regard to contact between seal and walrus. The natives always change their clothing or strip naked before eating seal during the walrus season.

The soul of the salmon is considered to be very powerful. Salmon must not be cooked in a pot that has been used for boiling other kinds of meat. It is always cooked at some distance from the hut. Boots that were used while hunting walrus must not be worn when fishing salmon, and no work on boot-legs is allowed until the first salmon has been caught and placed on a boot-leg.

The fact that these taboos are not restricted to caribou and walrus suggests that the mythical explanation given above does not account for the origin of these customs, but must be considered as a later effort to explain their existence.

The transgressions of taboos do not affect the souls of game alone. It has already been stated that the sea mammals see their effect upon man also, who appears to them of a dark color, or surrounded by a vapor which is invisible to ordinary man. This means, of course, that the transgression also affects the soul of the evil-doer. It becomes attached to it and makes him sick. The shaman is able to see, by the help of his guardian spirit, these attachments, and is able to free the soul from them. If this is not done the person must die. In many cases the transgressions become attached also to

persons who come in contact with the evil-doer. This is especially true of children, to whose souls the sins of their parents, and particularly of their mothers, become readily attached. Therefore when a child is sick the shaman, first of all, asks its mother if she has transgressed any taboos. The attachment seems to have a different appearance, according to the taboo that has been violated. A black attachment is due to removing oil-drippings from under the lamp. As soon as the mother acknowledges the transgression of a taboo, the attachment leaves the child's soul and the child recovers.

The souls of the deceased stay with the body for three days. If a taboo is violated during this time the transgression becomes attached to the soul of the deceased. The weight of the transgression causes the soul pain, and it roams about the village, endeavoring to free itself of its burden. It seeks to harm the people who, by their disobedience to custom, are causing its sufferings. It causes heavy snows to fall and brings sickness and death. Such a soul is called a tupilak. Toward the middle of autumn it hovers around the doors of the huts. When a shaman discovers the tupilak he advises the people, who assemble, and prepare to free it of its burden. All the shamans go in search of it, each a knife in hand. As soon as they find it, they stab it with their knives, and thus cut off the transgressions. Then the tupilak becomes a soul again. The knives with which it was stabbed are seen by the people to be covered with blood.

The Central Eskimo believe that man has two souls. One of these stays with the body, and may enter temporarily the body of a child which is given the name of the departed. The other soul goes to one of the lands of the souls. Of these there are several. There are three heavens, one above another, of which the highest is the brightest and best. Those who die by violence go to the lowest heaven. Those who die by disease go to Sedna's house first, where they stay for a year. Sedna restores their souls to full health and then she sends them up to the second heaven. Those who die by drowning go to the third heaven. People who commit suicide go to a place in which it is always dark and where they go about with their tongues lolling. Women who have had premature births go to Sedna's abode and stay in the lowest world.

The other soul stays with the body. When a child has been named after the deceased, the soul enters its body and remains there for about four months. It is believed that its presence strengthens the child's soul, which is

very light and apt to escape from the body. After leaving the body of the infant, the soul of the departed stays nearby, in order to re-enter its body in case of need. When a year has elapsed since the death of the person, his soul leaves the grave temporarily and goes hunting, but returns frequently to the grave. When the body has entirely decayed it may remain away for a long time.

Evidently the Eskimo also believe in the transmigration of souls. There is one tradition in which it is told how the soul of a woman passed through the bodies of a great many animals, until finally it was born again as an infant. In another story it is told how a hunter caught a fox in a trap and recognized in it the soul of his departed mother. In still another tale the soul of a woman, after her death, entered the body of a huge polar bear in order to avenge wrongs done to her during her lifetime.

Almost the sole object of the religious ceremonies of the Eskimo is to appease the wrath of Sedna, of the souls of animals, or of the souls of the dead, that have been offended by the transgressions of taboos. This is accomplished by the help of the guardian spirits of the angakut. The most important ceremony of the Eskimo is celebrated in the fall. At this time of the year the angakut, by the help of their guardian spirits, visit Sedna and induce her to visit the village, and they endeavor to free her of the transgressions that became attached to her during the preceding year. One angakok throws her with his harpoon, another one stabs her, and by this means they cut off all the transgressions. The ceremony is performed in a darkened snow-house. After the ceremony the lamps are lighted again and the people see the harpoon and the knife that were used in the ceremony covered with blood. If the angakut should fail to free Sedna from the transgressions, bad weather and hunger would prevail during the ensuing winter. On the following day Sedna sends her servant, who is called Kailleteta, to visit the tribe. She is represented by a man dressed in a woman's costume and wearing a mask made of seal-skin. On this day the people wear attached to their hoods pieces of skin of that animal of which their first clothing was made after they were born. It seems that the skins of certain animals are used for this purpose, each month having one animal of its own. It is said that if they should not wear the skin of the proper animal, Sedna would be offended and would punish them.

The angakut also cure sick persons and make good weather with the help of their guardian spirits. They discover transgressions of taboos and other causes of ill luck. One of the most curious methods of divination applied by the angakut is that of 'head-lifting.' A thong is placed around the head of a person who lies down next to the patient. The thong is attached to the end of a stick which is held in hand by the angakok. Then the latter asks questions as to the nature and outcome of the disease, which are supposed to be answered by the soul of a dead person, which makes it impossible for the head to be lifted if the answer is affirmative, while the head is raised easily if the answer is negative. As soon as the soul of the departed leaves, the head can be moved without difficulty.

Amulets are extensively used as a protection against evil influences and to secure good luck. Pregnant women wear the teeth of wolves on the backs of their shirts. These same teeth are fastened to the edge of the infant's hood. The string which passes under the large hood of the woman who carries her child on her back is fastened at one end to a bear's tooth, which serves to strengthen the child's soul. When the child begins to walk about, this string and the bear's tooth are attached to its shirt and worn as amulets. Pyrites, when thrown upon a spirit, are believed to drive it away.

As compared with the beliefs of the Greenlanders, the beliefs of the Central Eskimo are characterized by the great importance of the Sedna myth and the entire absence of the belief in a powerful spirit called Tonarssuk, which seems to have been one of the principal features of Greenland beliefs. There is an evident tendency among the Central Eskimo to affiliate all customs and beliefs with the myth of the origin of sea animals. This tendency seems to have been one of the principal causes that molded the customs and beliefs of the people into the form in which they appear at the present time.

MENTAL ENERGY.^E

BY EDWARD ATKINSON.

^E Presented before the New York meeting of the American Association for the Advancement of Science.

ACCORDING to the common conception, political economy is held to deal with material forces only; with land, labor and capital; with the production, distribution and consumption of the materials of human existence. These are food, clothing and shelter. It, therefore, bears the aspect of a purely material study of material forces. Yet no more purely metaphysical science exists, and there can be, in my view of the subject, no more ideal conceptions than those which are derived from the study of these purely material forces. Many of the errors commonly presented under the name of the 'claims of labor' have arisen from the limited and partial conception of the function of economic science.

We have become accustomed to deal with the so-called material forces of nature and with the physical work and labor of man under the general term of 'Energy'. What man does by his own labor or physical energy is to convert the products of land and sea, of mine and forest, into new forms from which he derives shelter, food and clothing. In a material sense all that any one can get in or out of life, be he rich or poor, is what we call our board and clothing. Such being the fact, what a man consumes is his cost to the community; what he spends yields to others the means of buying the supplies for their own wants; their consumption is then their cost to the community.

The physical forces of nature are limited. The earth is endowed with a fixed quantity of materials that we call gaseous, liquid and solid. It receives

a certain amount of heat from the sun which, for all practical purposes, may be considered a fixed quantity of energy, even if in eons it may be exhausted. The physical energy of man is devoted to the transformation of these physical forces under the law of conservation; he can neither add to nor diminish the quantity. He can transform solid into gas and gas into liquid. He can, according to common speech, consume some of these products, but his consumption is only another transformation. His own body is but one of the forms of physical energy on the way toward another form. These elements of nature, formerly limited to earth, air and water, are now listed under many titles of what are called elements; I believe over sixty that have not yet been differentiated, but all may yet be resolved into a unit of force.

You will observe that in our arithmetic we have ten numerals which can be divided into fractions. In our music we deal with seven notes and their variants. In our alphabet we have twenty-six letters. These factors correspond in some measure to what we call elements in nature. There is a limit to the number of combinations that can be made of the numerals and their fractions, to the notes of music and their variants, and of the letters of the alphabet; but in each case this limit is so remote as to be negligible, like the exhaustion of the heat of the sun. May we not deal with the elements of nature in the same way? Can any one prescribe a limit to their conversion and reconversion to the use of mankind? Is it not in these processes of conversion that we derive our subsistence?

We make nothing. All that we can do is to move something. We move the soil and we move the seed; nature gives the harvest. We direct the currents of falling water, of heat and of steam; nature imparts the force or energy to which man has only given a new direction. We are now imparting new directions to the force that we call electricity, and to what we call cold. What is the force from which we derive this power of transforming physical energy? May we not call it mental energy? Is not mental energy the factor in mankind by which he is differentiated from the beast? Does not man only accumulate experience, and is there any limit to the power of mind over matter?

If these points are well taken, mental energy is the fourth and paramount factor in providing for material existence, and the science of political economy, which deals with land, labor and capital, becomes a

purely metaphysical science when we admit the force of mental energy into the combination.

We deal, as I have said, with sixty elements, so-called, more or less, but the unity of nature is the most important fact ever proved by science; the correlation of all forms of physical energy leading logically from the idea of manifold forces or gods to the unity of creation, necessarily ending in the conception of unity of a creator, or the one God. This modern development of mental science is but the Hebrew concept of the creation in a new form. The Hebrew race was the first one of the historic races with whom the unity of creation and the unity of the creator became an article of faith. I doubt not that it was in that concept and the power derived from it that the Hebrew intellect asserted its preëminence in the history of the world. According to that concept, to man is given “dominion over the fish of the sea and over the fowl of the air and over every living thing that moveth upon the earth.” By what force does man hold dominion unless it is through his mental energy and his capacity to accumulate experience?

All the industrial arts are antedated by the industries of animals. The tailor finds his prototype in the tailor bird; the mason in the wasp; the farmer in the agricultural ant; the bridge-builder in the spider; the weaver in the weaver bird; the creator of water power in the beaver, and so on. Yet no other animal except man has developed or extended any of these arts. No other animal except man has learned to make and use fire and not to run away from it.

If, then, man by his power of mental energy converts the original and crude forces with which the earth is endowed into new forms, and by giving them new direction increases his power of production of the means of his own subsistence and enjoyment of life, *does it not follow that creation is a continuous procession in which man is a factor?* “There is a divinity that shapes our ends, rough hew them as we may.” The ideal of ‘an honest God the noblest concept of man’ becomes the converse of an honest man the noblest work of God—honest in a broad sense in his dealings with the forces of nature; true to his function.

There is a painful side to statistical and economic study. The penalty of being able to read what is written between the lines and the columns of the figures is the conclusion that after we have all done the best work that the

present conditions of science will permit, the entire product barely suffices to keep mankind in existence; his fixed capital, so-called, is at the mercy both of time and of the inventor who substitutes better methods which at less cost of effort or labor yield more abundance to the community as a whole. But on the other hand, no matter how hard the struggle for existence may be, we find the promise of future abundance even in the insufficient product which has been derived from the application of science and invention up to date. Witness the relative progress of the last century as compared with all the previous centuries; then attempt to conceive what will be the condition of humanity a century hence, knowing, as we do, that the applications of science through mental energy now proceed in geometric progression, reversing the dogma of Malthus and leading to the concept of production unlimited, consumption limited.

If it be true that there is no conceivable limit to the power of mind over matter or to the number of conversions of force that can be developed, providing in increasing measure for the wants of the human body, it follows that pauperism is due to poverty of mental energy, not of material resources.

The next step in the development of this theory may be presented in this form: *No man is paid by the measure in time or physical effort, for the work or labor that he performs.* No man can claim payment in money or in kind on the ground that he has done a day's work of a greater or less number of hours. In all civilized countries we are members one of another; rich or poor; whether we work with our hands or our heads, or both combined. Material existence is supported by conversion of one form of physical energy into another. Social energy is maintained by the exchange of one form of service for another. The measure of compensation is not the number of hours of labor put into the product or service. *The standard by which services are measured is what the buyer is saved from doing, not what the seller does.* Each of us might possibly be able to house, clothe and feed ourselves if we were cast upon an island possessing sufficient natural resources. If a hundred persons representing all the classes in society were wrecked upon such an island, each adult or each person above ten years old would probably find a way to house, feed and clothe himself. Why do we not house, feed and clothe ourselves, and why would not the hundred representatives of different classes wrecked on an island each do his own part of the work for himself only? Simply for the reason that men are either

endowed from birth with different aptitudes, or different aptitudes are developed in their environment. Each one finds out that by delegating to another certain kinds of work he saves his own time and energy. Each one finds out what he can do for the next man, while the next man finds out what he can do for him.

There is in every transaction of life an unconscious cerebration or estimate of the services rendered to us, saving each of us mental or manual energy, whenever we buy any product or service from another. That unconscious cerebration affects the minds or habits or acts of both parties in every purchase and sale. There may be errors in regard to the service itself. The ignorant man will buy quack medicines that he had better let alone, but what he pays under the false impression of benefit to himself is his measure of what he hopes to save; while the quack medicine vender, taking advantage of the ignorance of others, filches from them the means of subsistence, even of wealth, under the pretext of service. As time goes on, however, false measures of service are eliminated with increasing intelligence, and true benefits constitute more and more the vast proportion of the exchanges.

The same ignorance which leads the masses of the people of every country to submit to military dictation, even in a bad cause, also leads to the wars of tariffs among nations by which prejudice and animosity are kept up. The false conception that in international commerce what one nation gains another must lose, is promoted by the advocates of protection, many of whom very honestly believe that through the exclusion of foreign goods domestic industry may be promoted, wholly ignoring the fact that arts and industries are developed by intelligence and not by legislation.

The advocates of bounties and of special legislation also ignore the fact that in this country, where mental energy is more nearly free in its action than in any other, manufactures and the mechanic arts develop in due proportion according to the age and the natural resources of the territory or state, nine-tenths or more of the occupations which are listed under these titles being free in the nature of things from any possibility of foreign competition through the import of a product of like kind.

There may be nothing new in this essay, but until my own observation had led me to the conclusion that land, labor and capital were alike inert and

incapable without the coordinating power of mental energy, the doubt continued to exist in my mind which is often expressed about the possibility of economic science having any real existence or right to the title. Also, until my own observation led me to the conclusion that the cost of a man to the community is what he consumes, and not what he secures in the way of income, the correlation of wealth and welfare had not been satisfactorily reconciled. I think that a very large part of what is written under the title of political economy would be greatly modified, and perhaps never have been written, had these concepts been derived by the writers from experience, as they have been in my own observation.

I have not much patience with abstract or *à priori* theories, my own method being one of observation, then referring to the various authorities in order to find out whether my observations or their abstract theories have been shallow and superficial.

Again, I find in the ideal of the continuous miracle of creation in which man is a factor the solution of many intellectual difficulties. In the face of such a perception of the methods of the universe, the larger part of the dogmas that have been put forth under the name of religion take their place with much of the historic rubbish which passes under the name of history. When it becomes plain that every man has his place in the progress of continuous creation, and is a factor in it; that nothing is constant but change; that there is no such thing as fixed capital; all the doubts and fears regarding the future of humanity vanish in the light of sure progress.

What greater stimulus can there be than for every man each in his own way rendering service for service, his objective point being only the welfare of himself and his family, when he attains the conviction that by so much as his mental energy adds to the sum of the utilities by which mankind lives, so may that part which he consumes and which represents his cost to the community be fully justified, even though it is earned with more apparent ease and less physical exertion than are called for from his poorer neighbors.

Incomplete as his studies were, I have always found in the 'Harmonies' of Frederic Bastiat the greatest encouragement and the greatest incentive to the work which I have undertaken under the name of political economy, leading more and more to the conviction that war and warfare, whatever

influence they may have had in developing progress in the past, are now due to ignorance and greed; the war of tariffs due to selfishness and stupidity; and the contest of labor and capital due to the errors of the ignorant workman and the ignorant capitalist alike. All interests are harmonious. The evolution of science and invention will surely bring them together on the lines of righteousness, peace and material abundance.

This essay has been condensed from a lecture prepared and given before a Clergymen's Club some months ago. In it I tried to show the necessary connection of religion and life as developed by economic study, the law of mutual service being the rule by which commerce lives and moves and has its being. This lecture has since been read to several clubs of very different types of men, and from the great interest excited I am led to think there is something in it fit for the student of facts and figures to say.

I may, therefore, venture to repeat the statement of two principles which are presented in this treatise, which I think have been seldom if ever fully developed in any of the standard works upon political economy. To my own mind these are basic principles which when applied may profoundly modify many of the concepts of students of economic science. I join in the view that the family is the unit of society, the home the center. The end of all production is consumption. Nothing is constant but change, and there is no such thing as fixed material capital of any long duration in the progress of time. The two principles which I have endeavored to enforce are as follows:

First. The cost of each person or head of the family is what he and his immediate dependents consume. His income, whether measured in terms of money or in products, is, therefore, no measure of his cost; what he distributes in payment for service rendered being expended by those who receive it in procuring the commodities which constitute their cost to the community.

Second. No person who is occupied or is in the employment or service of others is paid for what he does. His work may occupy long hours and may be applied to arduous manual labor, or it may be done in a short number of hours per day, with but little physical effort. Neither the hours nor the effort constitute any measure on which payment can be based. The

measure of payment is fixed by the measure of the work saved to him who makes the payment, consciously or unconsciously estimated.

These two precepts or principles, coupled with the theory that there is no conceivable limit to the power of mind over matter, or to the number of transformations of physical energy to which direction may be given in the material support of humanity, bring the visions of the Utopians within the scope of a law of progress in material welfare to which no limit can be put in time or space.



CHAPTERS ON THE STARS.

BY PROF. SIMON NEWCOMB, U. S. N.

VARIABLE STARS.

IT is a curious fact that the ancient astronomers, notwithstanding the care with which they observed the heavens, never noticed that any of the stars changed in brightness. The earliest record of such an observation dates from 1596, when the periodical disappearance of Omicron Ceti was noticed. After this, nearly two centuries elapsed before another case of variability in a star was recorded. During the first half of the nineteenth century Argelander so systematized the study of variable stars as to make it a new branch of astronomy. In recent years it has become of capital interest and importance through the application of the spectroscope.

Students who are interested in the subject will find the most complete information attainable in the catalogues of variable stars, published from time to time by Chandler in the 'Astronomical Journal.' His third catalogue, which appeared in 1896, comprises more than 300 stars whose variability has been well established, while there is always a long list of 'suspected variables'—whose cases are still to be tried. The number to be included in the established list is continually increasing at such a rate that it is impossible to state it with any approximation to exactness. The possibility of such a statement has been yet further curtailed by the recent discovery at the Harvard Observatory that certain clusters of stars contain an extraordinary proportion of variables. Altogether at the time of the latest publication, 509 such stars were found in twenty-three clusters. The total number of these objects in clusters, therefore, exceeds the number known in the rest of the sky. They will be described more fully in a subsequent chapter. For the present we are obliged to leave this rich field out of consideration and confine our study to the isolated variable stars which are found in every region of the heavens.

Variable stars are of several classes, which, however, run into each other by gradations so slight that a sharp separation cannot always be made between them. Yet there are distinguishing features, each of which marks so considerable a number of these stars as to show some radical difference in the causes on which the variations depend.

We have first to distinguish the two great classes of irregular and periodic stars. The irregular ones increase and diminish in so fitful a way that no law of their change can be laid down. To this class belong the so-called ‘new stars’, which, at various periods in history, have blazed out in the heavens, and then in a few weeks or months have again faded away. It is a remarkable fact that no star of the latter class has ever been known to blaze out more than once. This fact distinguishes new stars from other irregularly variable ones.

Periodic stars are those which go through a regular cycle of changes in a definite interval of time, so that, after a certain number of days, sometimes of hours, the star returns to the same brightness. But even in the case of periodic stars, it is found that the period is more or less variable, and, in special cases, the amount of the variation is such that it cannot always be said whether we should call a star periodic or irregular.

The periodic stars show wide differences, both in the length of the period and in the character of the changes they undergo. In most cases they rapidly increase in brightness during a few days or weeks, and then slowly fade away, to go through the same changes again at the end of the period. In other cases they blaze up or fade out, from time to time, like the revolving light of a lighthouse. Some stars are distinguished more especially by their maximum, or period of greatest brightness, while others are more sharply marked by minima, or periods of least brightness. In some cases there are two unequal minima in the course of a period.

Chandler’s third catalogue of variable stars gives the periods of 280 of these objects, which seem to have been fairly well made out. A classification of these periods, as to their length, will be interesting. There are, of periods:

Less than 50 days	63 Stars.
Between 50 and 100 days	6 ”

”	100	”	150	”	9	”
”	150	”	200	”	18	”
”	200	”	250	”	29	”
”	250	”	300	”	40	”
”	300	”	350	”	44	”
”	350	”	400	”	44	”
”	400	”	450	”	18	”
”	450	”	500	”	6	”
”	500	”	550	”	1	”
”	550	”	600	”	1	”
”	600	”	650	”	1	”

It will be seen from this that, leaving out the cases of very short period, the greater number of the periods fall between 300 and 400 days. From this value the number falls off in both directions. Only three periods exceed 500 days, and of these the longest is 610 days. We infer from this that there is something in the constitution of these stars, or in the causes on which their variation depends, which limits the period. This limitation establishes a well-marked distinction between the periodic stars and the irregular variables to be hereafter described.

Returning to the upper end of the scale, the contrast between the great number of stars less than fifty days, and the small number between fifty and one hundred, seems to show that we have here a sharp line of distinction between stars of long and those of short period. But, when we examine the matter in detail, the statistics of the periods do not enable us to draw any such line. About eight periods are less than one day, and the number of this class known to us is continually increasing. About forty are between one and ten days, and from this point upwards they are scattered with a fair approach to equality up to a period of one hundred days. There is, however, a possible distinction, which we shall develop presently.

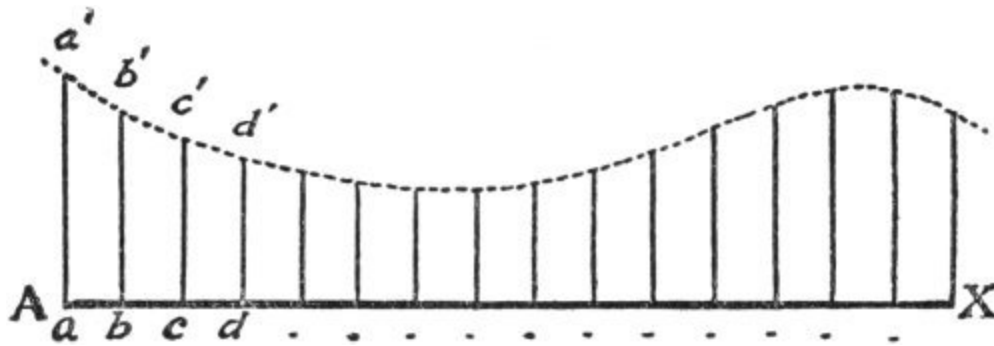


FIG. 1. THE LAW OF CHANGE IN A VARIABLE STAR.

The law of change in a variable star is represented to the eye by a curve in the following way. We draw a straight horizontal line $A X$ to represent the time. A series of equidistant points, a, b, c, d , etc., on this will represent moments of time. One of the spaces, a, b, c , etc., may represent an hour, a day, or a month, according to the rapidity of change. We take a to represent the initial moment, and erect an ordinate aa' , of such length as to represent the brightness of the star on some convenient scale at this moment. At the second moment, b , which may be an hour or a day later, we erect another ordinate bb' , representing the brightness at this moment. We continue this process as long as may be required. Then we draw a curve, represented by the dotted line, through the ends of all the ordinates. In the case of a periodic star it is only necessary to draw the curve through a single period, since its continuation will be a repetition of its form for any one period.

We readily see that if a star does not vary, all the ordinates will be of equal length, and the curve will be a horizontal straight line. Moreover, the curve will take this form through any portion of time during which the light of the star is constant.

There are three of the periodic stars plainly visible to the naked eye at maximum, of which the variations are so wide that they may easily be noticed by any one who looks for them at the right times, and knows how to find the stars. These stars are:

Omicron Ceti, called also *Mira Ceti*.

Beta Persei, or Algol.

Beta Lyræ.

It happens that each of these stars exemplifies a certain type or law of variations.

Omicron Ceti. On August 13, 1596, David Fabricius noticed a star in the constellation Cetus, which was not found in any catalogue. Bayer, in his 'Uranometria', of which the first edition was published in 1601, marked the star Omicron, but said nothing about the fact that it was visible only at certain times. Fabricius observed the star from time to time, until 1609, but he does not appear to have fully and accurately recognized its periodicity. But so extraordinary an object could not fail to command the attention of astronomers, and the fact was soon established that the star appeared at intervals of about eleven months, gradually fading out of sight after a few weeks of visibility. Observations of more or less accuracy having been made for more than two centuries, the following facts respecting it have been brought to light:

Its variations are somewhat irregular. Sometimes, when at its brightest, it rises nearly or quite to the second magnitude. This was the case in October, 1898, when it was about as bright as Alpha Ceti. At other times its maximum brightness scarcely exceeds the fifth magnitude. No law has yet been discovered by which it can be predicted whether it shall attain one degree of brightness or another at maximum.

Its minima are also variable. Sometimes it sinks only to the eighth magnitude; at other times to the ninth or lower. In either case it is invisible to the naked eye.

As with other stars of this kind, it brightens up more rapidly than it fades away. It takes a few weeks from the time it becomes visible to reach its greatest brightness, whatever that may be. It generally retains this brightness for two or three weeks, then fades away, gradually at first, afterward more rapidly. The whole time of visibility will, therefore, be two or three months. Of course, it can be seen with a telescope at any time.

The period also is variable in a somewhat irregular way. If we calculate when the star ought to be at its greatest brightness on the supposition that the intervals between the maxima ought to be equal, we shall find that sometimes the maximum will be thirty or forty days early, and at other times thirty or forty days late. These early or late maxima follow each other year after year, with a certain amount of regularity as regards the

progression, though no definable law can be laid down to govern them. Thus, during the period from 1782 to 1800 it was from thirteen to twenty-four days late. In 1812 it was thirty-nine days late. From 1845 to 1856 it was on the average about a month too early. Several recent maxima, notably those from 1895 to 1898, again occurred late. Formulæ have been constructed to show these changes, but there is no certainty that they express the actual law of the case. Indeed, the probability seems to be that there is no invariable law that we can discover to govern it.

Argelander fixed the length of the period at 331.9 days. More recently, Chandler fixed it at 331.6 days. It would seem, therefore, to have been somewhat shorter in recent times. It was at its maximum toward the end of October, 1898. We may, therefore, expect that future maxima will occur in July, 1901; June, 1902; May, 1903; April, 1904, and so on, about a month earlier each year. During the few years following 1903 the maxima will probably not be visible, owing to the star being near conjunction with the sun at the times of their occurrence. The most plausible view seems to be that changes of a periodic character, involving the eruption of heated matter from the interior, of the body to its surface, followed by the cooling of this matter by radiation, are going on in the star.

The star *Algol*, or *Beta Persei*, as it is commonly called in astronomical language, may, in northern latitudes, be seen on almost any night of the year. In the early summer we should probably see it only after midnight, in the northeast. In late winter it would be seen in the northwest. From August until January one can find it at some time in the evening by becoming acquainted with the constellations. It is nearly of the second magnitude. One might look at it a score of times without seeing that it varied in brilliancy. But at certain stated intervals, somewhat less than three days, it fades away to nearly the fourth magnitude for a few hours, and then slowly recovers its light. This fact was first discovered by Goodrick in 1783, since which time the variations have been carefully followed. The law of variation thus defined is expressed by a curve of the following form:



**FIG. 2. LAW OF VARIATION OF A STAR OF THE
ALGOL TYPE.**

The idea that what we see in the star is a partial eclipse caused by a dark body revolving round it, was naturally suggested even to the earliest observers. But it was impossible to test this theory until recent times. Careful observation showed changes in the period between the eclipses, which, although not conclusive against the theory, might have seemed to make it somewhat unlikely. The application of the spectroscope to the determination of radial motions, enabled Vogel, of Potsdam, in 1889, to set the question at rest. His method of reasoning and proceeding was this:

If the fading out which we see is really due to an eclipse by a dark body, that body must be nearly or quite as large as the star itself, else it could not cut off so much of its light. In this case, it is probably nearly as massive as the star itself, and therefore would affect the motion of the star. Both bodies would, in fact, revolve around their common center of gravity. Therefore, when after the dark body has passed in front of the star, it has made one-fourth of a revolution, which would require about seventeen hours, the star would be moving towards us. Again, seventeen hours before the eclipse, it ought to be moving away from us.

The measurement of six photographs of the spectrum, of which four were taken before the eclipses and two afterward, gives the following results:

Before eclipses: Velocity *from* the sun equals 39 km. per second.

After eclipses: Velocity *toward* the sun equals 47 km. per second.

These results show that the hypothesis in question is a true one, and afforded the first conclusive evidence of a dark body revolving around a distant star. A study of the law of diminution and recovery of the light

during the eclipse, combined with the preceding motions, enabled Vogel to make an approximate estimate of the size of the orbit and of the two bodies. The star itself is somewhat more than a million of miles in diameter; the dark companion a little less. The latter is about the size of our sun. Their distance apart is somewhat more than three millions of miles; the respective masses are about one-half and one-fourth that of the sun. These results, though numerically rather uncertain, are probably near enough to the truth to show us what an interesting system we here have to deal with. We can say with entire certainty that the size and mass of the dark body exceed those of any planet of our system, even Jupiter, several hundred fold.

The period of the star is also subject to variations of a somewhat singular character. These have been attributed by Chandler to a motion of the whole system around a third body, itself invisible. This theory is, however, still to be proved. Quite likely the planet which causes the eclipse is not the only one which revolves around this star. The latter may be the center of a system like our solar system, and the other planets may, by their action, cause changes in the motion of the body that produces the eclipses. The most singular feature of the change is that it seems to have taken place quite rapidly, about 1840. The motion was nearly uniform up to near this date; then it changed, and again remained nearly uniform until 1890. Since then no available observations have been published.

It is found that several other stars vary in the same way as Algol; that is to say, they are invariable in brightness during the greater part of the time, but fade away for a few days at regular intervals. This is a kind of variation which it is most difficult to discover, because it will be overlooked unless the observer happens to notice the star during the time when an eclipse is in progress, and is thoroughly aware of its previous brightness. One might observe a star of this kind very accurately a score of times, without hitting upon a moment when the partial eclipse was in progress. On the principle that like effects are due to like causes, we are justified in concluding that in the cases of all stars of this type, the eclipses are caused by the revolution of a dark body, now called 'Algol variables,' round the principal star.

A feature of all the Algol variables is the shortness of the periods. The longest period is less than five days, while three are less than one day. This is a result that we might expect from the nature of the case. The nearer a dark planet is to the star, the more likely it will be to hide its light from an

observer at a great distance. If, for example, the planet Jupiter were nearly as large as the sun, the chances would be hundreds to one against the plane of the orbit being so nearly in the line of a distant observer that the latter would ever see an eclipse of the sun by the planet. But if the planet were close to the sun, the chances might increase to one in ten, and yet farther to almost any extent, according to the nearness of the two bodies.

Still, we cannot set any definite limit to the period of stars of this type; all we can say is that, as the period we seek for increases, the number of stars varying in that period must diminish. This follows not only from the reason just given, but from the fact that the longer the interval that separates the partial eclipses of a star of the Algol type, the less likely they are to be detected.

STARS OF THE BETA LYRÆ TYPE.

The star Beta Lyræ shows variations quite different in their nature from those of Algol, yet having a certain analogy to them. Anyone who looks at the constellation Lyræ a few nights in succession and compares Beta with Gamma, a star of nearly the same brightness in its neighborhood, will see that while on some evenings the stars are of equal brightness, on others Beta will be fainter by perhaps an entire magnitude.

A careful examination of these variations shows us a very remarkable feature. On a preliminary study, the period will seem to be six and one-half days. But, comparing the alternate minima, we shall find them unequal. Hence the actual period is thirteen days. In this period there are two unequal minima, separated by equal maxima. That is to say, the partial eclipses at intervals of six and one-half days are not equal. At the alternate minima the star is half as bright again as at the intermediate minima.

It is impossible to explain such a change as this merely by the interposition of a dark body, and this for two reasons. Instead of remaining invariable between the minima, the variation is continuous during the whole period, like the rising and falling of a tide. Moreover, the inequality of the alternating minima is against the theory.

Pickering, however, found from the doubling of the spectral lines that there were two stars revolving round each other. Then Prof. G. W. Myers, of Indiana, worked out a very elaborate mathematical theory to explain the variations, which is not less remarkable for its ingenuity than for the curious nature of the system it brings to light. His conclusions are these:

Beta Lyræ consists of two bodies, gaseous in their nature, which revolve round each other, so near as to be almost touching. They are of unequal size. Both are self-luminous. By their mutual attraction they are drawn out into ellipsoids. The smaller body is somewhat darker than the other. When we see the two bodies laterally, they are at their brightest. As they revolve, however, we see them more and more end on, and thus the light diminishes. At a certain point one begins to cover the other and hide its light. Thus the combined light continues to diminish until the two bodies move across our line of sight. Then we have a minimum. At one minimum, however, the smaller and darker of the two bodies is projected upon the brighter one, and thus diminishes its light. At the other minimum, it is hiding behind the other, and therefore we see the light of the larger one alone.

This theory receives additional confirmation from the fact, shown by the spectroscope, that these stars are either wholly gaseous, or at least have self-luminous atmospheres. Some of Professor Myers's conclusions respecting the magnitudes are summarized as follows:

The larger body is about 0.4 as bright as the smaller.

The flattening of the ellipsoidal masses is about 0.17.

The distance of centers is about $1\frac{7}{8}$ the semi-major axis of the larger star, or about 50,000,000 kilometers (say 30,000,000 miles).

The mass of the larger body is about twice that of the smaller, and $9\frac{1}{2}$ times the mass of the sun.

The mean density of the system is a little less than that of air.^E

^E 'Astrophysical Journal', Vol. VII, January, 1898.

It should be remarked that these numbers rest on spectroscopic results, which need further confirmation. They are, therefore, liable to be changed by subsequent investigation. What is most remarkable is that we have here to deal with a case to which we have no analogy in our solar system, and which we should never have suspected, had it not been for observations of this star.

The gap between the variable stars of the Algol type and those of the Beta Lyrae type is, at the present time, being filled by new discoveries in such a way as to make a sharp distinction of the two classes difficult. It is characteristic of the Algol type proper that the partial eclipses are due to the interposition of a dark planet revolving round the bright star. But suppose that we have two nearly equal stars, *A* and *B*, revolving round their common center of gravity in a plane passing near our system. Then, *A* will eclipse *B*, and, half a revolution later, *B* will eclipse *A*, and so on in alternation. But, when the stars are equal, we may have no way of deciding which is being eclipsed, and thus we shall have a star of the Algol type, so far as the law of variation is concerned, yet, as a matter of fact, belonging rather to the Beta Lyrae type. If the velocity in the line of sight could be measured, the question would be settled at once. But only the brightest stars can, so far, be thus measured, so that the spectroscope cannot help us in the majority of cases.

The most interesting case of this kind yet brought to light is that of Tau Cygni. The variability of this star, ordinarily of the fourth magnitude, was discovered by Chandler in December, 1886. The minima occurred at intervals of three days. But in the following summer he found an apparent period of 1 d. 12 h., the alternate minima being invisible because they occurred during daylight, or when the star was below the horizon. With this period the times of minima during the summer of 1888 were predicted.

It was then found that the times of the alternate minima, which, as we have just said, were the only ones visible during any one season, did not correspond to the prediction. The period seemed to have greatly changed. Afterward, it seemed to return to its old value. After puzzling changes of this sort, the tangle was at length unraveled by Dunér, of Lund, who showed that the alternate periods were unequal. The intervals between minima were one day nine hours, one day fifteen hours, one day nine hours, one day fifteen hours, and so on, indefinitely. This law once established, the cause of

the anomaly became evident. Two bright stars revolve round their common center of gravity in a period of nearly three days. Each eclipses the other in alternation. The orbit is eccentric, and, in consequence, one-half of it is described in a less time than the other half. If we could distinguish the two stars by telescopic vision, and note their relative positions at the four cardinal points of their orbit, we should see the pair alternately single and double, as shown in the following diagrams:

	A	B
Position (1), stars at pericenter	*	*
Interval, 16 hours.		
Position (2), A eclipses B		*
Interval, 20 hours.		
	B	A
Position (3), stars at apocenter	*	*
Interval, 20 hours.		
Position (4), B eclipses		*
Interval, 16 hours.		
Position (1) is repeated	*	*

U Pegasi is a star which proved as perplexing as *Tau Cygni*. It was first supposed to be of the Algol type, with a period of about two days. Then it was found that a number of minima occurred during this period, and that the actual interval between them was only a few hours. The great difficulty in the case arises from the minuteness of the variation, which is but little more than half a magnitude between the extremes. The observations of Wendell, at the Harvard Observatory, with the polarizing photometer, enabled Pickering to reach a conclusion which, though it may still be open to some doubt, seems to be the most likely yet attainable. The star is of the Beta Lyræ type; its complete period is 8 hours 59 minutes 41 seconds, or 19 seconds less than nine hours; during this period it passes through two equal maxima, each of magnitude 9.3, and two unequal minima 9.76 and 9.9, alternately.

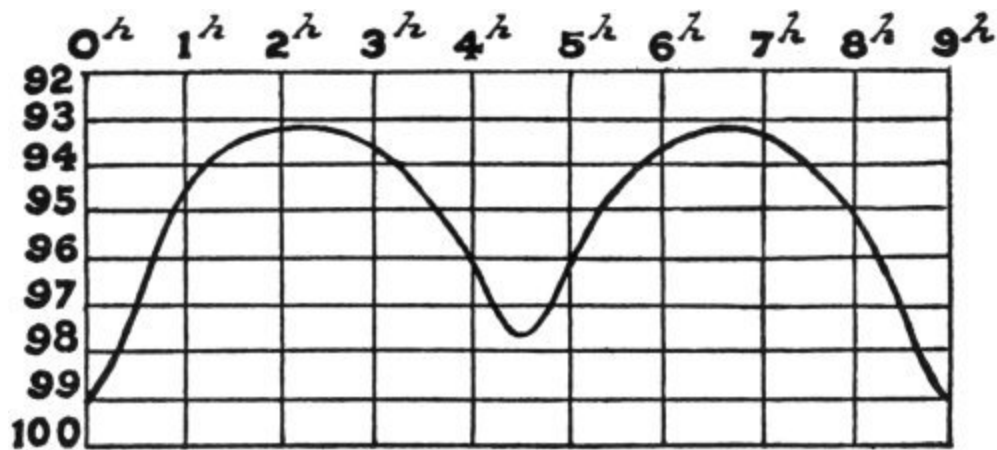


FIG. 3. LIGHT CURVE OF U PEGASI, OF THE BETA LYRÆ TYPE, FROM OBSERVATIONS BY WENDELL AT THE HARVARD OBSERVATORY. MAGNITUDE AT MAXIMUM, 9.32; AT PRINCIPAL MINIMUM, 9.90; AT SECONDARY MINIMUM, 9.76. PERIOD, 9 HOURS.

The difference of these minima, 0m. 14, is less than the errors which really ordinarily affect measures of a star's magnitude with the best photometers. Some skepticism has, therefore, been felt as to the reality of the difference which, if it does not exist, would reduce the periodic time below four and one-half hours, the shortest yet known. But Pickering maintains that, in observations of this kind upon a single star, the precision is such that the reality of the difference, small though it be, is beyond reasonable doubt.

Taking Pickering's law of change as a basis, Myers has represented the light-curve of *U* Pegasi on a theory similar to that which he constructed for Beta Lyræ. His conclusion is that, in the present case, the two bodies which form the visible star are in actual contact. A remarkable historic feature of the case is that Poincaré has recently investigated, by purely mathematical methods, the possible forms of revolving fluid masses in a condition of equilibrium, bringing out a number of such forms previously unknown. One of these, which he calls the apiodal form, consists of two bodies joined into one, and it is this which Myers finds for *U* Pegasi.

Quite similar to these two cases is that of Zeta Herculis. This star, ordinarily of the seventh magnitude, was found, at Potsdam, in 1894, to diminish by about one magnitude. Repeated observations elsewhere indicate a period of very nearly four days. Actually it is now found to be only ten minutes less than four days. The result was that during any one season of observation the minima occur at nearly the same hour every night or day. To an observer situated in such longitude that they occur during the day, they would, of course, be invisible.

Continued observations then showed a secondary minimum, occurring about half-way between the principal minima hitherto observed. It was then found that these secondary minima really occur between one and two hours earlier than the mid-moment, so that the one interval would be between forty-six and forty-seven hours and the other between forty-nine and fifty. The time which it takes the star to lose its light and regain it again is about ten hours. More recent observations, however, do not show this inequality, so that there is probably a rapid motion of the pericenter of the orbit.

It will be seen that this star combines the Algol and Beta Lyræ types. It is an Algol star in that its light remains constant between the eclipses. It is of the Beta Lyræ type in the alternate minima being unequal.

From a careful study, Seliger and Hartwig derived the following particulars respecting this system:

Diameter of principal star, 15,000,000 kilometers.

Diameter of smaller star, 12,000,000 kilometers.

Mass of the larger star, 172 times sun's mass.

Mass of the smaller star, 94 times sun's mass.

Distance of centers, 45,000,000 kilometers.

Time of revolution, 3d. 23h. 49m. 32.7s.

It must be added that the data for these extraordinary numbers are rather slender and partly hypothetical.

Beta Lyræ is always of the same brightness at the same hour of its period, and Algol has always the same magnitude at minimum. It is true that the length of the period varies slowly in the case of these stars. But this may arise from the action of other invisible bodies revolving around the visible stars. This general uniformity is in accord with the theory which attributes

the apparent variations to the various aspects in which we see one and the same system of revolving stars.

Another variable star showing some unique features is Eta Aquilæ. What gives it special interest is that spectroscopic observations of its radial motion show it to have a dark body revolving round it in a very eccentric orbit, and in the same time as the period of variation. It might therefore be supposed that we have here a star of the Algol or Beta Lyræ type. But such is not the case. There is nothing in the law of variation to suggest an eclipsing of the bright star, nor does it seem that the variations can readily be represented by the varying aspects of any revolving system.

The orbit of this star has been exhaustively investigated by Wright from Campbell's observations of the radial motion. The laws of change in the system are shown by the curves below, which are reproduced, in great part, from Wright's paper in the 'Astrophysical Journal.'

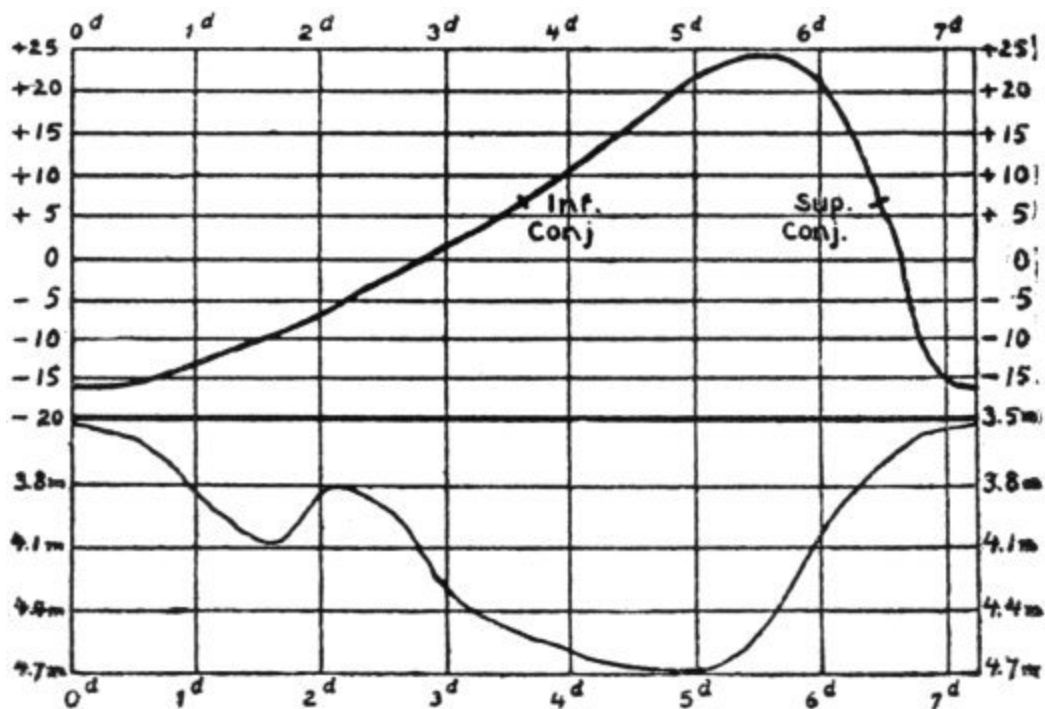


FIG. 4. LIGHT-CURVE AND RADIAL VELOCITY OF ETA AQUILÆ.

The lower curve is the light-curve of the star during a period of 7.167 days. Starting from a maximum of 3.5 mag., it sinks, in the course of 5 days, to a minimum of 4.7m. It was found by Schwab that the diminution is not progressive, but that a secondary maximum of 3.8m. is reached at the end of the second day. After reaching the principal minimum it rises rapidly to the principal maximum in $2\frac{1}{4}$ days.

The upper curve shows the radial velocity of the star during the period of variation. It will be seen that the epoch of greatest negative velocity, which referred to the center of mass of the system, is 16.2 km. per second, occurs at the time of maximum brightness. The greatest positive velocity, 23.9 km., occurs during the sixth day of the period just after the time of minimum brightness.

Finally, the moments of inferior and superior conjunction of the dark body with the bright one are neither of them an epoch of minimum brightness, which takes place half-way between the two.

The most plausible conclusion we can draw is that the light of the star is affected by the action of the dark body during its revolution. But how the change may be produced we cannot yet say.

CLASSIFICATION OF VARIABLE STARS.

A classification of variable stars, based on the period of variation and the law of change, was proposed by Pickering. It does not, however, seem that a hard and fast line can yet be drawn between different types and classes of these bodies, one type running into another, as we have found in the case of the Algol and Beta Lyræ types. Yet the discovery of the cause of the variation in these types makes it likely that a division into two great classes, dependent on the cause of variation, is possible. We should then have:

(1) Stars, or systems, constituting to vision a single star, of which the apparent variability arises from the rotation of the system as a whole, or from the revolution of its components around each other.

(2) Stars of which the changes arise from other and as yet unknown causes.

The main feature of the stars of the first class is that we are under no necessity of supposing any actual change in the amount of light which they emit. Their apparent variations are purely the effect of perspective, arising from the various aspects which they present to us during their revolution round each other. If we could change our point of view so that the plane of the orbit of Algol's planet no longer passed near our system, Algol would no longer be a variable star. Under the same circumstances the apparent variations in a star of the Beta Lyræ type would cease to be noticeable, if they did not disappear entirely.

The stars of this class are also distinguished by the uniformity and regularity with which they go through their cycle of change.

The stars of the other class, which we may call the Omicron Ceti type, are different not only in respect to the length of the period, but in the character of the variation. There are certain general laws of variation and irregularities of brightness which stars of this class go through. Starting from the time of the minimum, the increase of light is at first very slow. It grows more and more rapid as the maximum is approached, in which time there may be as great an increase in two or three days as there formerly was in a month. The diminution of light is generally slower than the increase. The magnitude at corresponding times in different periods may be very different. Thus, as we have already remarked, Omicron Ceti is ten times as bright at some maxima as it is at others. The periods also, so far as they have been made out, vary more widely than those of stars of the other type.

The idea has sometimes been entertained that these variations of light are due to a revolution of the star on its axis. A very little consideration will, however, show that this explanation cannot be valid. However bright a star might be on one side, or however dark on the other, any one region of its surface would be visible to us half the time and a change of brightness from different degrees of brilliancy on different sides would be gradual and regular.

It is not impossible that the variability may be in some way connected with the action of a body revolving round the star. This seems to be the case with Eta Aquilæ. The radial motion of this object shows the existence of a

dark body revolving round it in the same period as that of the star's variation.

From what has been said, it will be seen that, although a sharp line cannot be drawn, there seems to be some distinction between the stars of short and long periods. The number of stars which have been known to belong to the first class is quite small, only about fifteen, all told. On the other hand, there are still left some stars having a period less than ten days, which are otherwise not distinguishable from the Omicron Ceti type. It seems quite likely that the variations in the periods of these stars are, in some way, connected with the revolution of bright or dark bodies round them.

They also vary more widely than those of stars of the other two types. This might easily happen in the case of stars really variable through a cycle of changes going on in consequence of the action of interior causes.

The periodic stars of short period, which have not been recognized as of the Algol or Beta Lyræ type, form an interesting subject of study. Although the separation between them and the stars of long period is not sharp, it seems likely to have some element of reality in it. But no conclusions on the subject can be reached until the light-curves of a large number of them are carefully drawn; and this requires an amount of patient and accurate observation which cannot be carried out for years to come.

SUSPECTED VARIATIONS IN THE COLOR OF STARS.

The question whether certain stars vary in color without materially changing their brightness has sometimes been raised. This was at one time supposed to be the case with one of the stars of Ursa Major. This suspected variation has not, however, been confirmed, and it does not seem likely that any such changes take place in the color of stars not otherwise variable.

POSSIBLE SECULAR VARIATIONS IN THE BRILLIANCY OF STARS.

All the variations we have hitherto considered take place with such rapidity that they can be observed by comparisons embracing but a short interval of time—a few days or months at the outside. A somewhat different question of great importance is still left open. May not individual stars be subject to a secular variation of brilliancy, meaning by this term a change which would not be sensible in the course of only one generation of men, but admitting of being brought out by a comparison of the brightness of the stars at widely distant epochs? Is it certain that, in the case of stars which we do not recognize as variable, no change has taken place since the time of Hipparchus and Ptolemy? This question has been investigated by C. S. Pierce and others. The conclusion reached is that no real evidence of any change can be gathered. The discrepancies are no greater than might arise from errors of estimates.

There is, however, an analogous question which is of great interest and has been much discussed in recent times. In several ancient writings the color of Sirius is described as red. This fact would, at first sight, appear to afford very strong evidence that, within historic times, the color of the brightest star in the heavens has actually changed from red to a bluish white.

Two recent writers have examined the evidence on this subject most exhaustively and reached opposite conclusions. The first of these was Dr. T. J. J. See, who collated a great number of cases in which Sirius was mentioned by ancient writers as red or fiery, and thus concluded that the evidence was in favor of a red color in former times. Shortly afterwards, Schiaparelli examined the evidence with equal care and thoroughness and reached an opposite conclusion, showing that the terms used by the ancient authors, which might have indicated redness of color, were susceptible of other interpretations; they might mean fiery, blazing, etc., as well as red in color, and were therefore probably suggested by the extraordinary brightness of Sirius and the strangeness with which it twinkled when near the horizon. In this position a star not only twinkles, but changes its color rapidly. This change is not sensible in the case of a faint star, but if one watches Sirius when on the horizon, it will be seen that it not only changes in appearance, but seems to blaze forth in different colors.

It seems to the writer that this conclusion of Schiaparelli is the more likely of the two. From what we know of the constitution of the stars, a

change in the color of one of these bodies in so short a period of time as that embraced by history is so improbable as to require much stronger proofs than any that can be adduced from ancient writers. In addition to the possible vagueness or errors of the original writers, we have to bear in mind the possible mistakes or misinterpretations of the copyists who reproduced the manuscripts.

THE PARALLAXES OF THE STARS.

It needs only the most elementary conceptions of space, direction and motion to see that, as the earth makes its vast swing from one extremity of its orbit to the other, the stars, being fixed, must have an apparent swing in the opposite direction. The seeming absence of such a swing was in all ages before our own one of the great stumbling blocks of astronomy. It was the base on which Ptolemy erected his proof that the earth was immovable in the center of the celestial sphere. It was felt by Copernicus to be a great difficulty in the reception of his system. It led Tycho Brahe to suggest a grotesque combination of the Ptolemaic and Copernican systems, in which the earth was the center of motion, round which the sun revolved, carrying the planets with it.

With every improvement in their instruments, astronomers sought to detect the annual swing of the stars. Each time that increased accuracy in observations failed to show it, the difficulty in the way of the Copernican system was heightened. How deep the feeling on the subject is shown by the enthusiastic title, *Copernicus Triumphans*, given by Horrebow to the paper in which, from observations by Roemer, he claimed to have detected the swing. But, alas, critical examination showed that the supposed inequality was produced by the varying effect of the warmth of the day and the cold of the night upon the rate of the clock used by the observer, and not by the motion of the earth.

Hooke, a contemporary of Newton, published an attempt to determine the parallax of the stars, under the title of "An Attempt to Prove the Motion of the Earth," but his work was as great a failure as that of his predecessors. Had it not been that the proofs of the Copernican system had accumulated until they became irresistible, these repeated attempts might have led men

to think that perhaps, after all, Ptolemy and the ancients were somehow in the right.

The difficulty was magnified by the philosophic views of the period. It was supposed that Nature must economize in the use of space as a farmer would in the use of valuable land. The ancient astronomers correctly placed the sphere of the stars outside that of the planets, but did not suppose it far outside. That Nature would squander her resources by leaving a vacant space hundreds of thousands of times the extent of the solar system was supposed contrary to all probability. The actual infinity of space; the consideration that one had only to enlarge his conceptions a little to see spaces a thousand times the size of the solar system look as insignificant as the region of a few yards round a grain of sand, does not seem to have occurred to anyone.

Considerations drawn from photometry were also lost sight of, because that art was still undeveloped. Kepler saw that the sun might well be of the nature of a star; in fact, that the stars were probably suns. Had he and his contemporaries known that the light of the sun was more than ten thousand million times that of a bright star, they would have seen that it must be placed at one hundred thousand times its present distance to shine as a bright star. If, then, the stars are as bright as the sun, they must be one hundred thousand times as far away, and their annual parallax would then have been too small for detection with the instruments of the time. Such considerations as this would have removed the real difficulty.

The efforts to discover stellar parallax were, of course, still continued. Bradley, about 1740, made observations on γ Draconis, which passed the meridian near his zenith, with an instrument of an accuracy before unequalled. He thus detected an annual swing of 20" on each side of the mean. But this swing did not have the right phase to be due to the motion of the earth; the star appeared at one or the other extremity of its swing when it should have been at the middle point, and *vice versa*. What he saw was really the effect of aberration, depending on the ratio of the velocity of the earth in its orbit to the velocity of light. It proved the motion of the earth, but in a different way from what was expected. All that Bradley could prove was that the distances of the stars must be hundreds of thousands of times that of the sun.

An introductory remark on the use of the word parallax may preface a statement of the results of researches now to be considered.

In a general way, the change of apparent direction of an object arising from a change in the position of an observer is termed *parallax*. More especially, the parallax of a star is the difference of its direction as seen from the sun and from that point of the earth's orbit from which the apparent direction will be changed by the greatest amount. It is equal to the angle subtended by the radius of the earth's orbit, as seen from the star. The simplest conception of an arc of one second is reached by thinking of it as the angle subtended by a short line at a distance of two hundred and six thousand times its length. To say that a star has a parallax of 1" would therefore be the same thing as saying that it was at a distance of a little more than two hundred thousand times that of the earth from the sun. A parallax of one-half a second implies a distance twice as great; one of one-third, three times as great. A parallax of 0"20 implies a distance of more than a million times that of our unit of measure.

The first conclusive result as to the extreme minuteness of the parallax of the brighter stars was reached by Struve, at Dorpat, about 1830. In the high latitude of Dorpat the right ascension of a star can be determined with great precision, not only at the moment of its transit over the meridian, but also at transit over the meridian below the pole, which occurs twelve hours later. He, therefore, selected a large group of stars which could be observed twice daily in this way at certain times of the year, and made continuous observations on them through the year. It was not possible, by this method, to certainly detect the parallax of any one star. What was aimed at was to determine the limit of the average parallax of all the stars thus observed. The conclusion reached was that this limit could not exceed one-tenth of a second and that the average distance of the group could not, therefore, be much less than two million times the distance of the sun; if, perchance, some stars were nearer than this, others were more distant.

By a singular coincidence, success in detecting stellar parallax was reached by three independent investigators almost at the same time, observing three different stars.

To Bessel is commonly assigned the credit of having first actually determined the parallax of a star with such certainty as to place the result

beyond question. The star having the most rapid proper motion on the celestial sphere, so far as known to Bessel, was 61 Cygni, which is, however, only of the fifth magnitude. This rapid motion indicated that it was probably among the stars nearest to us, much nearer, in fact, than the faint stars by which it is surrounded.

After several futile attempts, he undertook a series of measurements with a heliometer, the best in his power to make, in August, 1837, and continued them until October, 1838. The object was to determine, night after night, the position of 61 Cygni, relative to certain small stars in its neighborhood. Then he and his assistant, Sluter, made a second series, which was continued until 1840. All these observations showed conclusively that the star had a parallax of about $0''.35$.

While Bessel was making these observations, Struve, at Dorpat, made a similar attempt upon Alpha Lyrae. This star, in the high northern latitude of Dorpat, could be accurately observed throughout almost the entire year. It is one of the brightest stars near the Pole and has a sensible proper motion. There was, therefore, reason to believe it among the nearest of the stars. The observations of Struve extended from 1835 to August, 1838, and were, therefore, almost simultaneous with the observations made by Bessel on 61 Cygni. He concluded that the parallax of Alpha Lyrae was about one-fourth of a second. Subsequent investigations have, however, made it probable that this result was about double the true value of the parallax.

The third successful attempt was made by Henderson, of England, astronomer at the Cape of Good Hope. He found from meridian observations that the star Alpha Centauri had a parallax of about $1''$. This is a double star of the first magnitude, which, being only 30° from the south celestial pole, never rises in our latitudes. Its nearness to us was indicated not only by its magnitude, but also by its considerable proper motion.

Although subsequent investigation has shown the parallax of this body to be less than that found by Henderson, it is, up to the time of writing, the nearest star whose distance has been ascertained. The extreme difficulty of detecting movements so slight as those we have described, when they take six months to go through their phases, will be obvious to the reader. He would be still more impressed with it when, looking through a powerful telescope at any star, he sees how it flickers in consequence of the continual

motions going on in the air through which it is seen and how difficult it must be to fix any point of reference from which to measure the change of direction.

The latter is the capital difficulty in measuring the parallax. How shall we know that a star has changed its direction by a fraction of a second in the course of six months? There must be for this purpose some standard direction from which we can measure.

The most certain of these standard directions is that of the earth's axis of rotation. It is true that this direction varies in the course of the year, but the amount of the variation is known with great precision, so that it can be properly allowed for in the reduction of the observations. The angle between the direction of a star and that of the earth's axis, the latter direction being represented by the celestial pole, can be measured with our meridian instruments. It is, in fact, the north polar distance of the star, or the complement of its declination. If, therefore, the astronomer could measure the declination of a star with great precision throughout the entire year, he would be able to determine its parallax by a comparison of the measures. But it is found impossible in practice to make measures of so long an arc with the necessary precision. The uncertain and changing effect of the varying seasons and different temperatures of day and night upon the air and the instrument almost masks the parallax. After several attempts with the finest instruments, handled with the utmost skill, to determine stellar parallax from the declinations of the stars, the method has been practically abandoned.

The method now practiced is that of relative parallax. By this method the standard direction is that of a small star apparently alongside one whose parallax is to be measured, but, presumably, so much farther away that it may be regarded as having no parallax. In this assumption lies the weak point of the method. Can we be sure that the smaller stars are really without appreciable parallax? Until recent times it was generally supposed that the magnitude of the stars afforded the best index to their relative distances. If the stars were of the same intrinsic brilliancy, the amount of light received from them would, as already pointed out, have been inversely as the square of the distance. Although there was no reason to suppose that any such equality really existed, it would still remain true that, in the general average, the brighter stars must be nearer to us than the fainter ones. But when the

proper motions of stars came to be investigated, it was found that the amount of this motion afforded a better index to the distance than the magnitude did.

The diversity of actual or linear motion is not so wide as that of absolute brilliancy. Stars have, therefore, in recent times, been selected for parallax very largely on account of their proper motion, without respect to their brightness. It is now considered quite safe to assume that the small stars without proper motion are so far away that their parallax is insensible.

Ever since the time of Bessel the experience of practical astronomers has tended toward the conclusion that the best instrument for delicate measurements like these is the heliometer. This is an equatorial telescope of which the object glass is divided along a diameter into two semicircles, which can slide along each other. Each half of the object glass forms a separate image of any star at which the telescope may be pointed. By sliding the two halves along each other, the images can be brought together or separated to any extent. If there are two stars in proximity, the image of one star made by one-half of the glass can be brought into coincidence with that of the other star made by the other half. The sliding of the two halves to bring about this coincidence affords a scale of measurement for the angular distance of the two stars.

The most noteworthy forward steps in improving the heliometer are due to the celebrated instrument-makers of Hamburg, the Messrs. Repsold, aided by the suggestions of Dr. David Gill, astronomer at the Cape of Good Hope. The latter, in connection with his coadjutor, Elkin, made an equally important step in the art of managing the instrument and hence in determining the parallax of stars. The best results yet attained are those of these two observers and of Peter, of Germany.

Yet more recently, Kapteyn, of Holland, has applied what has seemed to be the unpromising method of differences of right ascension observed with a meridian circle. This method has also been applied by Flint, at Madison, Wis. Through the skill of these observers, as well as that of Brünnow and Ball, in applying the equatorial telescope to the same purposes, the parallax of nearly 100 stars has been measured with some approach to precision.

A rival method to that of the heliometer has been discovered in the photographic telescope. The plan of this instrument, and its application to such purposes as this, are extremely simple. We point a telescope at a star and set the clock-work going, so that the telescope shall remain pointed as exactly as possible in the direction of the star. We place a sensitized plate in the focus and leave it long enough to form an image both of the particular star in view and of all the stars around it. The plate being developed, we have a permanent record of the relative positions of the stars which can be measured with a suitable instrument at the observer's leisure. The advantage of the method consists in the great number of stars which may be examined for parallax, and in the rapidity with which the work can be done.

The earliest photographs which have been utilized in this way are those made by Rutherford in New York during the years 1860 to 1875. The plates taken by him have been measured and discussed principally by Rees and Jacoby, of Columbia University. Before their work was done, however, Pritchard, of Oxford, applied the method and published results in the case of a number of stars.

One of the pressing wants of astronomy at the present time is a parallactic survey of the heavens for the purpose of discovering all the stars whose parallax exceeds some definable limit, say $0''.1$. Such a survey is possible by photography, and by that only. A commencement, which may serve as an example of one way of conducting the survey, has been made by Kapteyn on photographic negatives taken by Donner at Helsingfors.

These plates cover a square in the Milky Way about two degrees on the side, extending from $35^{\circ} 50'$ in declination to $36^{\circ} 50'$, and from 20h. 1m. in R. A. to 20h. 10m. 24s. Three plates were used, on each of which the image of each star is formed twelve times. Three of the twelve impressions were made at the epoch of maximum parallactic displacement, six at the minimum six months later, and three at the following maximum. The parallaxes found on the plates can only be relative to the general mean of all the other stars, and must therefore be negative as often as positive. The following positive parallaxes, amounting to $0''.1$, came out with some consistency from the measures:

Star,	Mag.	R.A.	Dec.	Par.
B. D.,	8.6	20h.,	$+35^{\circ}.5$	$+0''.11$

3972			2m. 0s.		
Star,	Mag.	R.A.	Dec.	Par.	
B. D.,	7.1	20h.,	+36°.1	+0".18	
3883		2m. 3s.			
Star,	Mag.	R.A.	Dec.	Par.	
B. D.,	9.2	20h.,	+35°.4	+0".10	
4003		4m. 58s.			
Star,	Mag.	R.A.	Dec.	Par.	
B. D.,	7.0	20h.,	+36°.3	+0".10	
3959		9m. 14s.			

Against these are to be set negative parallaxes of -0".09, -0".08 and several a little smaller, which are certainly unreal.

The presumption in favor of the actuality of one or more of the above positive values, which is created by their excess over the negative values, is offset by the following considerations: The area of the entire sky is more than 40,000 square degrees, or 10,000 times the area covered by the Helsingfors plates. We cannot well suppose that there are 1,000 stars in the sky with a parallax of 0".10, or more without violating all the probabilities of the case. The probabilities of the case are therefore against even one star with such a parallax being found on the plates. Yet the cases of these four stars are worthy of further examination, if any of them are found to have a sensible proper motion.

On an entirely different plan is a survey just concluded by Chase with the Yale heliometer. It includes such stars having an annual proper motion of 0".05 or more as had not already been measured for parallax. The results, in statistical form, are these:

- 2 stars have parallaxes between +9".20 and +0".25.
- 6 stars have parallaxes between +0".15 and +0".20.
- 11 stars have parallaxes between +0".10 and +0".15.
- 24 stars have parallaxes between +0".05 and +0".10.

34	stars have parallaxes between $+0''.00$ and $+0''.05$.
8	stars have parallaxes between $-0''.05$ and $0''.00$.
5	stars have parallaxes between $-0''.10$ and $-0''.05$.
<u>2</u>	stars have parallaxes between $-0''.15$ and $-0''.10$.
92,	total number of stars.

It will be understood that the negative parallaxes found for fifteen of these stars are the result of errors of observation. Assuming that an equal number of the smaller positive values are due to the same cause, and subtracting these thirty stars from the total number, we shall have sixty-two stars left of which the parallax is real and generally amounts to $0''.05$, more or less. The two values approximating to $0''.25$ seem open to little doubt. We might say the same of the six next in the list. The first two belong to the stars 54 Piscium and Weisse, 17h., 322.

DISCUSSION AND CORRESPONDENCE.

THE MEETINGS OF THE AMERICAN ASSOCIATION.

The American Association for the Advancement of Science has a membership ranging from 1,900 to 2,000. Of this number probably at no one time was there an aggregate of 300 persons present at the recent annual meeting in New York.

When the Association meets in an Eastern city the attendance is generally twice if not three times as large as when it convenes in the West. So little was made of the recent meeting, locally or officially, that an intelligent resident of the city remarked: "Why, I intended to have attended some of the meetings, but seeing no reference in the daily papers, it entirely escaped my mind."

Of the 2,000 members, about 800 are fellows; the 1,200 and more registered as members are, presumably, persons devoting little or no time to independent research along scientific lines, but persons who while not actively so engaged are more than ordinarily interested in the discussion of scientific topics. These have in the past paid dues and attended the meetings of the Association with more or less regularity. It is a question in the minds of some of the 1,200 if their attendance at the meetings is desired. Their membership, so far as it relates to the five dollars initiation fee and three dollars dues, is without question acceptable, and to persons reading papers in the various sections their presence is preferable to empty seats, but in view of the fact that during recent years the management of the Association has eliminated, so far as possible, the popular features of the general programme, the question is reasonably asked: "Does the management desire the attendance of the 1,200, or is their financial support all that is desired?"

It was stated some years ago that the purpose of the Association was to furnish not only an occasion for scientists to present original papers, but also to interest the public by holding the meetings annually in different parts of the country; but if attendance is not secured (by preparation and publication of interesting features of a programme) no great interest will be awakened by a meeting held in any part of the country.

I should like to suggest the following ways of increasing the interest of the meetings:

The general daily sessions might be made occasions of rare interest by the introduction of prominent men of science who would make at least brief remarks. This would make it possible for those who have limited time to become familiar with the faces of those whom they would like to know, and the little 'sample' of scientific thought thrown out would doubtless awaken desire for more.

It will be objected that the meetings of the council immediately preceding the general session prevent holding an official meeting at that hour. The public and the 1,200 would care little whether the session were official or unofficial so it were interesting and instructive.

The officers of the several sections could easily secure distinguished representatives of their respective sciences to give brief addresses followed by discussion, and thus the morning hour would prove an attraction to citizens and others who might be unable to attend the sessions following.

Again, citizens, where the meetings are held, would be pleased to provide excursions to points of local interest and extend social courtesies, if they were given in return the mental food in digestible form, with which the Association is so amply supplied.

It remains with the management to decide whether attendance shall be restricted to the few actively engaged in scientific pursuits, or whether it shall include the 1,200 and more who would be glad to avail themselves of the benefits of a programme suited to average scholarship and intellectual capacity.

There is no better medium for discussion of the above views than through the widely read pages of THE POPULAR SCIENCE MONTHLY.

M. E. D. TROWBRIDGE.

Detroit, Mich.

[The questions brought up by our correspondent have been carefully considered by all those who are interested in the American Association for the Advancement of Science. When the Association was founded fifty years ago there was no division into sections; the papers and discussions were intelligible and interesting to all members. At that time there were but few members, the scientific life of the country was small, and it was a privilege for a city to entertain the Association. But fifty years have brought changes in many directions. Specialization in science has become essential for its further progress, and it has been necessary to divide the Association into numerous sections and to found special societies. Hospitality can now only be provided at great expense, and Eastern cities no longer regard it as a privilege to entertain the numerous societies that gather within their hotels. The newspapers do not regard a meeting of the Association as an important event and will not devote space to it.

The Association must do the best it can to adapt itself to existing conditions. The recent meeting in New York had perhaps the largest attendance of scientific men of any in the history of the Association with the exception of the anniversary meeting two years ago, but New York City, especially in the month of June, is not a desirable place for social functions. It is not reasonable for a member interested in science as an amateur to expect to purchase for three dollars a week's entertainment. His dues secure reduced railway and hotel rates; he can meet his friends and become acquainted with scientific men; he can always find on the programme papers that are of interest; he receives the annual volume of 'Proceedings' and the weekly journal, 'Science,' the cost of which is five dollars per year. But apart from these direct returns, he is surely repaid for membership by knowing that he is one of those who are united for the advancement of science in America.—EDITOR, POPULAR SCIENCE MONTHLY.]

THE COLOR RED.

TO THE EDITOR OF THE POPULAR SCIENCE MONTHLY: Mr. Havelock Ellis, in your August number, in 'The Psychology of Red,' says, 'A great many different colors are symbolical of mourning ... but so far as I am aware, red never.' The following may possibly be of interest in this connection:

"Our English Pliny, Bartholomew Glantville, who says after Isydorus, 'Reed clothes ben layed upon deed men in remembrance of theyr hardynes and boldnes, whyle they were in theyr bloudde.' On which his commentator, Batman, remarks: 'It appereth in the time of the Saxons that the manner over their dead was a red cloath, as we now use black. The red of valiauncie, and that was over kings, lords, knights and valyaunt souldiers; white over cleargie men, in token of their profession and honest life, and over virgins and matrons.'"—(Dr. Furness's Variorum. Merchant of Venice, p. 56.)

CHAS. E. DANA.

University of Pennsylvania.

SCIENTIFIC LITERATURE.

MENTAL AUTOMATISM.

A recent work by Prof. Th. Flournoy, entitled ‘Des Indes à la Planète Mars,’^G contains an account of a remarkable case of mental automatism, or sub-conscious personality. The subject is a young woman of about thirty years, apparently in good health, but always of a nervous and imaginative type. She developed tendencies towards lapses of consciousness, hallucinations and automatic actions; and these developed later, under the inspiration of spiritualistic séances, into a series of cycles, or automatic dramas, in which the medium speaks or writes and acts under the influence of several diverse subordinate personalities. In one of these cycles—which, it must be understood, are continued from one sitting to another, although in her intermediate normal life she knows nothing of what she has said or done in the trance—she becomes Marie Antoinette, and is said to act the part with unusual dramatic skill. In another and far more elaborate cycle the scene is transferred to the planet Mars, and the houses, scenery, plants and animals, peoples, customs and goings-on of the planet are described; sketches are made, and reproduced in the volume, of these extra-mundane appearances. Still more remarkable is the appearance of the Martian language, which in successive séances the subject hears, speaks, sees before her in space, and, in the end, even writes. From the mystery of Mars we are taken to the equally mysterious Hindu cycle; here the medium becomes an Indian princess of the fifteenth century, reveals her history and that of her associates in the Oriental life, tells of herself as Simandini; of Sivrouka, her prince, who reigned over Kanara and built in 1401 the fortress of Tschandraguiri. Wonderful to relate, these names are not fictitious, but are mentioned by one De Marlès in a volume published in 1828; the author, however, does not enjoy a high reputation as a historian. When occasional

utterances of the Hindu princess are taken down, they are found in part to have close resemblance to Sanskrit words; while in her normal condition the medium is as ignorant of Sanskrit as she is of any language except French, and is entirely ignorant of both De Marlès and the people of India five hundred years ago. Surely this is a tale, bristling with mystery and improbability, which, if told carelessly or with a purpose, we should dismiss as a willful invention! M. Flournoy has been unusually successful in revealing the starting points of the several automatisms and of connecting them with intelligible developments of the medium's mental life; and the manifestations, though they remain as remarkable examples of unconscious memory and elaboration of ideas, nowhere transcend these limitations. The sketches of Martian scenery are clearly Japanesque or vaguely Oriental; the Martian language is pronounced an 'infantile' production, and is clearly modeled after the French, the characters being the result of an attempt to make them as oddly different from our own as possible; the Sanskrit goes no farther than what one could get from a slight acquaintance with a Sanskrit grammar; and while there is a copy of De Marlès in the Geneva Library (where the medium lives), no connection can be established between either De Marlès or the grammar and the subject of this study. Most of this knowledge of these remarkable sub-conscious states would have been impossible were it not for 'spirit control' of one Leopold, who, in accordance with the doctrine of reincarnation which permeates the several cycles, was in his life the famous Cagliostro. By suitable suggestion, Leopold can be induced to make the entranced subject speak, write, draw, or interpret her strange messages from other worlds; and where Leopold says 'nay' all progress is stopped. This case has many analogies with other cases that have been recorded, but goes beyond most of them in the complexity and bizarre character of the unconscious elaborations and in the feats of memory and creative imagination which it entails. These accomplishments, it should be well understood, never appeared suddenly or fully developed, but only after a considerable period of subliminal preparation, and then only hesitatingly, and little by little, just as is the case with the acquisitions of normal consciousness; and all these acquisitions bear unmistakable marks of belonging to the same person. The special value of this account thus lies in the accuracy of the description and the success with which the account has been made thoroughly intelligible and significant.

[G](#) The book has just been published by the Harpers in an English version, under the title 'She Lived in Mars.'

THE MOSQUITOES OF THE UNITED STATES.

Dr. L. O. Howard, the entomologist of the United States Department of Agriculture, has just published a bulletin entitled, "Notes on the Mosquitoes of the United States: Giving some Account of their Structure and Biology, with Remarks on Remedies." The author has, for some years, been interested in the general subject of the biology of mosquitoes and of remedies to be used against them, and has brought together in this bulletin all the published and unpublished notes which he has been collecting during this period. The bulletin contains synoptic tables of all North American mosquitoes, prepared by Mr. D. W. Coquillett, and gives detailed facts regarding the geographical distribution of the different species mentioned. All the five North American genera are illustrated and full, illustrated accounts are given of the life history of the two principal genera, *Culex* and *Anopheles*, as studied in *Culex pungens* and *Anopheles quadrimaculatus*. The author calls special attention to the two genera of large mosquitoes, *Psorophora* and *Megarhinus*, and urges the importance of the study of these two genera, especially by physicians in the South, in regard to their possible relation to the spread of malaria. Considerable space is given to the subject of remedies, the principal ones considered being kerosene on breeding pools, the introduction of fish in fishless ponds, the artificial agitation of water and general community work. It is clearly shown not only that the mosquito may be, in many localities, readily done away with at comparatively slight expense, but that by careful work many malarious localities may be made healthy. The subject of mosquitoes and malaria is not discussed in the bulletin, which contains simply references to available papers on this subject, like the article by Dr. Patrick Manson, published in *THE POPULAR SCIENCE MONTHLY* for July, the aim of the author being to bring together all available facts about the mosquitoes of the United States, in

order to assist physicians who are studying the malarial relation from the point of view of local conditions.



THE PROGRESS OF SCIENCE.

The British, French and German Associations for the Advancement of Science have held their annual meetings in the course of the past month. In each of these countries and in most other European countries, as well as in America, there are migratory scientific congresses of the same general character. As these have grown up somewhat independently, they evidently meet a common need. Science cannot be advanced by a man working independently and in isolation. The printing press was essential to the beginnings of modern science, while at the same time it was usual for the scientific student to travel from place to place that he might learn and teach. Then in the seventeenth and eighteenth centuries, as the cultivation of science became more general, royal academies were founded. The Royal Society was established at London in 1660 under the patronage of Charles II., the Academy of Sciences at Paris in 1666 under Louis XIV., the Royal Academy at Berlin in 1700 under Frederick I., the Imperial Academy at St. Petersburg in 1724 under Peter the Great, and in other cities similar academies were founded under similar auspices. Then in the first half of the present century, as science continued to grow, the more democratic organizations for the advancement of science were established. The Society of German Scientific Men and Physicians was formed, chiefly through the efforts of Humboldt, in 1822; the Swiss Association in 1829, and the British Association in 1831. Our own Association was established in 1847, but was then the intergrowth of a society dating from 1840. These associations are significant of the spread of science among all the people. Science is no longer the concern of a few men under royal patronage, but the two great movements of the present century—the growth of democracy and the growth of science—have united for their common good.

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The British Association held its annual meeting at Bradford, beginning on September 5, under the presidency of Sir William Turner, professor of anatomy in the University of Edinburgh. We are able to publish, from a copy received in advance of its delivery, his presidential address, which traces the growth during the present century of knowledge regarding fundamental biological problems. The addresses of the presidents before the sections are usually written in a way that can be readily understood by those who are not specialists, and are consequently of greater interest to a general audience than some of the corresponding addresses before the American Association. The addresses at Bradford were: Before the section of mathematical and physical science Dr. Joseph Larmor discussed recent developments of physics with special reference to the extent to which explanation can be reduced purely to description; before the section of chemistry Prof. H. W. Perkin argued that radical changes should be made in the methods of teaching inorganic chemistry; before the section of geology Prof. W. J. Sollas spoke of the development of the earth, including the different critical periods in its history; before the section of zoölogy Dr. R. H. Traquair chose as his subject the bearing of fossil fishes on the doctrine of descent; before the section of geography Sir George Robertson considered certain geographical aspects of the British Empire and the changes brought about by improved means of intercommunication; before the section of economic science and statistics Major P. G. Craigie spoke of the use of statistics in agriculture; before the section of mechanical science Sir Alexander Binnie traced the historical development of science; before the section of anthropology Prof. John Rhys dealt with the ethnology of the British Isles, with special reference to language and folk-lore; before the section of botany Prof. Sidney H. Vines reviewed the development of botany during the present century. In addition to these addresses, evening discourses were given by Prof. Francis Gotch on 'Animal Electricity,' and by Prof. W. Stroud on 'Range Finders.' The usual lecture to workingmen was given by Prof. Sylvester P. Thompson, his subject being 'Electricity in the Industries.'

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Bradford is situated in the coal regions, and is an industrial center devoted especially to the manufacture of textiles. More attention was paid

to local interests than is usual at the meetings of the American Association. An exhibit was arranged to show the development of the elaborate fabrics from the unwashed fleeces, and another consisting of a collection of carboniferous fossils found in the neighborhood. A joint discussion was arranged between the sections of zoölogy and botany on the conditions which existed during the growth of the forests which supplied material for the coal, and there were a number of papers devoted to the coal measures and the fossils which they contain. Another subject connected with the place of meeting was the report of the committee on the underground water system in the carboniferous limestone. By the use of chemicals the course of the underground waters has been traced, including their percolation through rock fissures, and excursions were made to the site of the experiments. The local industries received treatment from several sides. Among other discussions of more than usual interest was that on 'Ions' before the physical section and on 'What is a Metal?' before the chemical section. Features of popular interest were accounts of adventures in Asia, Africa and the Antarctic regions, by Captain Deasy, Captain George and Mr. Borchgrevinck, respectively, and Major Ross's paper on 'Malaria and Mosquitoes.'

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The French Association met at Paris in the month of August, with the numerous other congresses. General Sebert, in his presidential address, reviewed the progress of the mechanical industries during the century and devoted the last third of his time to a discussion of international bibliography, but without mentioning the International Catalogue which now seems to be an accomplished fact. The secretary of the Association, in his review of the year, devoted special attention to the joint meetings of the British and French associations last summer at Dover and Calais. The treasurer was able to make a report that the treasurers of other national associations will envy. The capital is over \$250,000, and the income from all sources about \$17,000, of which about \$3,000 was awarded for the prosecution of research and to defray the cost of publication of scientific monographs. The national association for the advancement of science of Germany—the 'Gesellschaft deutscher Naturforscher und Aerzte'—held its annual meeting at Aachen toward the middle of September. An account of

the proceedings has not yet reached us, but the congresses are always largely attended and the combination of addresses of general interest, of special papers before the numerous sections and of social functions, is perhaps more effective than in any other society. It also appears to be a considerable advantage for medical men and scientific men to meet together.

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While from the scientific point of view the present century has been notable for the development of national associations for the advancement of science, its latter decades have witnessed a growth of international scientific meetings which may be expected to become dominant in the twentieth century. There are at least one hundred congresses, having more or less reference to science, meeting at Paris during the present summer. Perhaps the most noteworthy of these, from the point of view of the organization of science, is the International Association of Academies, which was established last year at a conference held at Wiesbaden. In this Association eighteen of the great academies of the world, including our own National Academy of Sciences, have been united to promote the interests of science. Literature is also included—of the eighteen academies, twelve include in their scope both science and literature, four are devoted to science only and two to literature only. It is planned to have a general meeting every three years, to which each academy will send as many delegates as it regards as desirable, though each academy will have but one vote. In the interval between the general meetings, the business of the Association is to be directed by a committee, on which each academy is represented. The object of the Association is to plan and promote scientific work of international interest which may be proposed by one of the constituent academies, and generally to promote scientific relations between different countries. The Royal Society has proposed the measurement, by international coöperation, of an extended arc of the meridian in the interior of Africa.

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The International Congress of Physics marked an advance owing to the fact that it met for the first time this year, and it appears that the proceedings were of unusual interest. This was in a large measure due to the

arrangements of the French Physical Society, which did not simply make up a programme from a mass of heterogeneous researches, but secured some eighty reports on the present condition of physical science. These were prepared by many of the leading physicists of the world and when published—as they are about to be in three volumes—will set forth the condition of the science with completeness and authority. There were in all seven sections. In the first, which was concerned with measurement, in addition to numerous reports several propositions were brought forward in regard to units, which, being international in character, are specially fitted for discussion at such a congress. As the members, however, were not in most cases delegates from governments and scientific bodies, no definite action was taken, though some recommendations were made. The decimalization of time was not recommended, nor was the proposal to give a name to units of velocity and acceleration. It was, however, decided that the ‘Barrie’ be adopted as the unit of pressure. The other sections were for mechanical physics, for optics, for electricity, for magneto-optics and radio-activity, for cosmical physics and for biological physics. Among the reports and papers of commanding interest only two can be mentioned—the introductory address by M. Poincaré, discussing the relations between experimental and mathematical physics, and one by Lord Kelvin on the waves produced in an elastic solid traversed by a body acting on it by attraction or repulsion, in which, from a strictly mathematical point of view, he advanced the hypothesis of a movable atom surrounded by an immovable ether. In addition to various receptions, a session was held at the Sorbonne, where Messrs. Becquerel and Curie gave demonstrations with radio-active substances, and one at the Ecole Polytechnique, where President Cornu showed apparatus which had been used in the determination of the velocity of light. At the close of the congress the foreign secretaries placed a crown on the tomb of Fresnel.

* * * * *

While a physical congress was meeting at Paris this year for the first time, the Geological Congress, which was one of the first international congresses to be organized, held its eighth session, beginning on August 16. America, in spite of the number and importance of the inventions it has given to the world, has not as yet done its share for the advancement of

physical science, but in geology it occupies a foremost place. It was natural, therefore, that while American physicists were scarcely represented on the programme of the Physical Congress, they occupied a prominent place on the programme of geological papers. Among the three hundred members present, the representation from America included Messrs. Stevenson, Hague, Osborn, Ward, Willis, White, Cross, Scott, Todd, Kunz, Choquette, Adams, Mathew and Rice, and they presented a number of the more important papers. M. Karpinsky, the retiring president, gave the opening address, which was followed by an address of welcome by M. Gaudry, the president of the congress. A geological congress can offer special attractions in the way of excursions, and these were admirably arranged on the present occasion—both the shorter excursions to the classic horizons in the neighborhood of Paris and the more extended ones that followed the close of the meeting. The guide for the twenty long excursions and numerous shorter trips, prepared by the leading French geologists, was an elaborately illustrated volume representing the present condition of our knowledge of French geology. The ninth geological congress will be held at Vienna three years hence.

* * * * *

The International Congress of Mathematics met for the second time at Paris, though there had been a preliminary meeting on the occasion of the Chicago Exposition. There were about two hundred and twenty-five mathematicians in attendance, including seventeen from the United States. M. Poincaré presided, and the vice-presidents, some of whom were not present, were Messrs. Czuber, Gordon, Greenhill, Lindelöf, Lindemann, Mittag-Leffler, Moore, Tikhomandritzky, Volterra, Zeuthen and Geiser. The sections and their presiding officers were as follows: (1) Arithmetic and Algebra: Hilbert; (2) Analysis: Painlevé; (3) Geometry: Darboux; (4) Mechanics and Mathematical Physics: Larmor; (5) Bibliography and History: Prince Roland Bonaparte; (6) Teaching and Methods: Cantor. Valuable papers were presented by M. Cantor on works and methods concerned with the history of mathematics, by Professor Hilbert on the future problems of mathematics and by Professor Mittag-Leffler on an episode in the life of Weierstrass, but the programme appears to have been not very full nor particularly interesting. Time was found for a half-day's

discussion of a universal language, but not to carry into effect the plans begun at Zurich three years ago for a mathematical bibliography. The next congress will meet four years hence in Germany, probably at Baden-Baden.

* * * * *

The untimely death of James Edward Keeler, director of the Lick Observatory, is a serious blow to astronomy and to science. Born at La Salle, Ill., forty-three years ago, he was educated at the Johns Hopkins University and in Germany. When only twenty-one years old he observed the solar eclipse of 1878, and drew up an excellent report. Three years later he was a member of the expedition to Mt. Whitney under Professor Langley, whose assistant he had become at the Allegheny Observatory, and whose bolometric investigations owe much to him. He became astronomer at the Lick Observatory while it was in course of erection, and in 1891 he succeeded Professor Langley as director of the Allegheny Observatory. He was called to the directorship of the great Lick Observatory in 1898. Keeler's work in astrophysics, including his photographs of the spectra of the red stars and his spectroscopic proof of the meteoric constitution of Saturn's rings, demonstrated what he could accomplish at a small observatory unfavorably situated. At Mt. Hamilton he was able in the course of only two years to organize thoroughly the work of the Observatory, and to adapt the Crossley reflector for his purpose, taking photographs of the nebulae that have never been equalled. His discovery that most nebulae have a spiral structure is of fundamental importance. It is not easy to overestimate what might have been accomplished by Keeler in the next twenty or thirty years, both by his own researches and by his rare executive ability, for it must be remembered that his genius as an investigator was rivaled by personal qualities which made his associates and acquaintances his friends.

* * * * *

Henry Sidgwick, late Knightbridge professor of moral philosophy at Cambridge, died on August 28, at the age of sixty-two years. There are usually not many events to record in the life of a university professor, but Sidgwick had an opportunity to prove his character when he resigned a fellowship in Trinity College because holding it implied the acceptance of

certain theological dogmas. Liberalizing influences, however, were at work, of which he himself was an important part, and he was later elected honorary fellow of the same college, and in 1883 became professor of moral philosophy in the University. Sidgwick published three large works—‘Methods of Ethics’ (1874), ‘Principles of Political Economy’ (1883) and ‘Elements of Politics’ (1891)—in addition to a great number of separate articles. All these works, especially the ‘Ethics,’ show an intellect to a rare degree both subtle and scientific. There was a distinction and a personal quality in what he wrote that made each book or essay a work of art, as well as a contribution to knowledge. Those who knew Professor Sidgwick—and the writer of the present note regards it as one of the fortunate circumstances of his life that he was for several years a student under him—realize that the qualities of the man were even more rare than those of the author. His hesitating utterance, always ending in exactly the right word, but represented the caution and correctness of his thought. Subtlety, sincerity, kindness and humor were as happily combined in his daily conversation as in his writings. It is said that he was never ‘entrapped into answering a question by yes or no,’ but his deeds and his influence were positive without qualification or limitation.

* * * * *

Friedrich Wilhelm Nietzsche, who died on almost the same day as Sidgwick, was also a writer on ethics and once a university professor, but the life and writings of the two men present a strange contrast. Where Sidgwick’s touch was light as an angel’s, Nietzsche trampled like a bull; the one was the embodiment of reason, caution, consideration and kindness, the other represented paradox, recklessness, violence and brute force. Still Nietzsche deserves mention here, as his ethical views, based on the Darwinian theory of the survival of the fit, are not unlikely to be urged hereafter by saner men, and to become an integral part of ethics when ethics becomes a science. As a matter of fact, after resigning his professorship at Zurich, and even while writing his remarkable books, Nietzsche suffered from brain disease, and during the past eleven years his reason was completely lost.

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Transcribers' Notes

Punctuation, hyphenation, and spelling were made consistent when a predominant preference was found in this book; otherwise they were not changed.

Simple typographical errors were corrected; occasional unbalanced quotation marks retained.

Ambiguous hyphens at the ends of lines were retained.

Text uses both “angakok” and “angakut”; both retained.

Page [561](#): Footnote ‘A’ was not referenced in the text; Transcriber attributed it to the title of the article.

Page [564](#): Transcriber’s transliteration of Greek text shown in {curly braces}.

Page [575](#): “milkrosomen” was printed that way.

Page [577](#): Text uses both “Quan-si” and “Quansi”; both retained.

Page [580](#): “grewsome” was printed that way.

Page [588](#): “where in 1861” was printed that way, but likely is a misprint, perhaps for “1681”, as the next paragraph says that the plague disappeared from Europe in the eighteenth century.

[Pages 601, 602](#): use both “Vallee” and “Vallée”; both retained.

Page [645](#): “Moreever” was printed that way.

[Pages 669](#), [672](#): “Wood’s Holl” was spelled that way when this issue of the magazine was published.

Page [672](#): Missing page reference “76” added to “Vickery” entry, based on examination of the May, 1900 issue.

Links in the Index to the other five issues of Volume 56 will not function on all reading devices. When they do work, the first use in each session may lead to the beginning of the other issue, rather than to the specified page, but subsequent uses of any link in the same session should lead to the specified page. Links in the Index were not checked for accuracy, and the external links were not checked for validity. Due to the way two of those eBooks were prepared, links to the first article in the first two issues lead to the second page of those articles.

The other five issues of Volume 57 also are available at Project Gutenberg. The eBook numbers are [47219](#), [47227](#), [47238](#), [47261](#), and [47281](#).

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